



*Mechanical
Engineering
Department*

**KONYA
TÜRKİYE**



ENGINEERING MECHANICS STATICS

| SIXTH EDITION |

J. L. MERIAM

L. G. KRAIGE



*Mechanical
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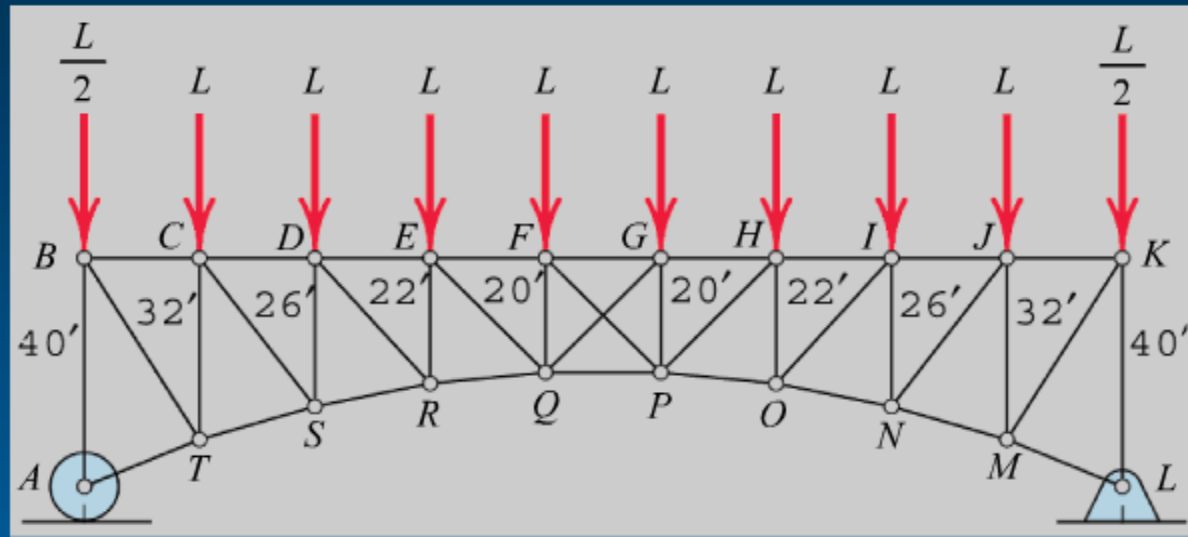
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4

STRUCTURES

4. Structures



Introduction

Frames & Machines

Plane Trusses

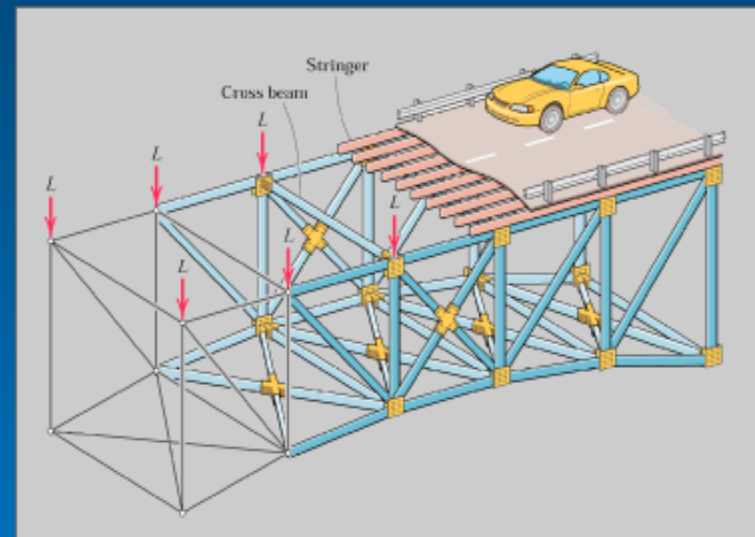
a. Method of Joints

b. Method of Sections

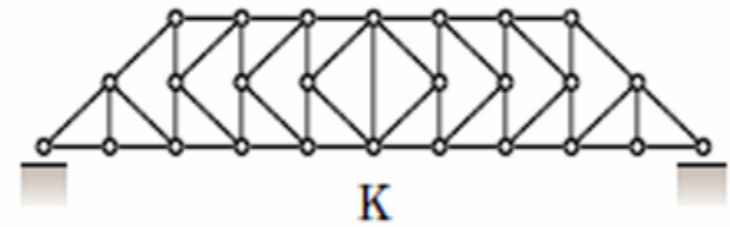
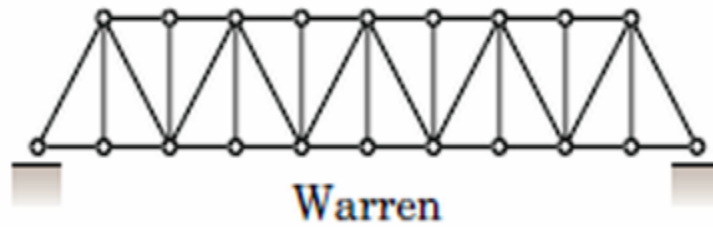
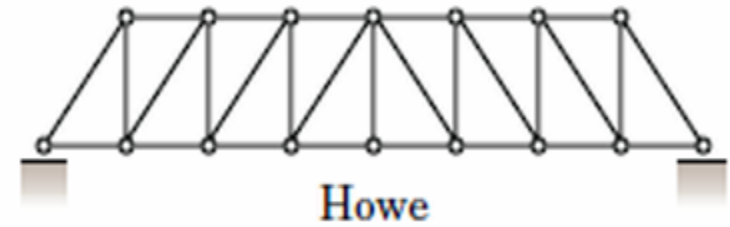
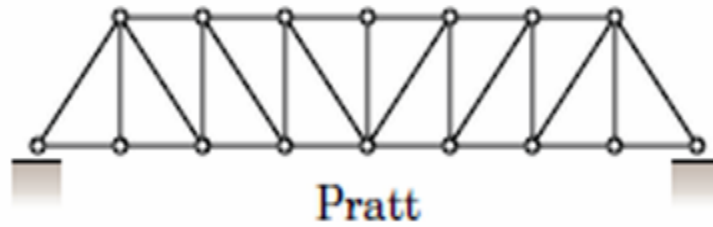
Introduction

Previously, we have been concerned with the *external* forces acting on a body. In this section, however, we study the forces *internal* to a structure. Determining the load on each internal member of a truss or frame allows one to then *design* that particular member.

As an example, consider the typical bridge structure shown in the figure. Each vertical side of the structure is a plane truss,

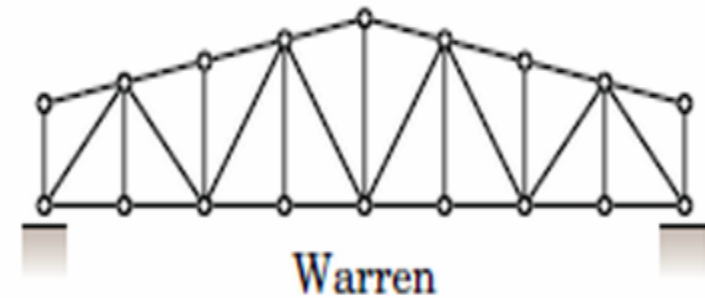
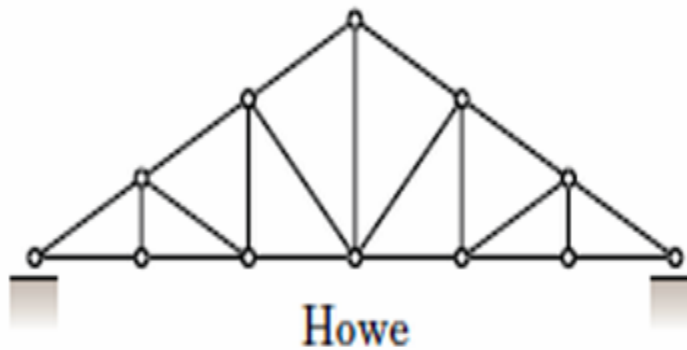
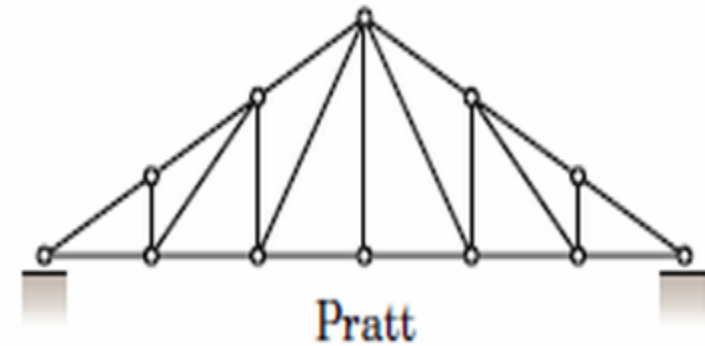
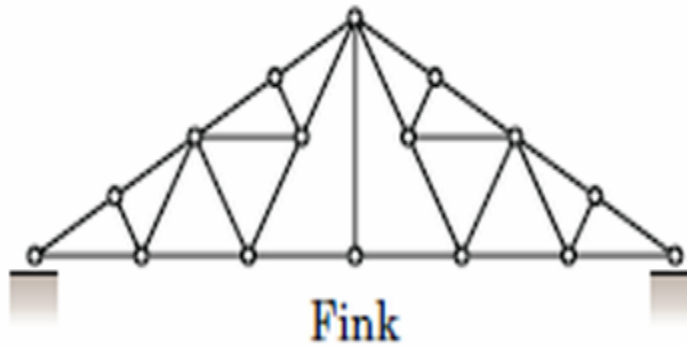


a truss being a framework composed of members joined at their ends. A simplified *model* of the structure is indicated at the left end of the illustration. The forces **L** represent the portions of the weights of the roadway, vehicles, stringers, and cross beams which are transferred to the truss joints. Note that the joints are modeled as simple pins, even though the joints of the actual structure may be gusset plates or welded connections.



Commonly Used Bridge Trusses



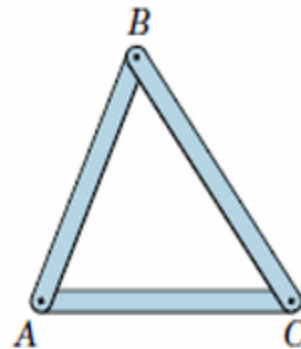


Commonly Used Roof Trusses

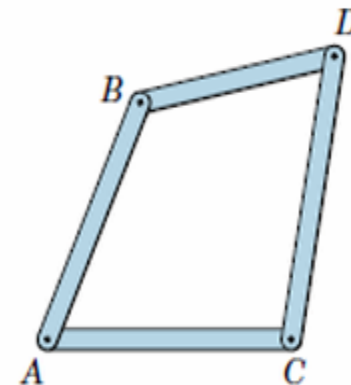


4/2 PLANE TRUSSES

A framework composed of members joined at their ends to form a rigid structure is called a *truss*. Bridges, roof supports, derricks, and other such structures are common examples of trusses. Structural members commonly used are I-beams, channels, angles, bars, and special shapes which are fastened together at their ends by welding, riveted connections, or large bolts or pins. When the members of the truss lie essentially in a single plane, the truss is called a *plane truss*.



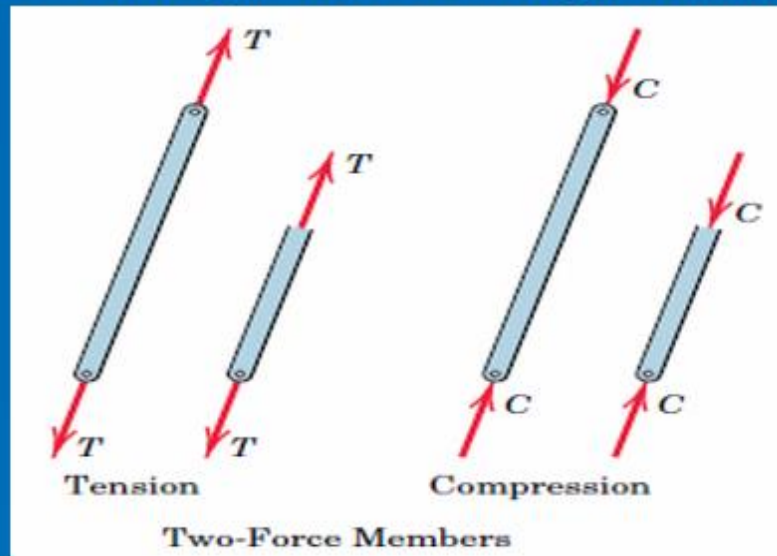
(a)



(b)

To design a truss we must first determine the forces in the various members and then select appropriate sizes and structural shapes to withstand the forces. Several assumptions are made in the force analysis of simple trusses. First, we assume all members to be *two-force members*. A two-force member is one in equilibrium under the action of two forces only, as defined in general terms with Fig. 3/4 in Art. 3/3. Each member of a truss is normally a straight link joining the two points of application of force. The two forces are applied at the ends of the member and are necessarily equal, opposite, and *collinear* for equilibrium.

The member may be in tension or compression, as shown in Fig. 4/4. When we represent the equilibrium of a portion of a two-force member, the tension T or compression C acting on the cut section is the same



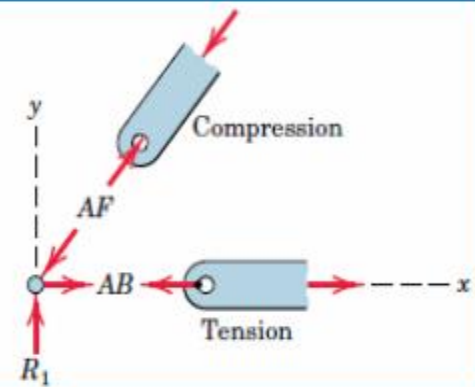
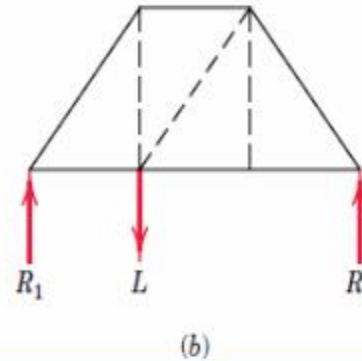
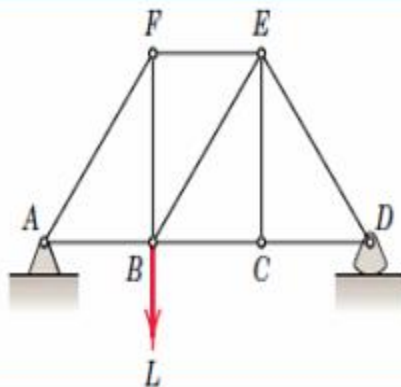
Truss Connections and Supports

When welded or riveted connections are used to join structural members, we may usually assume that the connection is a pin joint if the centerlines of the members are concurrent at the joint as in Fig. 4/5.

We also assume in the analysis of simple trusses that all external forces are applied at the pin connections. This condition is satisfied in most trusses. In bridge trusses the deck is usually laid on cross beams which are supported at the joints, as shown in Fig. 4/1.

For large trusses, a roller, rocker, or some kind of slip joint is used at one of the supports to provide for expansion and contraction due to temperature changes and for deformation from applied loads. Trusses and frames in which no such provision is made are statically indeterminate, as explained in Art. 3/3. Figure 3/1 shows examples of such joints.

Two methods for the force analysis of simple trusses will be given. Each method will be explained for the simple truss shown in Fig. 4/6*a*. The free-body diagram of the truss as a whole is shown in Fig. 4/6*b*. The external reactions are usually determined first, by applying the equilibrium equations to the truss as a whole. Then the force analysis of the remainder of the truss is performed.



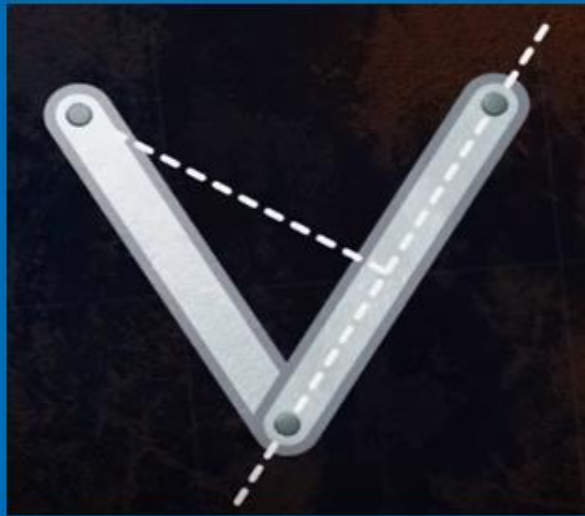
zero force members

only component in the
vertical direction



2/2



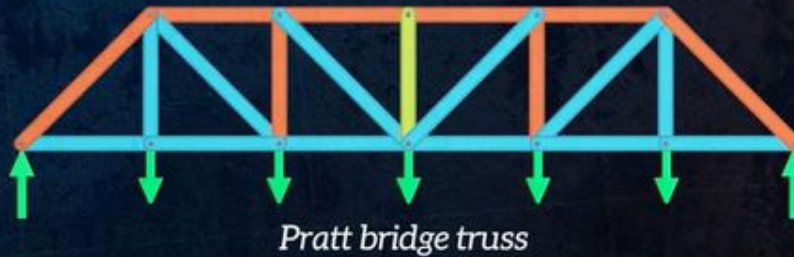


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- vertical members in tension
- diagonal members in compression
- not very cost-effective

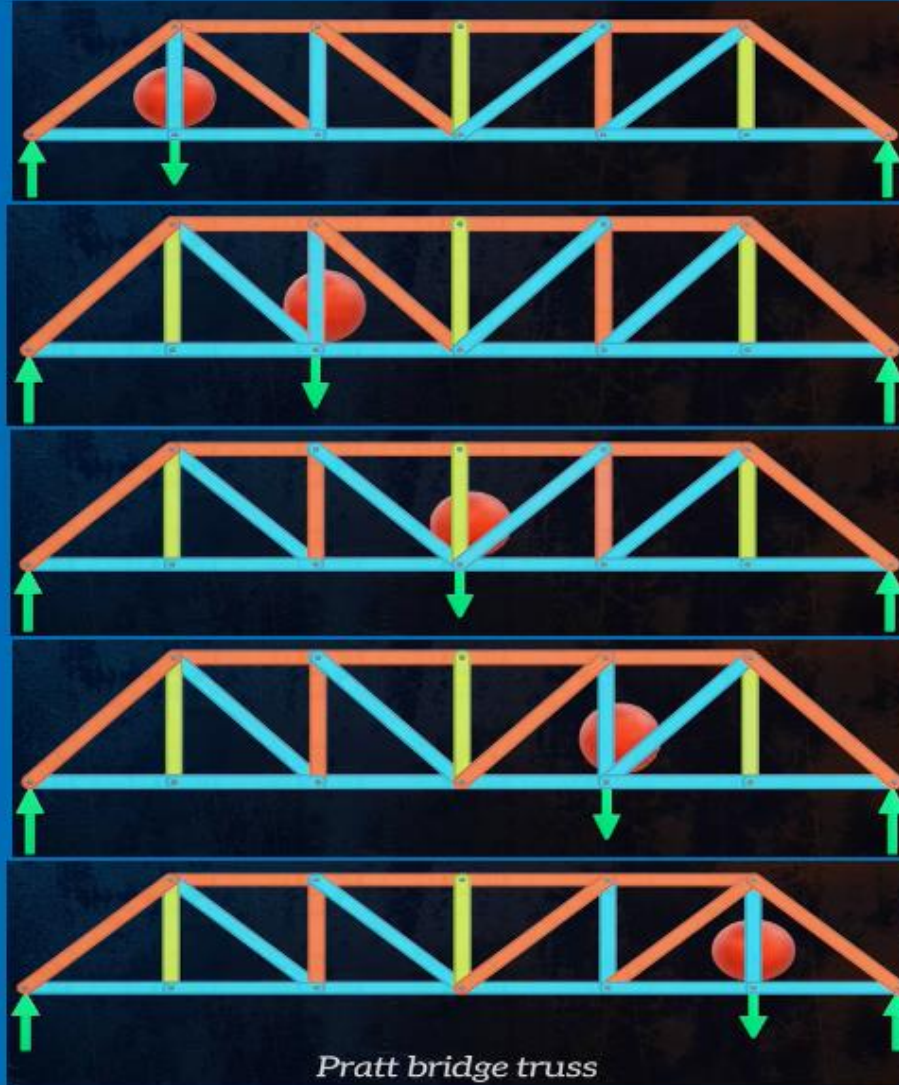


- inner vertical members in compression
- inner diagonal members in tension
- more cost-effective than Howe truss, as longer members are in tension



- uses equilateral triangles
- less members than Howe and Pratt trusses
- diagonal members alternate between tension and compression





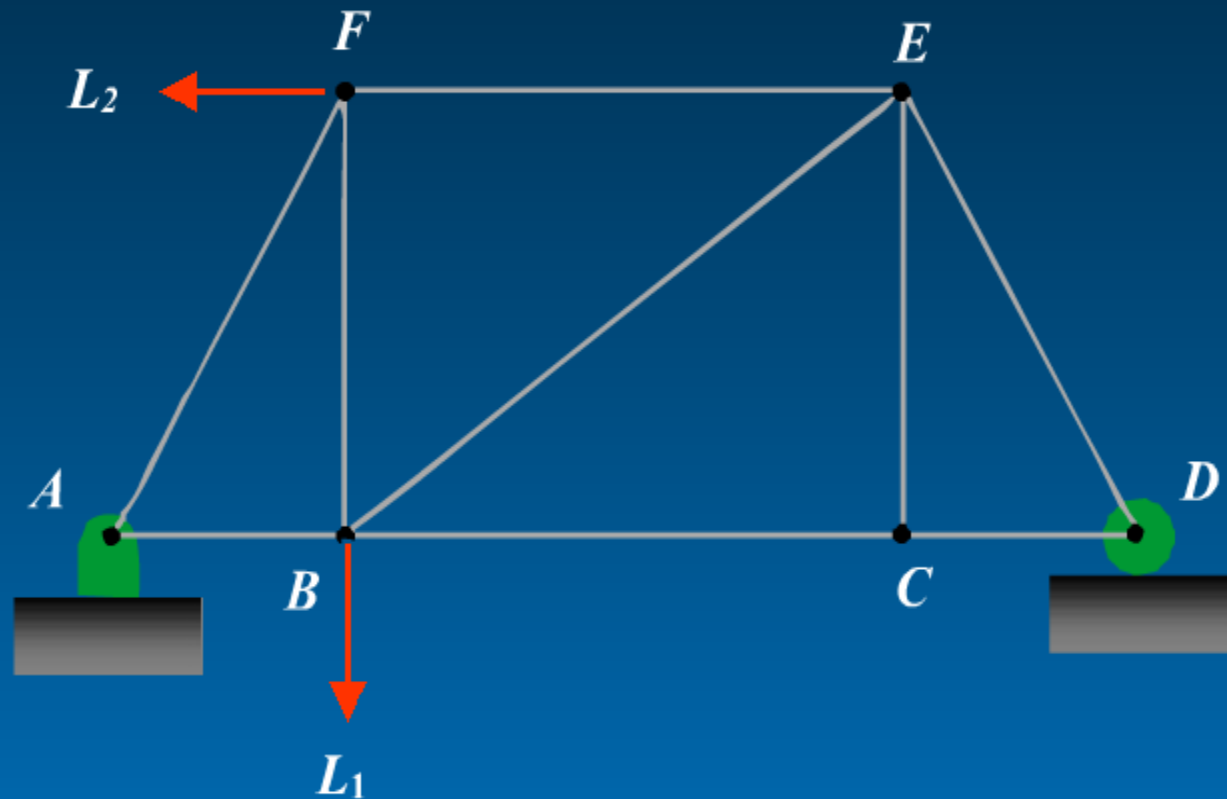
— Compression
— Zero Force Member
— Tension

Method of Joints

Procedure:

1. In general, find the external reactions first (may not be necessary).
2. Begin at a joint where there are 2 or fewer unknowns, and step through the truss joint by joint.
3. Draw a FBD of each joint and apply $\begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \end{cases}$
4. Record the force and its character (tension or compression) for each member.
5. The last joint may not always yield new information, but may serve as an equilibrium check.

General Concept of the Method of Joints



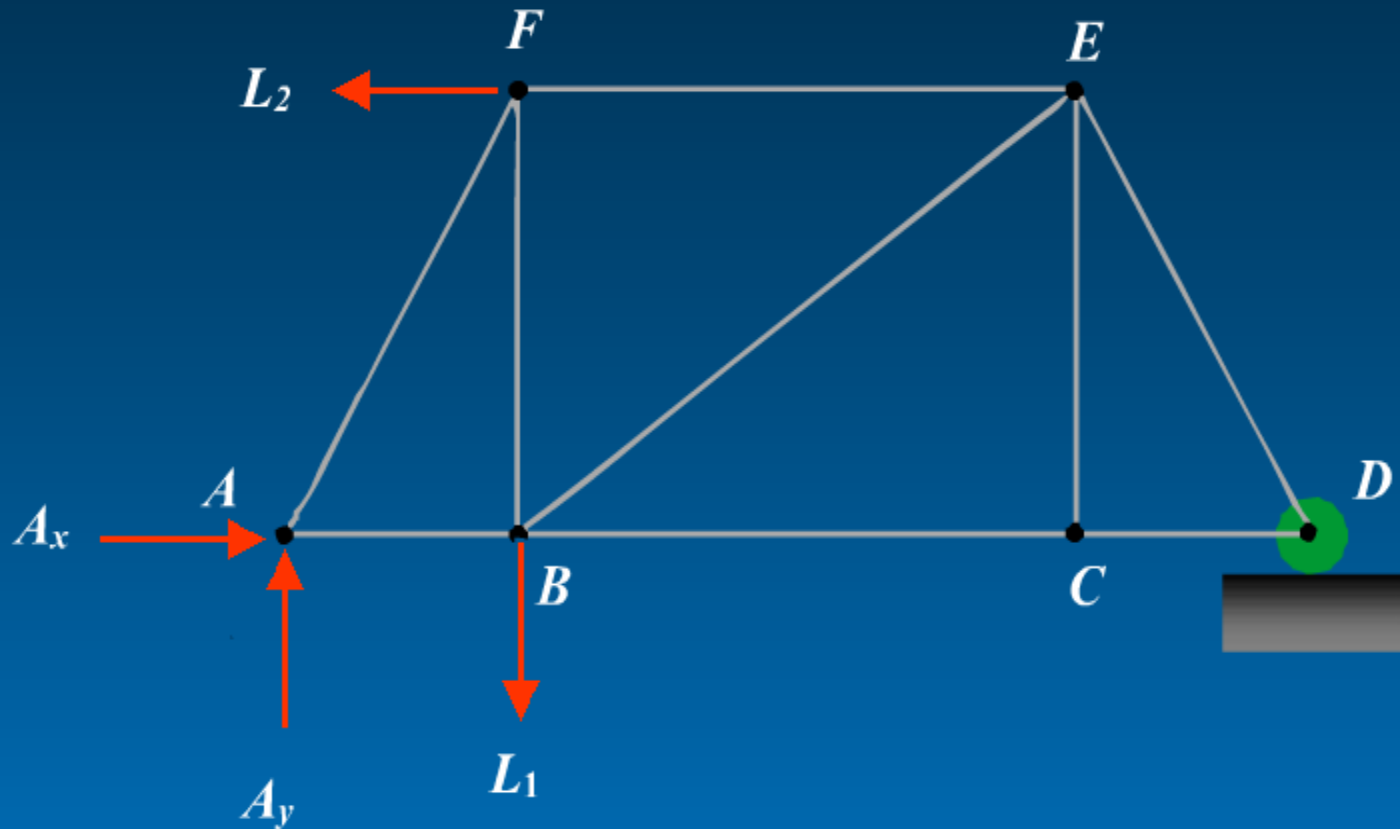
Click on support A and D



2/8



General Concept of the Method of Joints

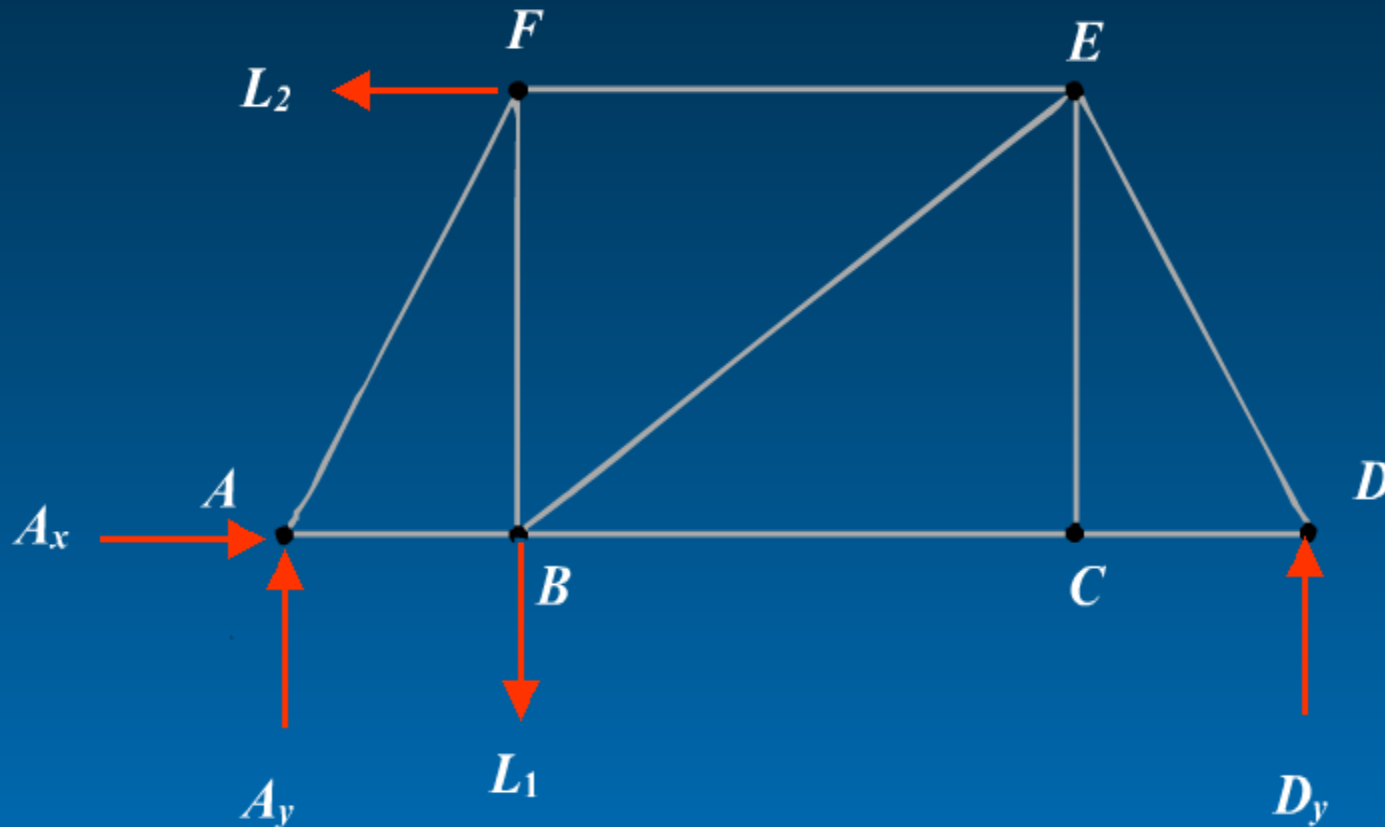


Click on support A and D



2/8



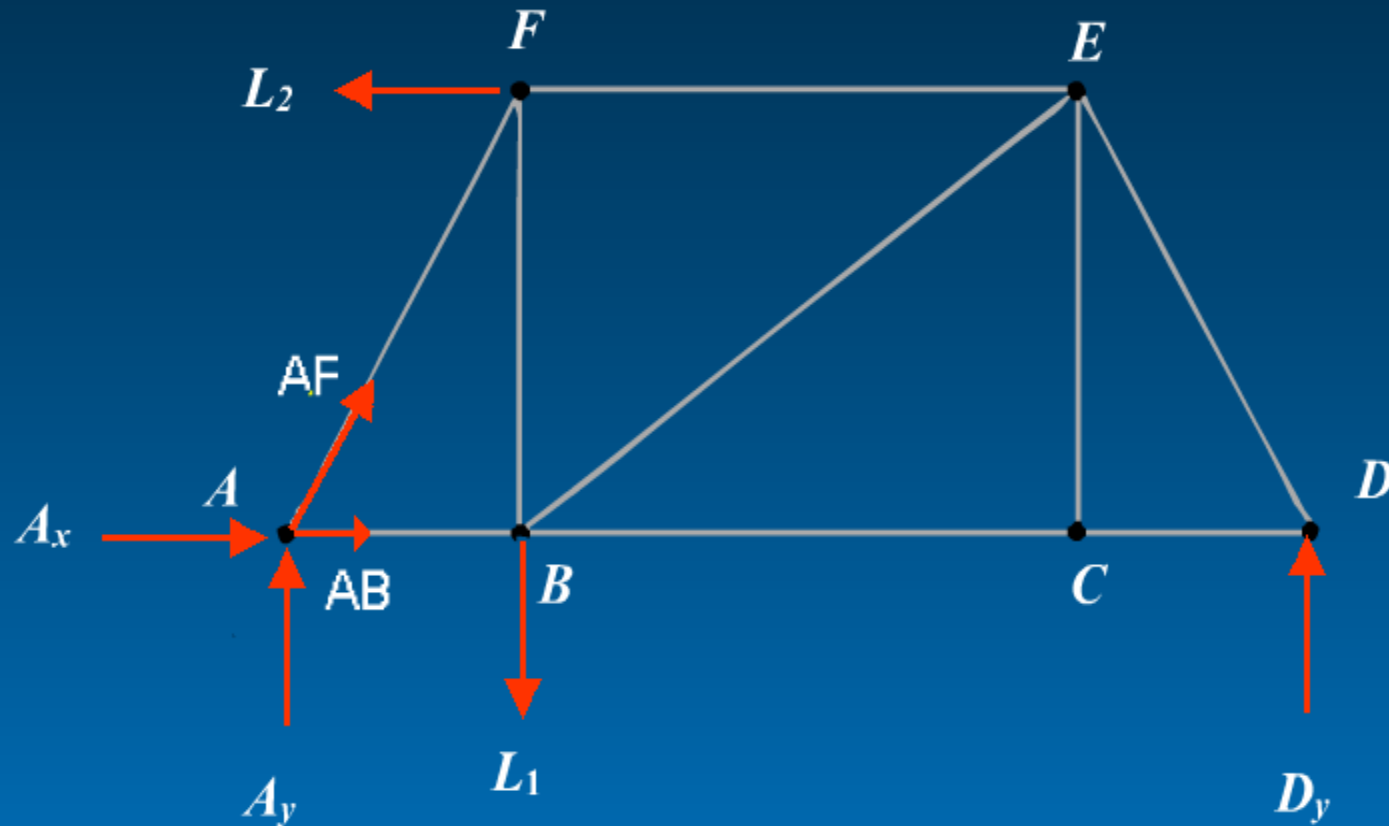


Click on each joint to show/hide forces.



2/8



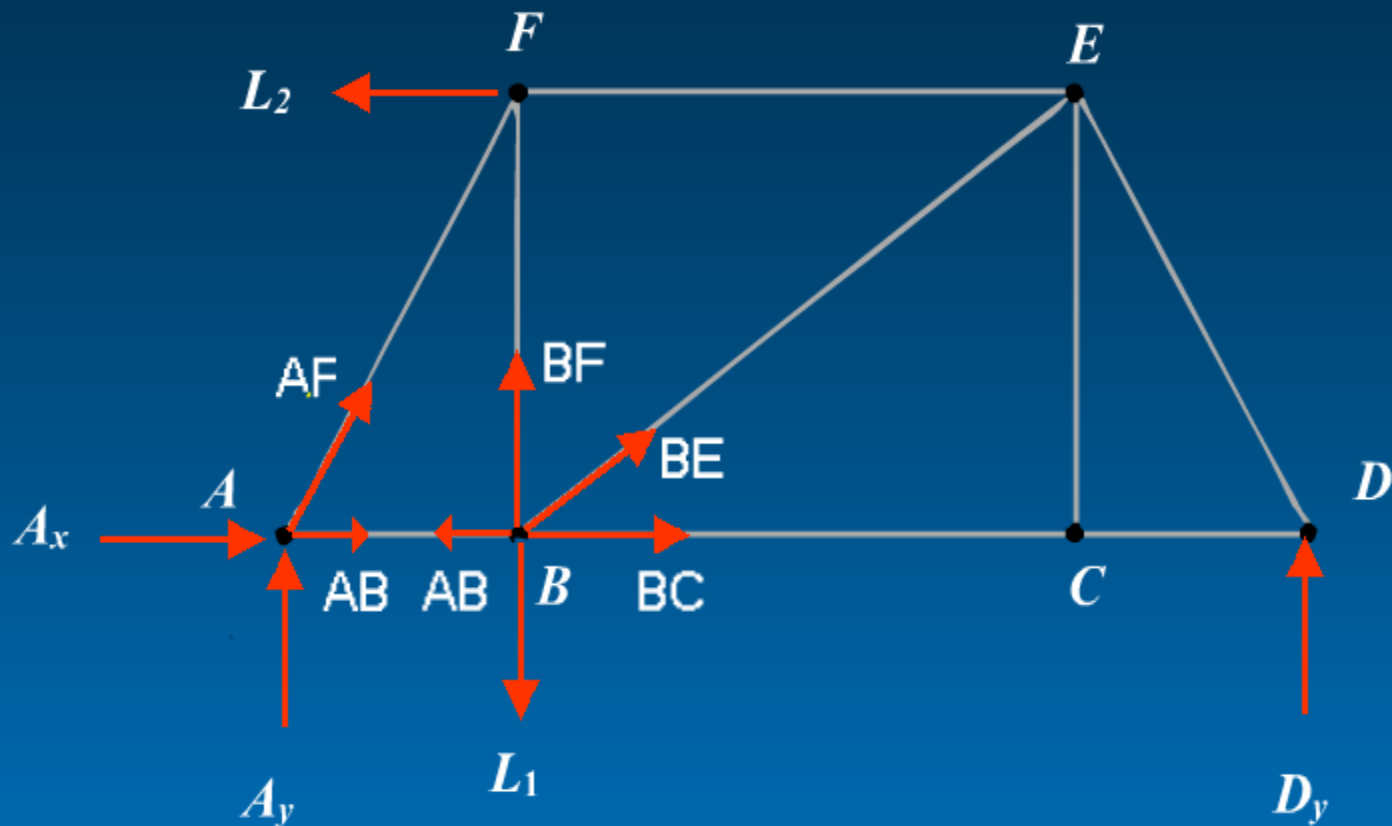


Click on each joint to show/hide forces.



2/8



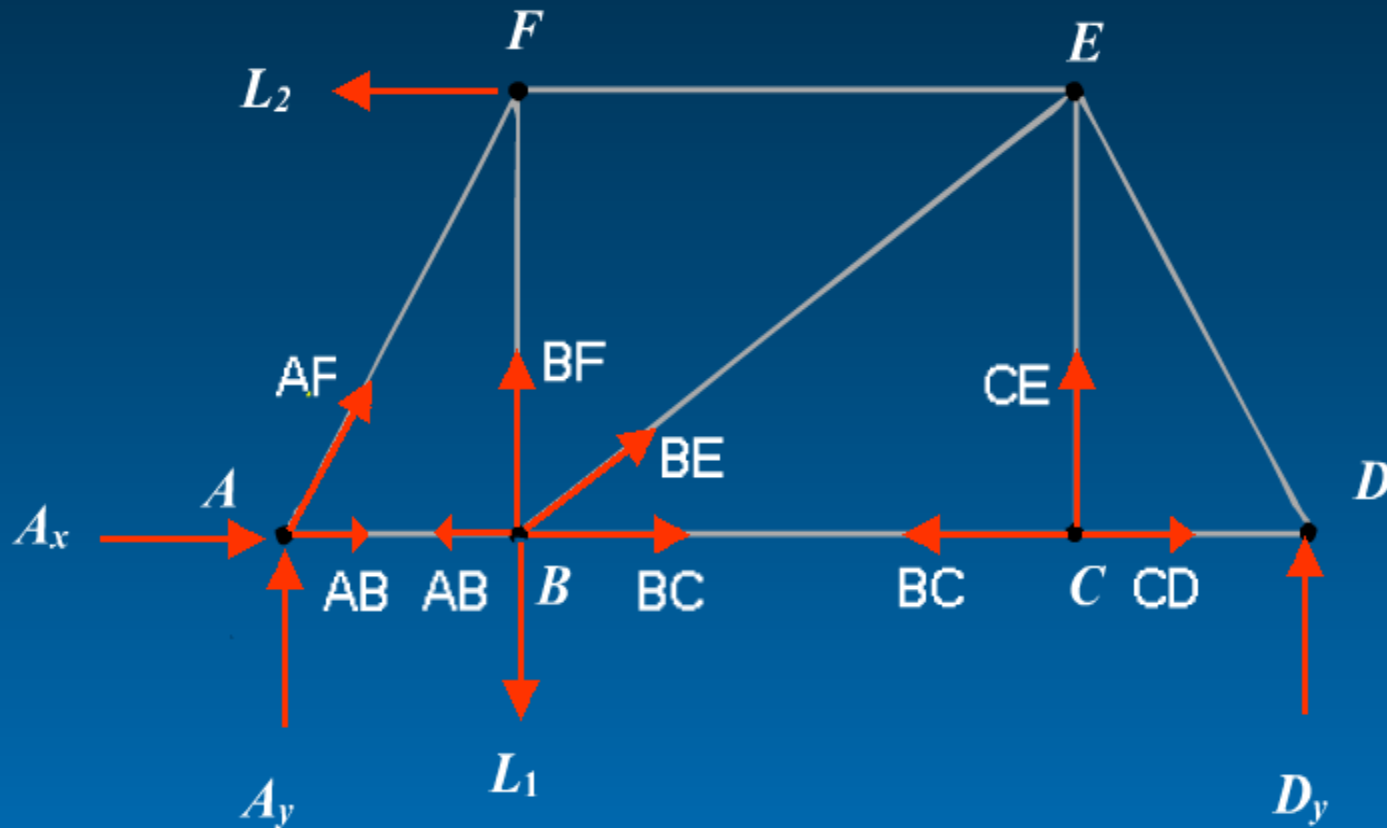


Click on each joint to show/hide forces.



2/8



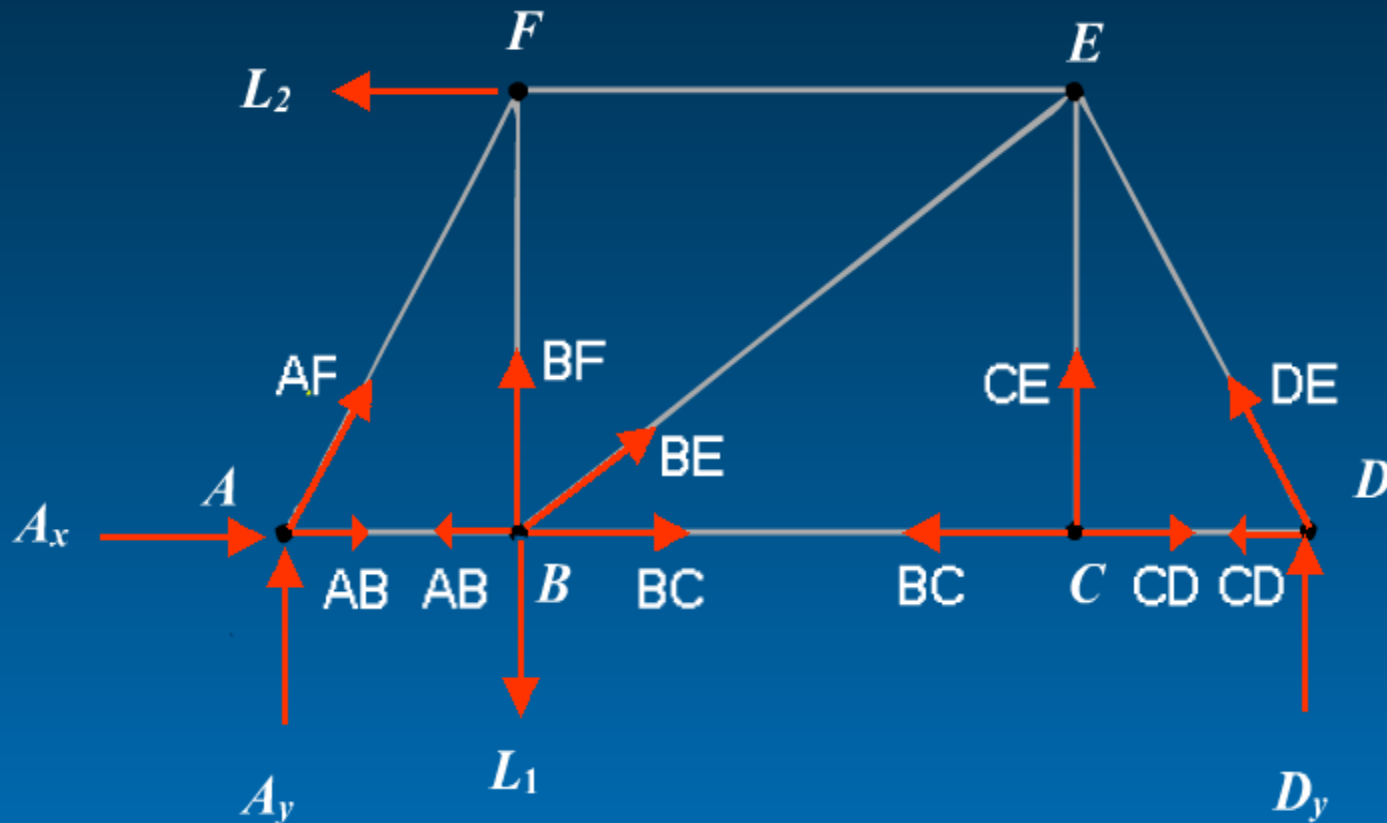


Click on each joint to show/hide forces.



2/8



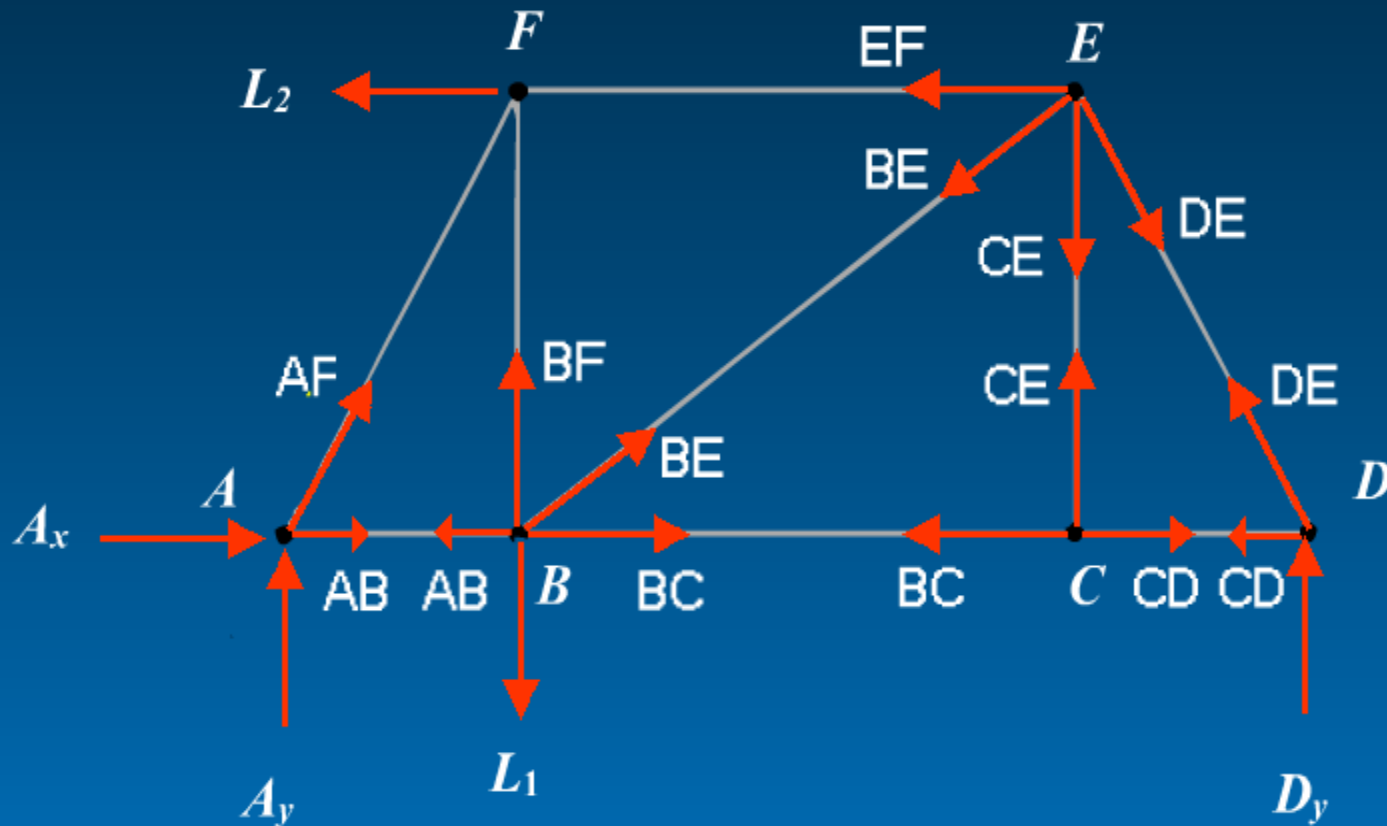


Click on each joint to show/hide forces.



2/8



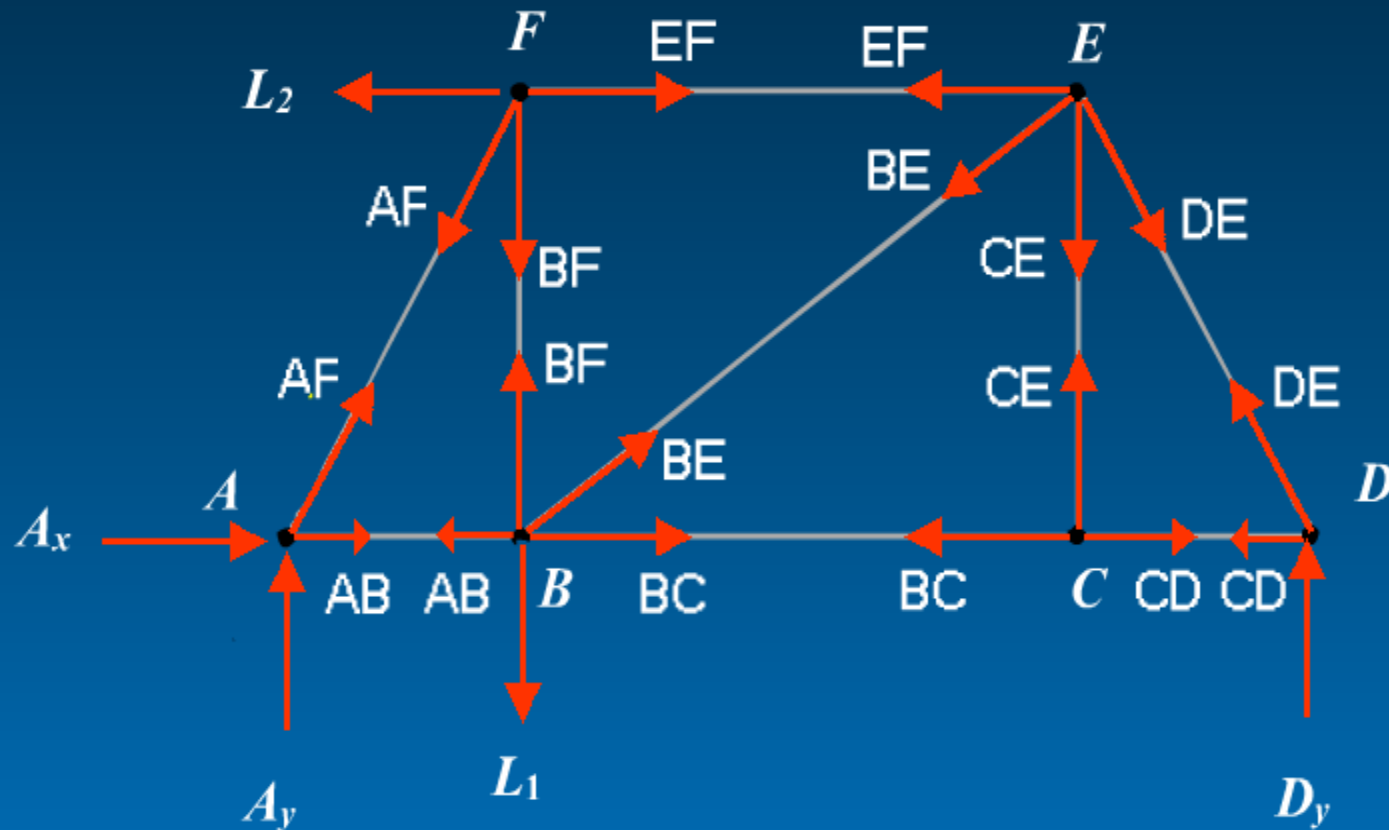


Click on each joint to show/hide forces.



2/8





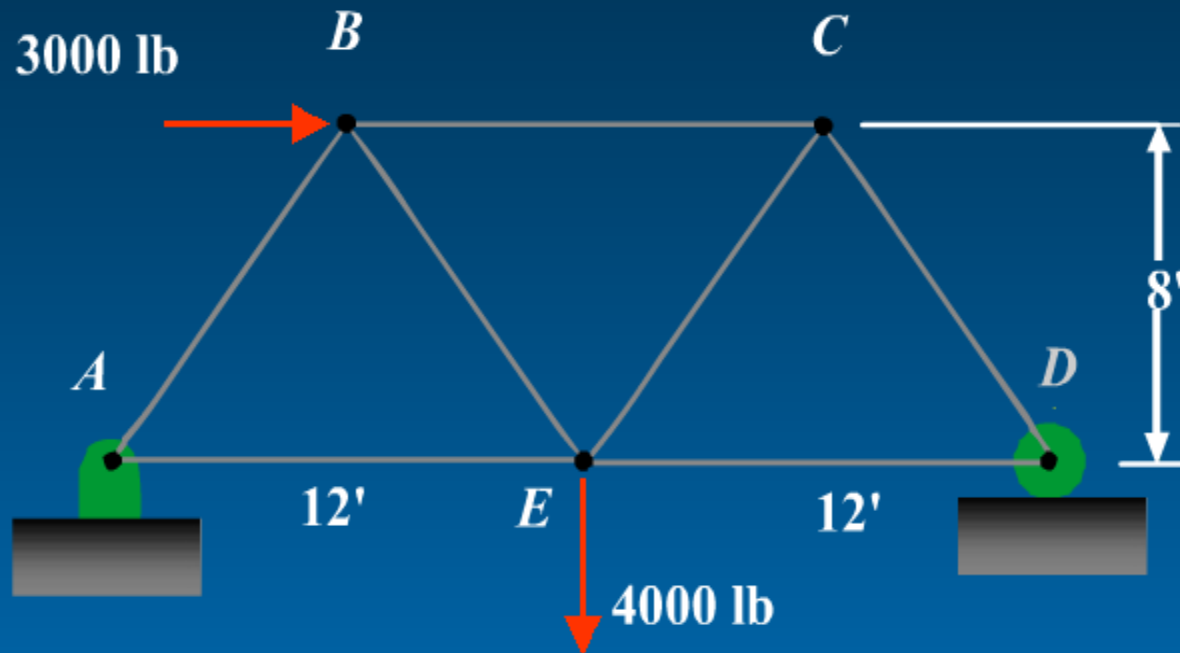
Click on each joint to show/hide forces.



2/8



Example: Method of Joints



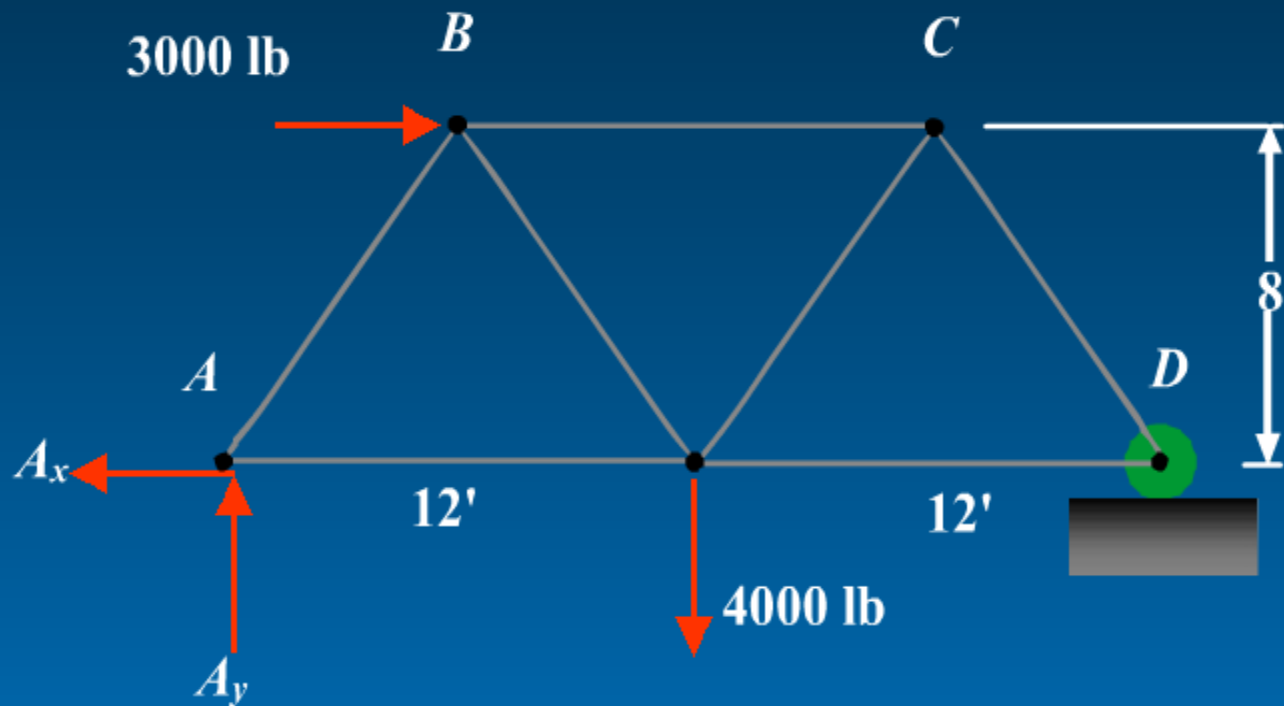
Click on support A and D



3/8



Example: Method of Joints



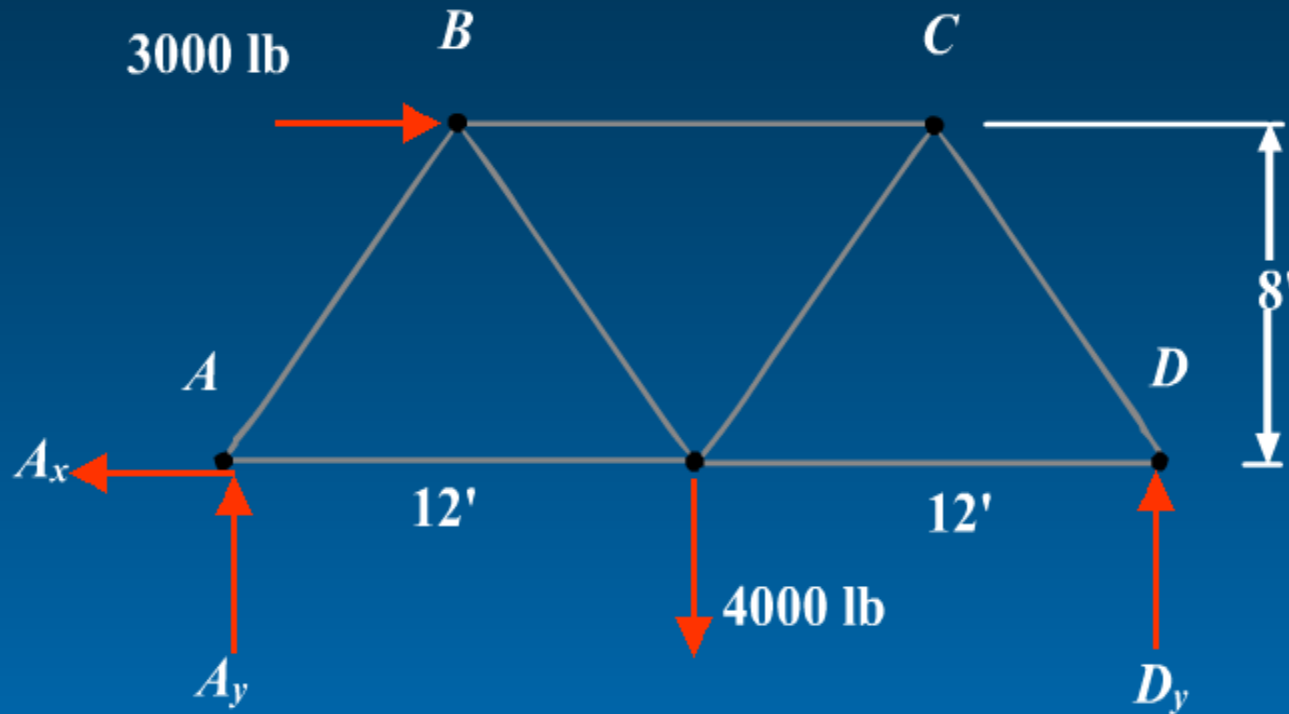
Click on support A and D



3/8



Example: Method of Joints

[Next](#)

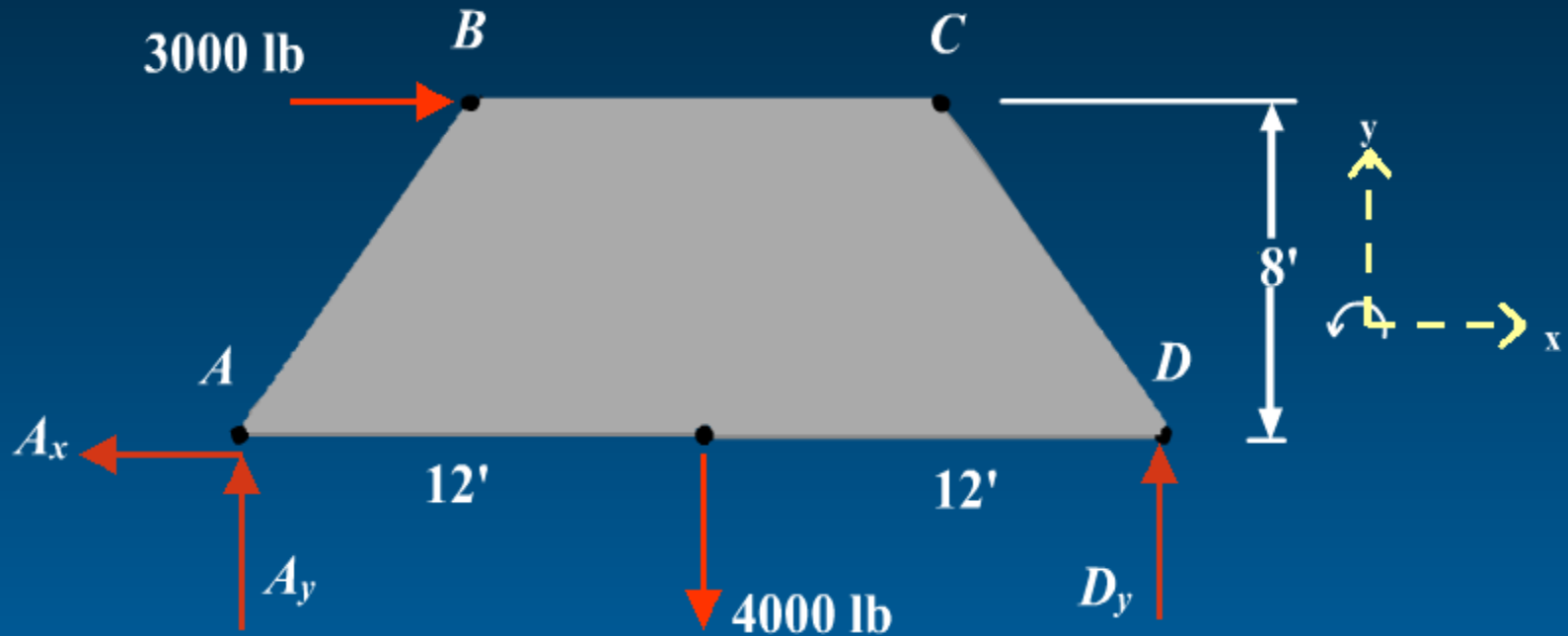
Click on "Next" to proceed



3/8



Example: Method of Joints



$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_A = 0 :$$

[Show solution](#)

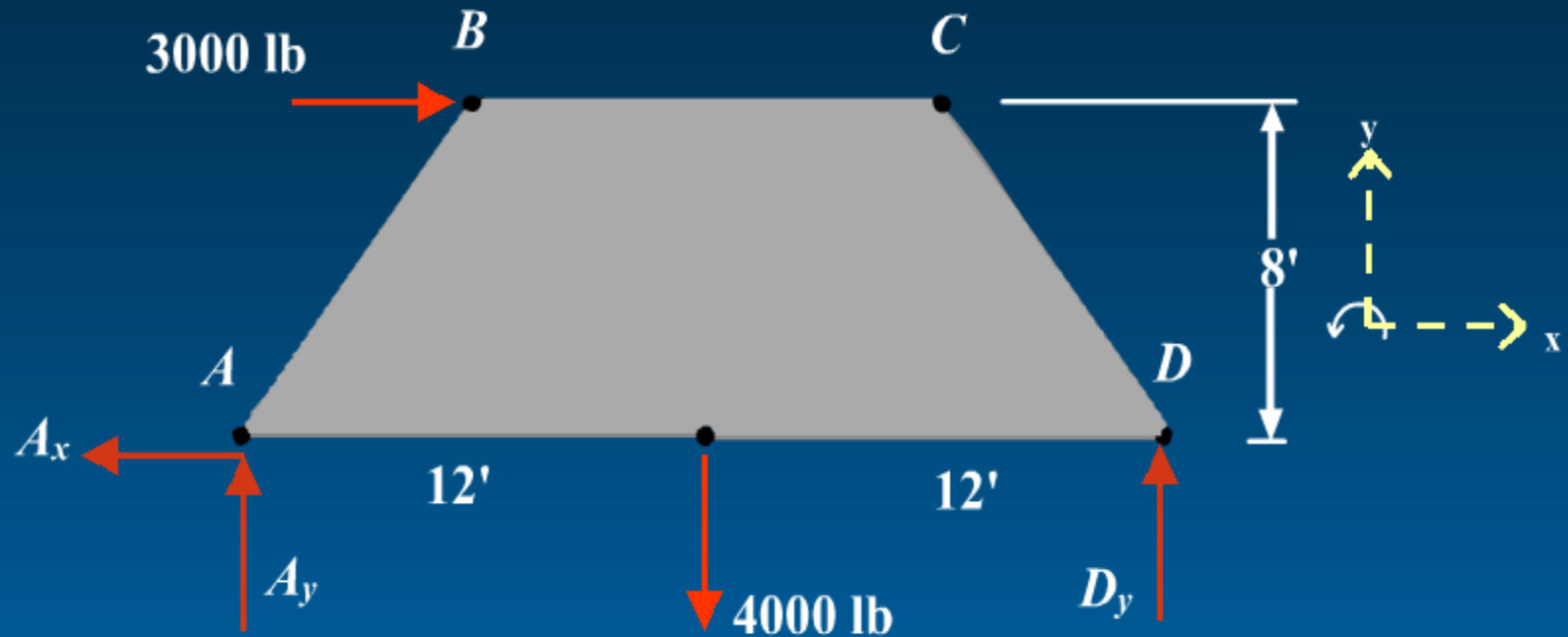
Click on the governing equations



3/8



Example: Method of Joints



$$\Sigma F_x = 0 : 3000 - A_x = 0$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_A = 0 :$$

[Show solution](#)

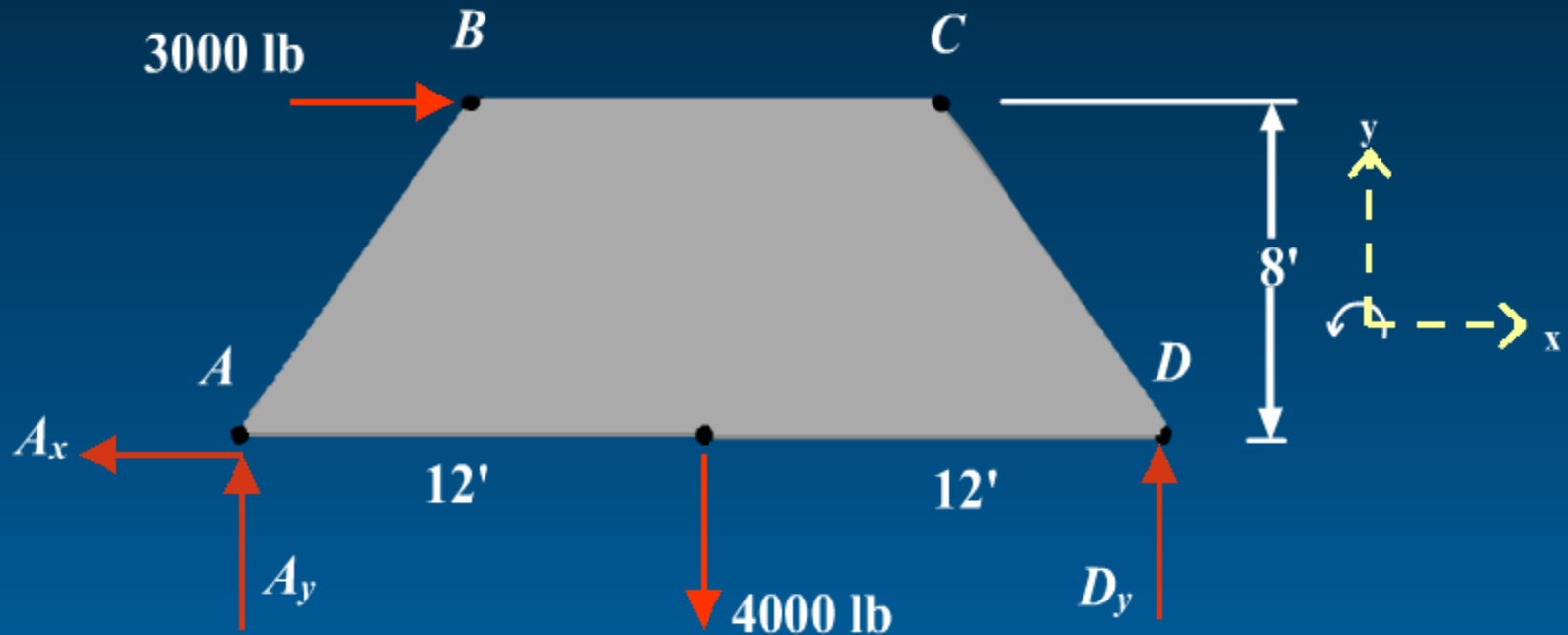
Click on the governing equations



3/8



Example: Method of Joints



$$\Sigma F_x = 0 : 3000 - A_x = 0$$

$$\Sigma F_y = 0 : A_y + D_y - 4000 = 0$$

$$\Sigma M_A = 0 :$$

[Show solution](#)

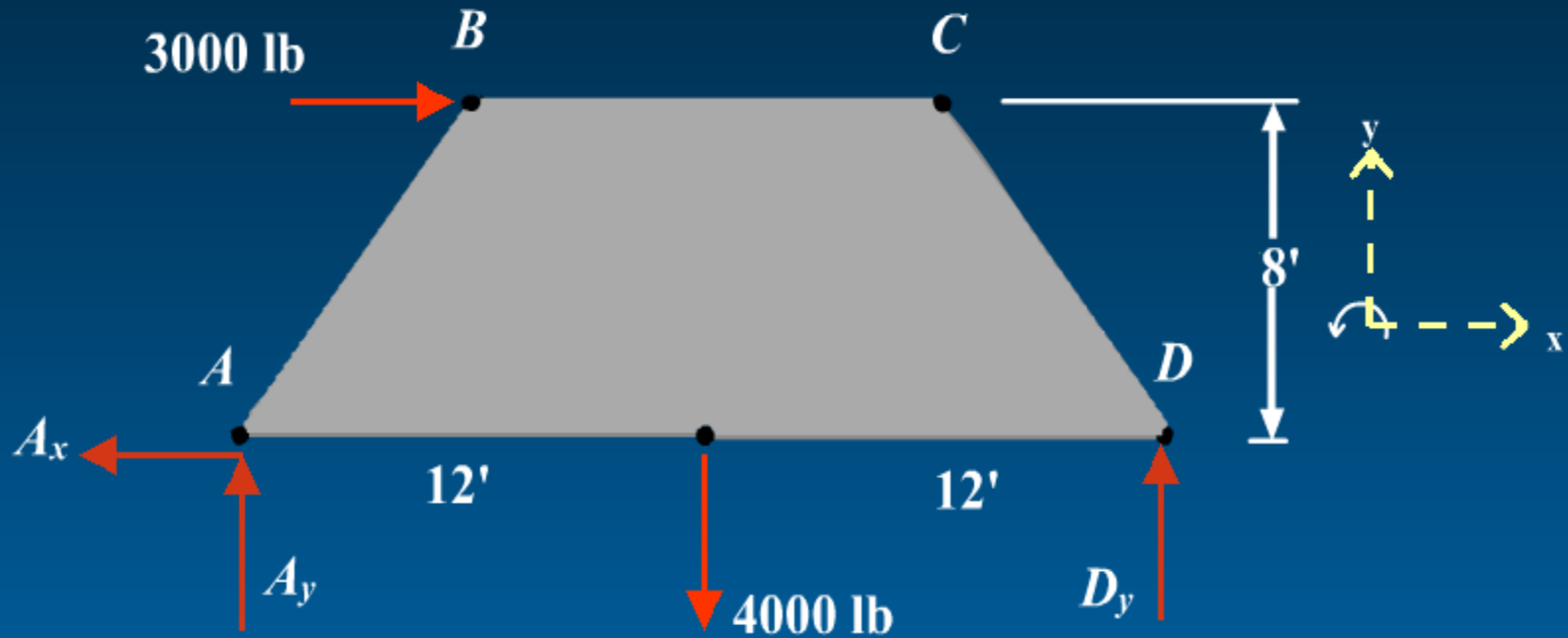
Click on the governing equations



3/8



Example: Method of Joints



$$\Sigma F_x = 0 : 3000 - A_x = 0$$

$$\Sigma F_y = 0 : A_y + D_y - 4000 = 0$$

$$\Sigma M_A = 0 : -3000(8) - 4000(12) + D_y(24) = 0$$

[Show solution](#)

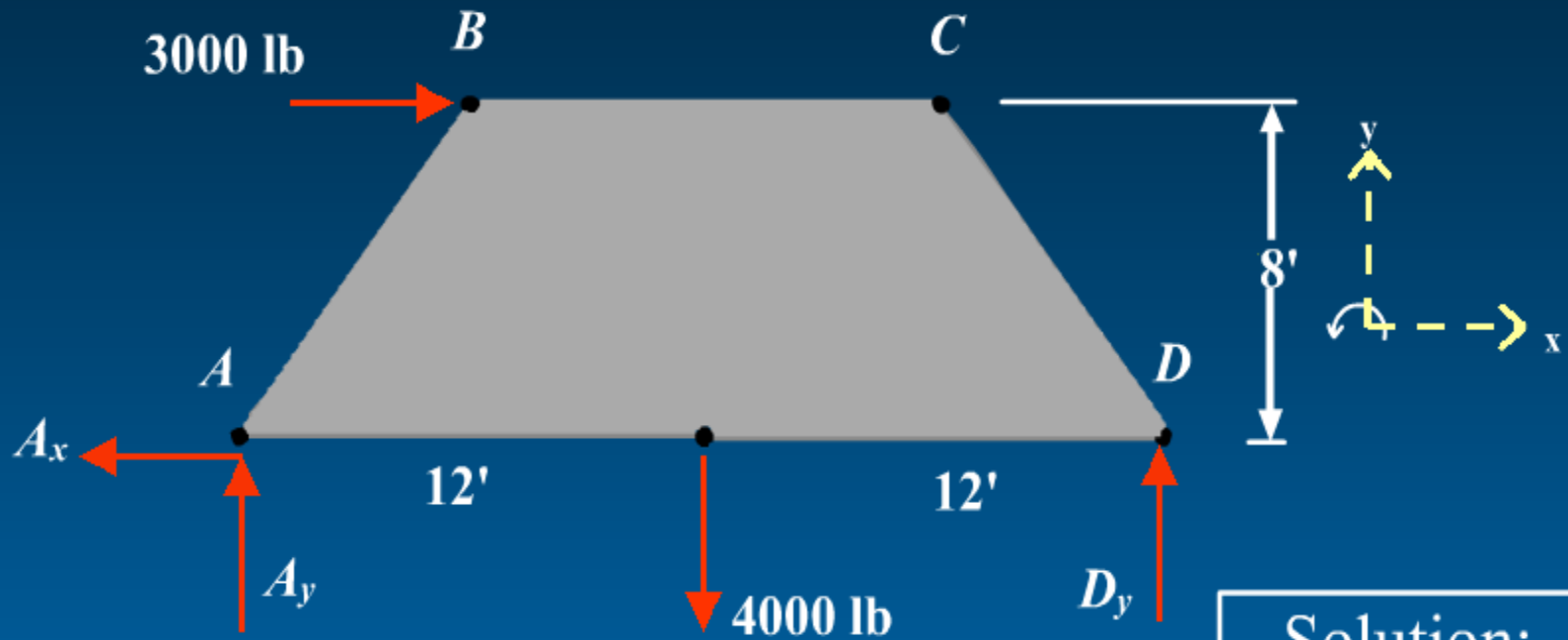
Click on the governing equations



3/8



Example: Method of Joints



$$\Sigma F_x = 0 : 3000 - A_x = 0$$

$$\Sigma F_y = 0 : A_y + D_y - 4000 = 0$$

$$\Sigma M_A = 0 : -3000(8) - 4000(12) + D_y(24) = 0$$

Solution:

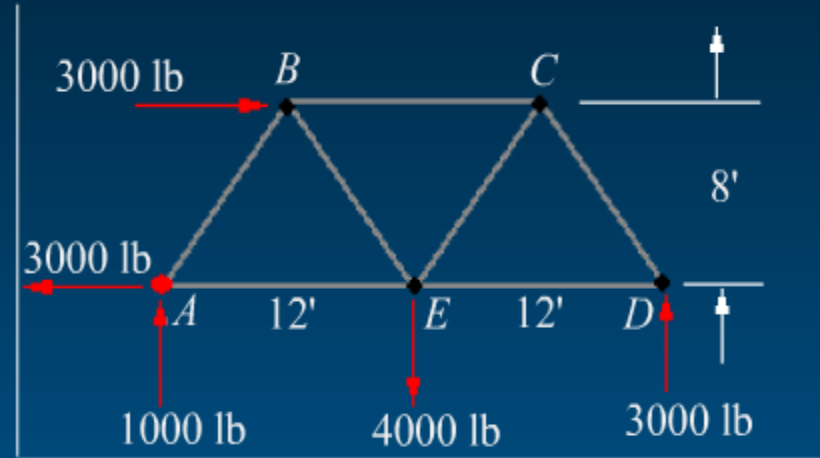
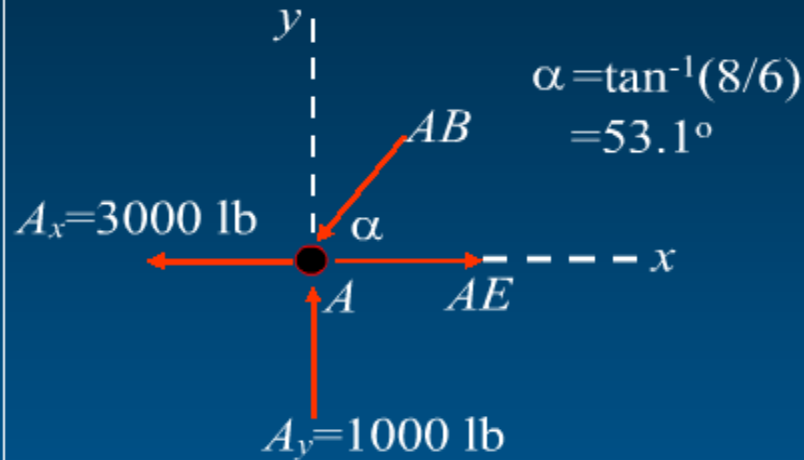
$$A_x = 3000 \text{ lb}$$
$$A_y = 1000 \text{ lb}$$
$$D_y = 3000 \text{ lb}$$



3/8



Joint A



$$\Sigma F_x = 0 : -3000 - AB \cos 53.1^\circ + AE = 0$$

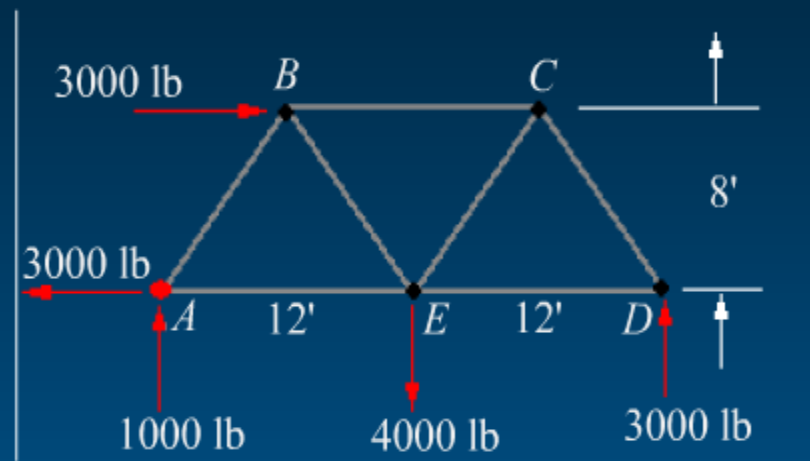
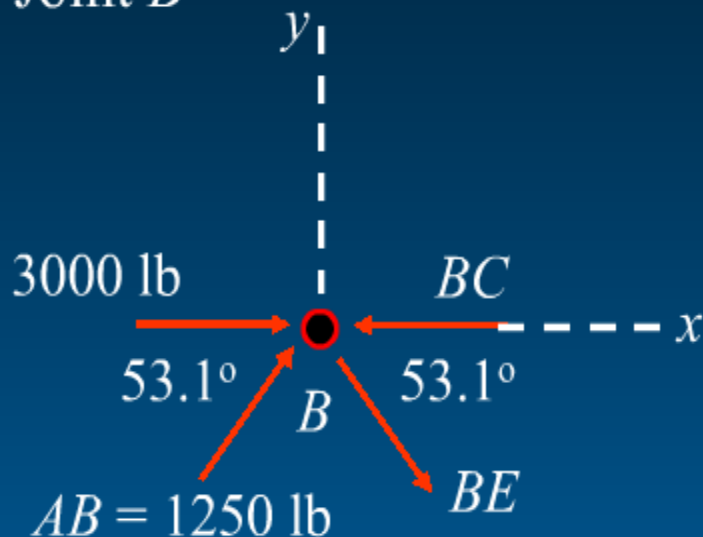
$$\Sigma F_y = 0 : 1000 - AB \sin 53.1^\circ = 0$$

Solution:

$$AB = 1250 \text{ lb C}$$

$$AE = 3750 \text{ lb T}$$

Joint B



$$\Sigma F_x = 0 : \quad 3000 + 1250 \cos 53.1^\circ + BE \cos 53.1^\circ - BC = 0$$

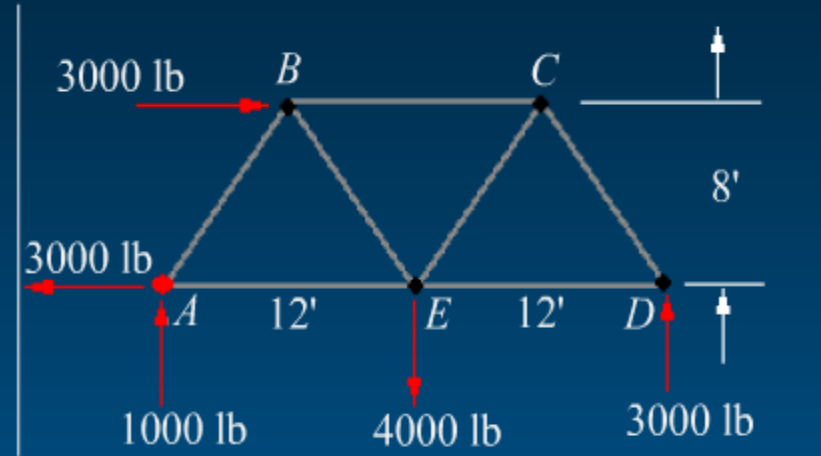
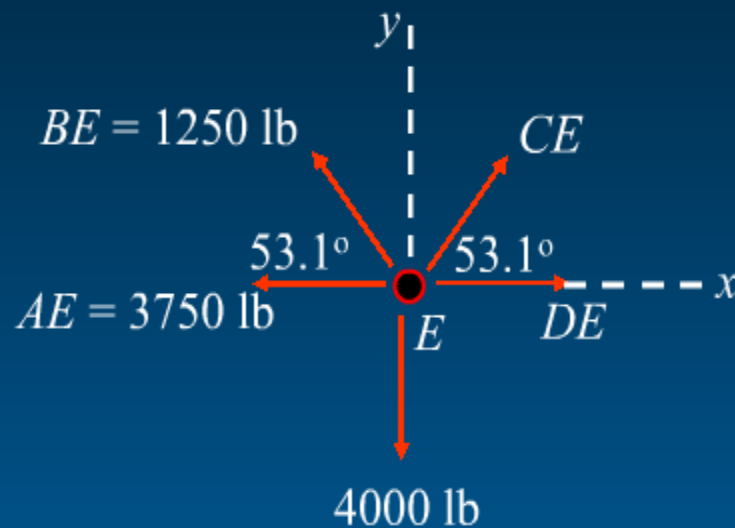
$$\Sigma F_y = 0 : \quad 1250 \sin 53.1^\circ - BE \sin 53.1^\circ = 0$$

Solution:

$$BE = 1250 \text{ lb T}$$

$$BC = 4500 \text{ lb C}$$

Joint E



$$\Sigma F_x = 0 : -3750 - 1250 \cos 53.1^\circ + CE \cos 53.1^\circ + DE = 0$$

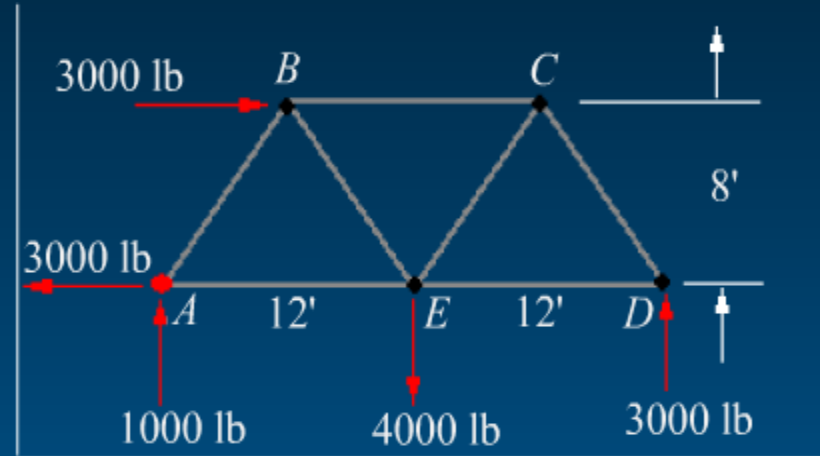
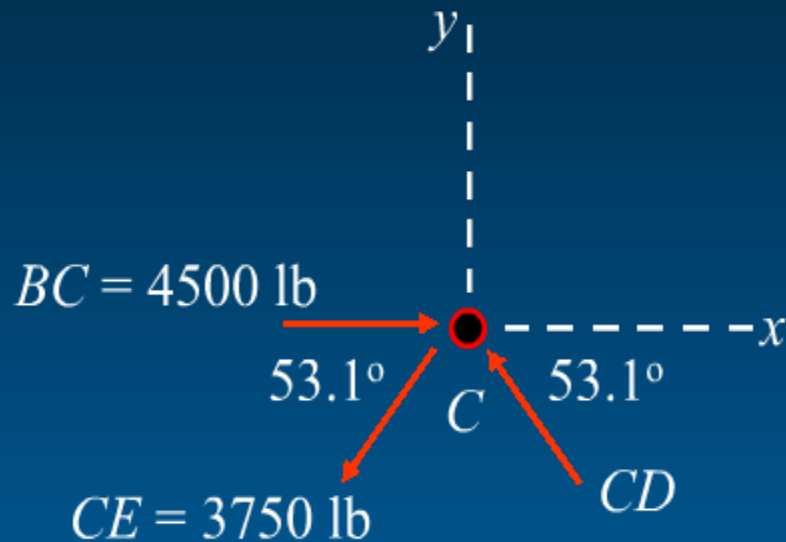
$$\Sigma F_y = 0 : 1250 \sin 53.1^\circ + CE \sin 53.1^\circ - 4000 = 0$$

Solution:

$$CE = 3750 \text{ lb T}$$

$$DE = 2250 \text{ lb T}$$

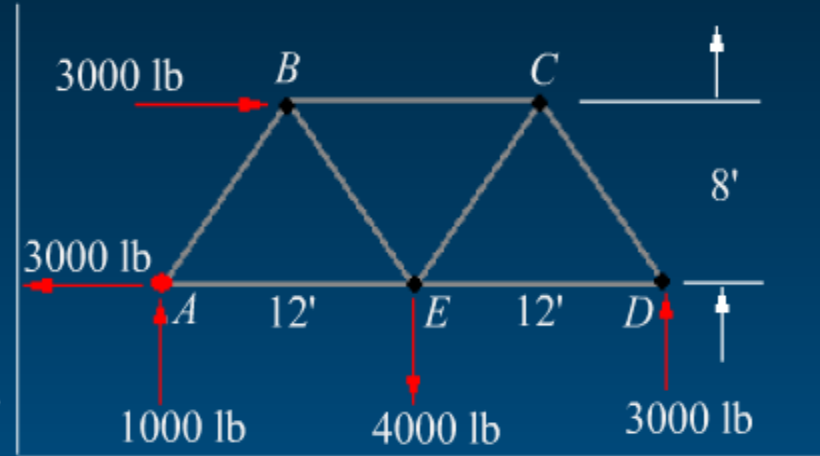
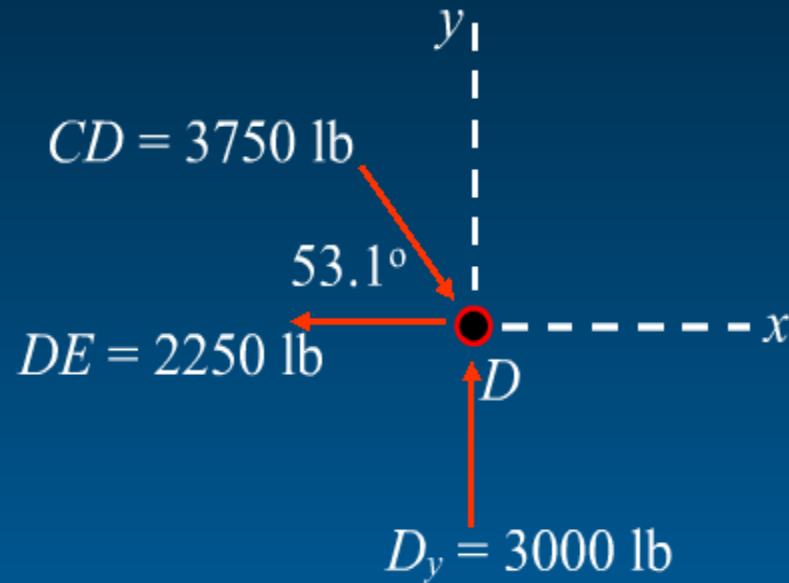
Joint C (could go to joint D)



$$\Sigma F_y = 0 : -3750 \sin 53.1^\circ + CD \sin 53.1^\circ = 0$$

Solution:
 $CD = 3750 \text{ lb C}$

Joint D (used as a check)



$$\Sigma F_x = 0 : -2250 + 3750 \cos 53.1^\circ = 0 \quad \checkmark$$

$$\Sigma F_y = 0 : -3750 \sin 53.1^\circ + 3000 = 0 \quad \checkmark$$

Click on the governing equations



8/8



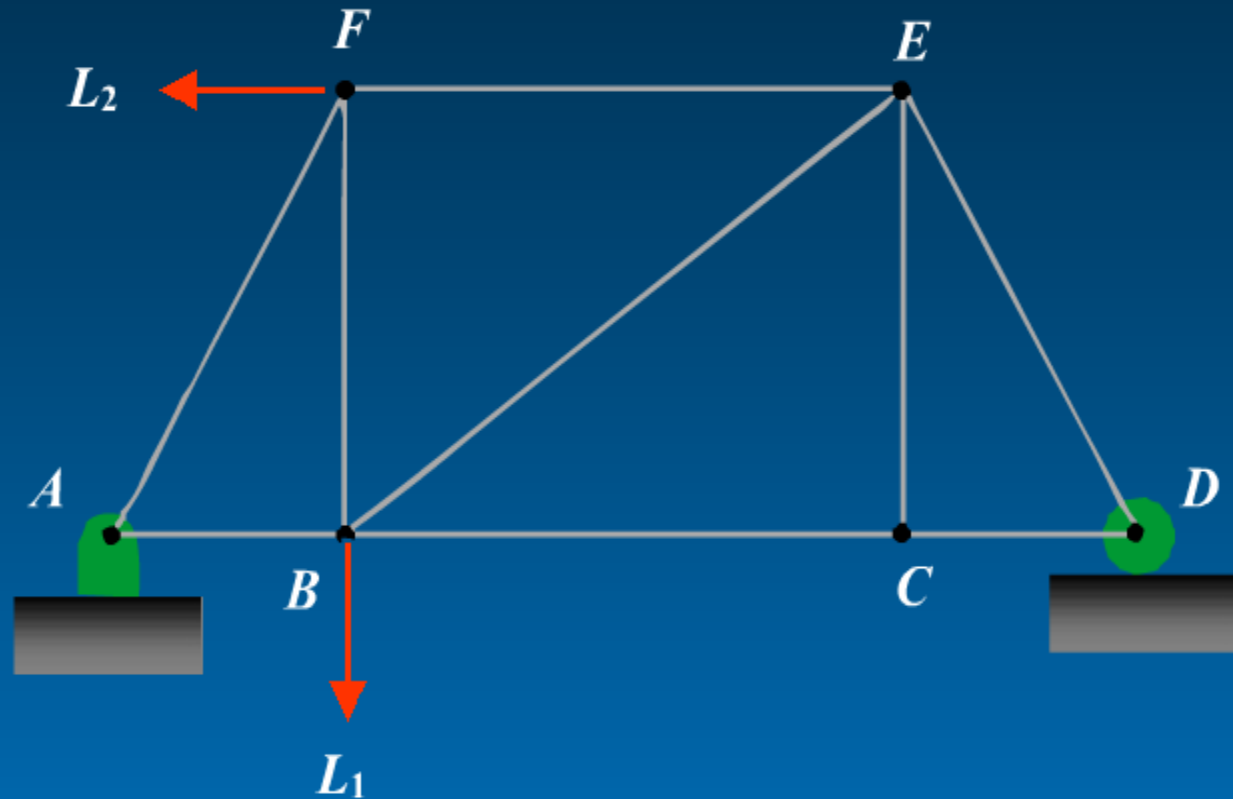
Method of Sections

Procedure:

1. In general, find the external reactions first (may not be necessary).
2. If possible, pass a section (cut) through the desired member and up to 2 other unknown members, isolating a portion of the truss.
3. Apply 2-D rigid-body equilibrium equations to the isolated truss portion; for example: $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M_o = 0$
4. Solve for the unknowns.
5. Note that the methods of sections and joints may be used in combination.



General Concept of the Method of Sections



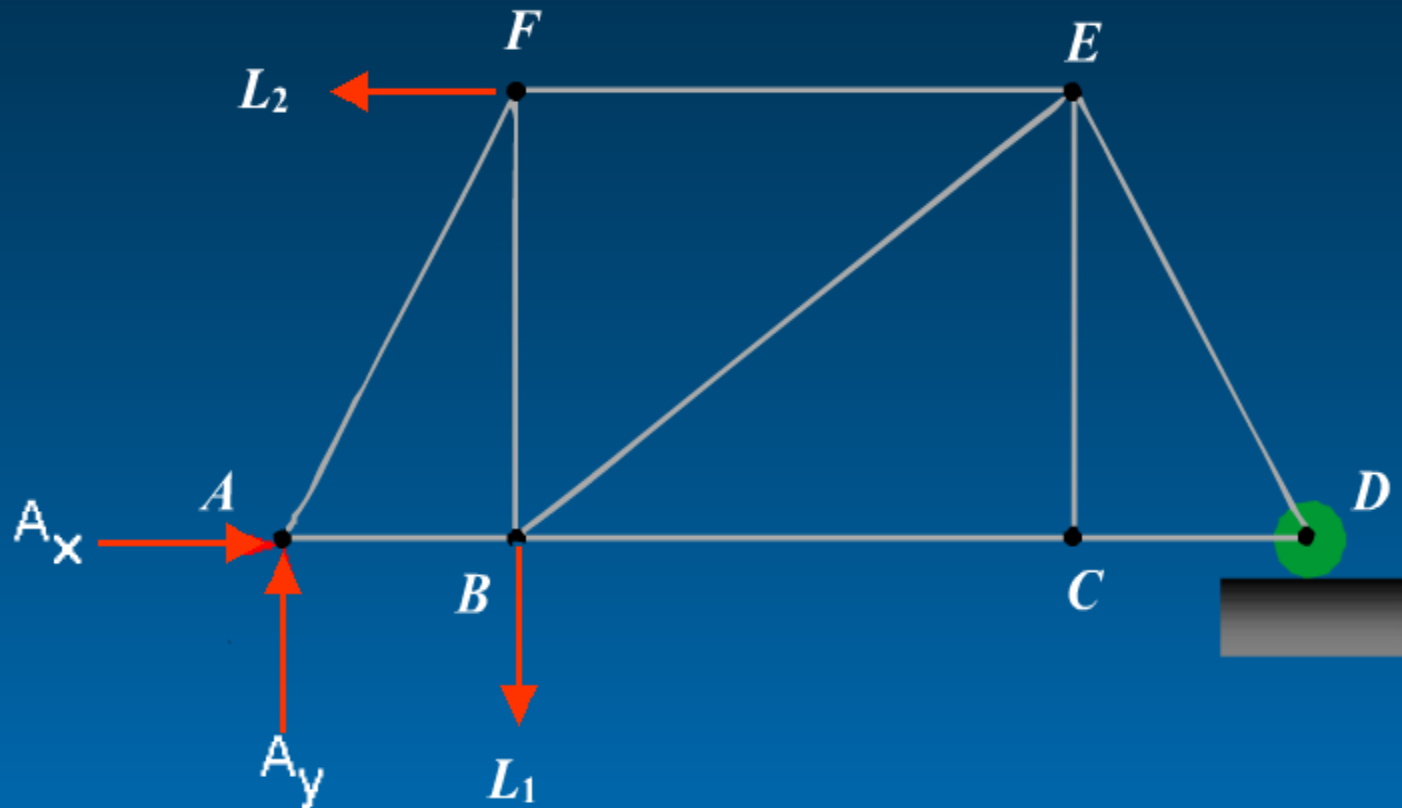
Click on support A and D



2/3



General Concept of the Method of Sections



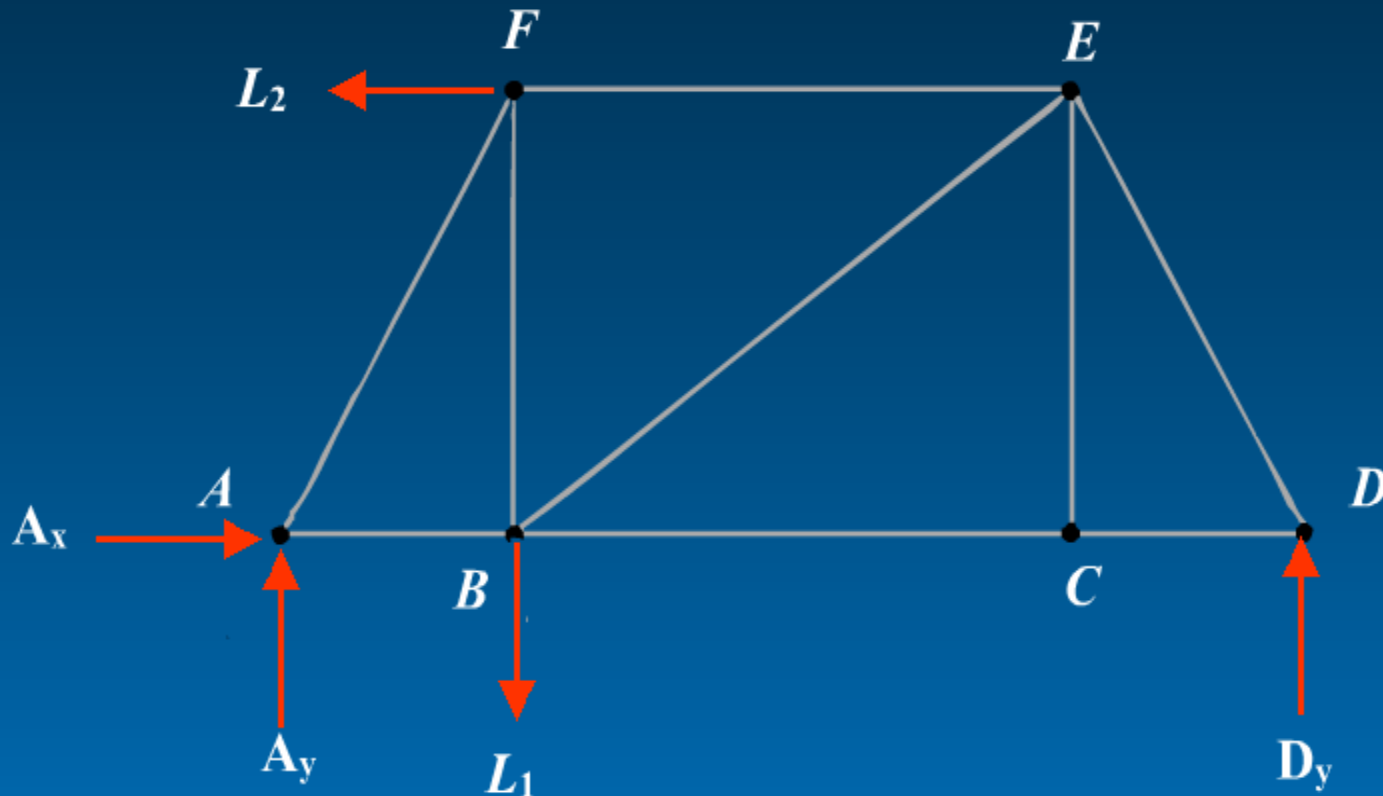
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2/3



General Concept of the Method of Sections



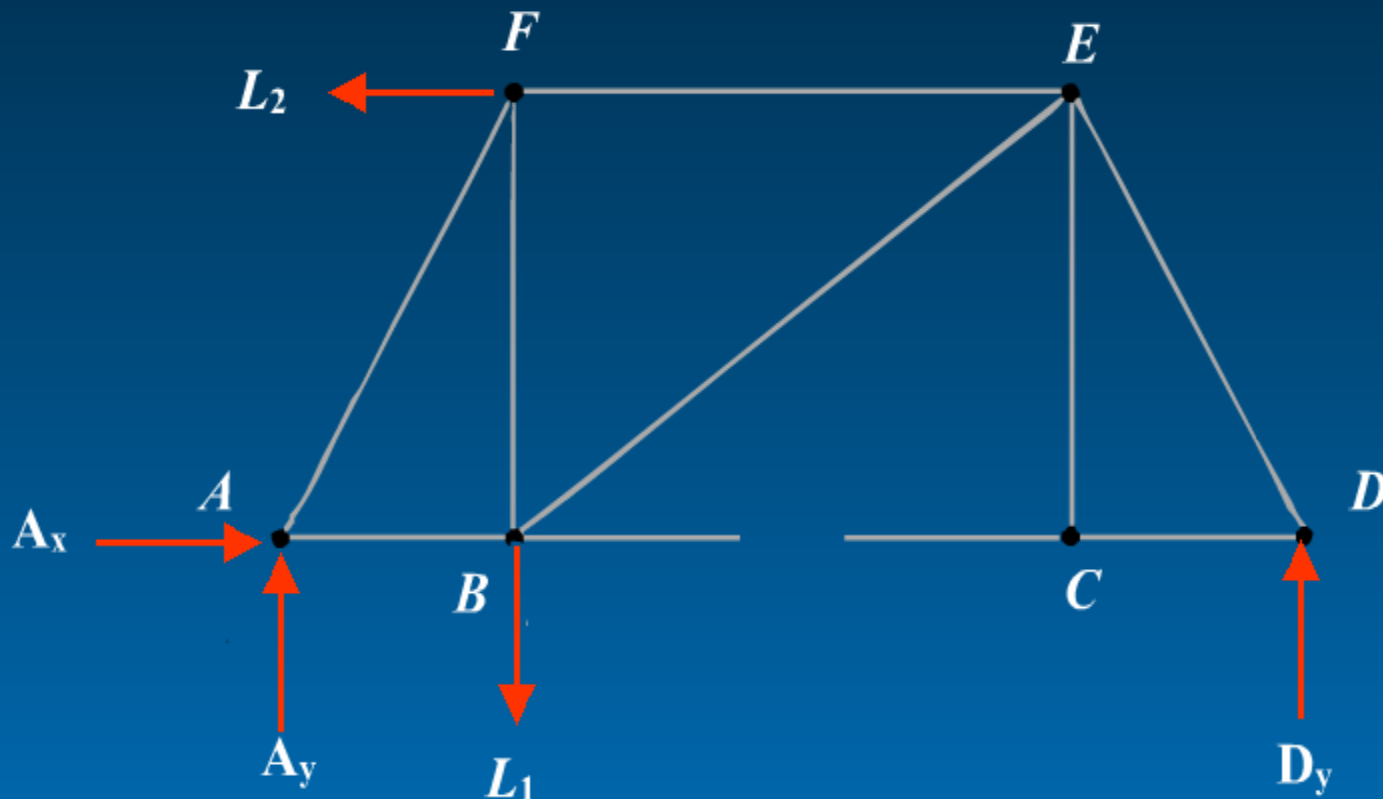
Click on the members to cut/uncut.



2/3



General Concept of the Method of Sections



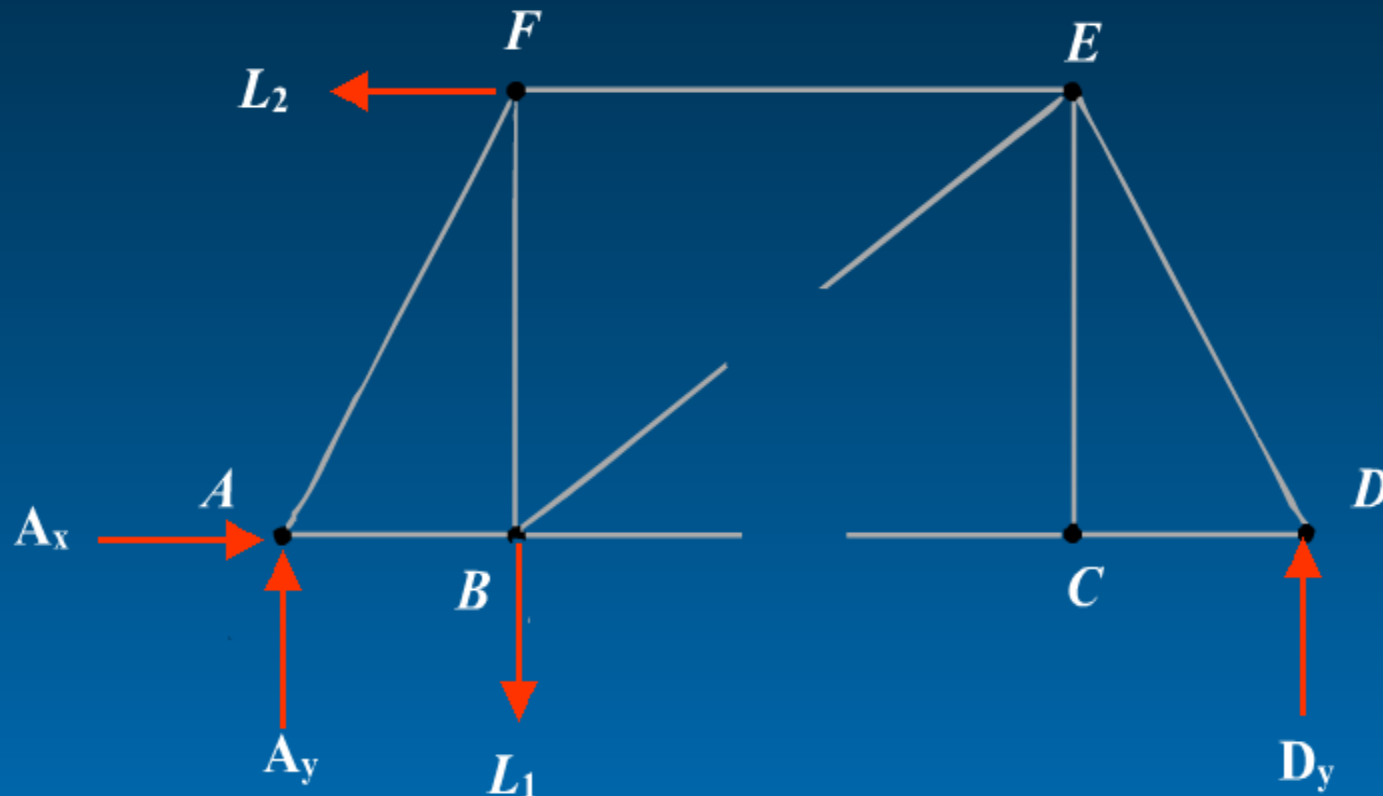
Click on the members to cut/uncut.



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General Concept of the Method of Sections



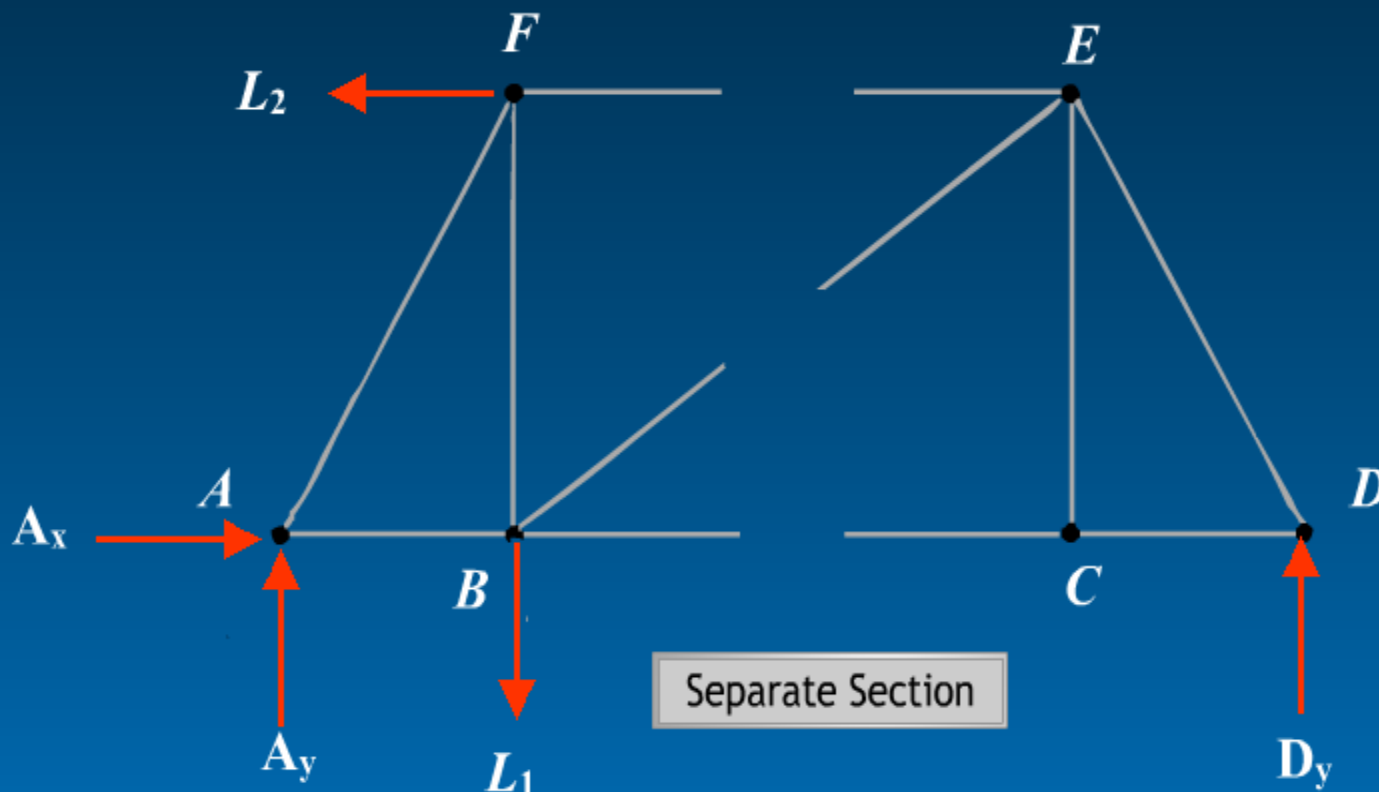
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General Concept of the Method of Sections

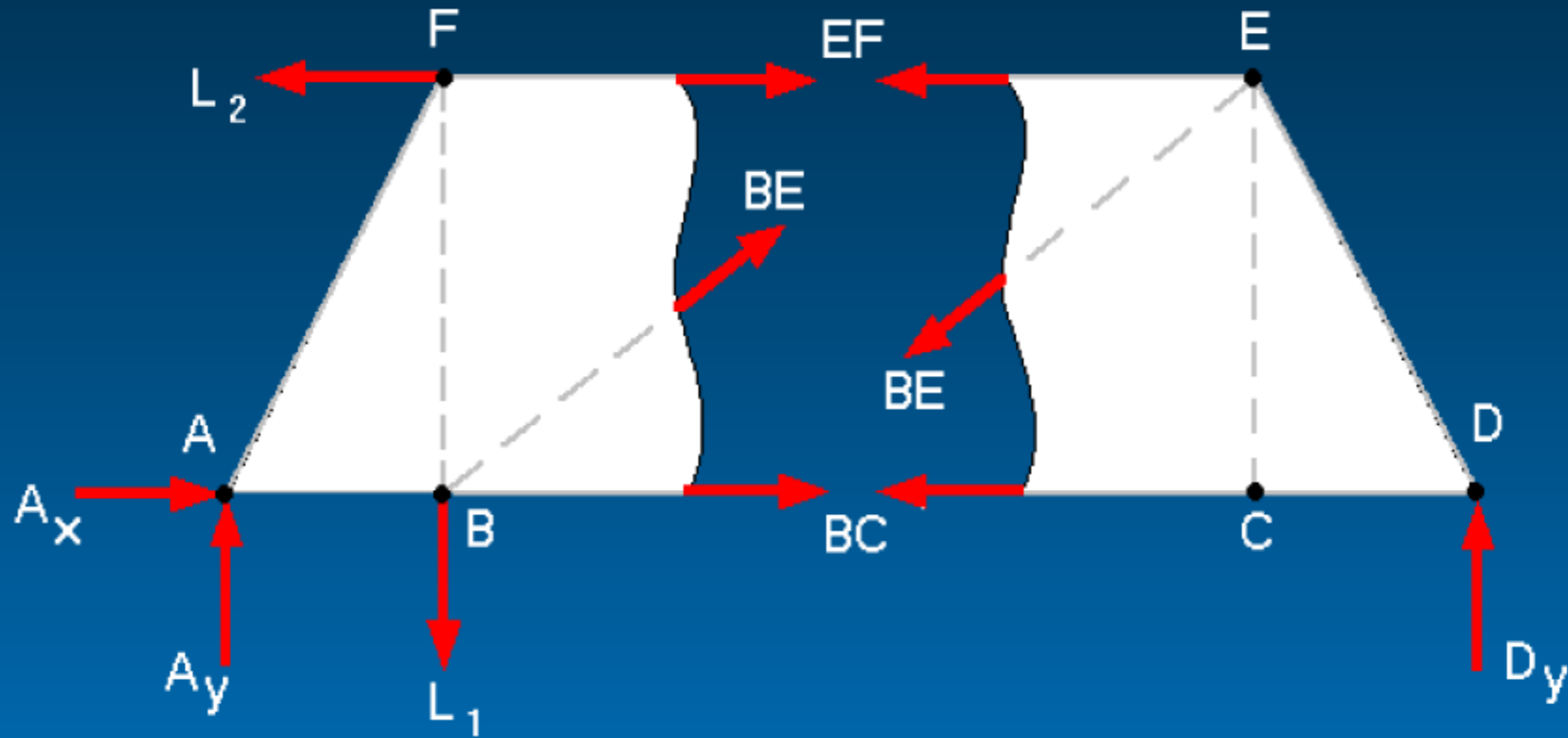


Click on the members to cut/uncut.



2/3



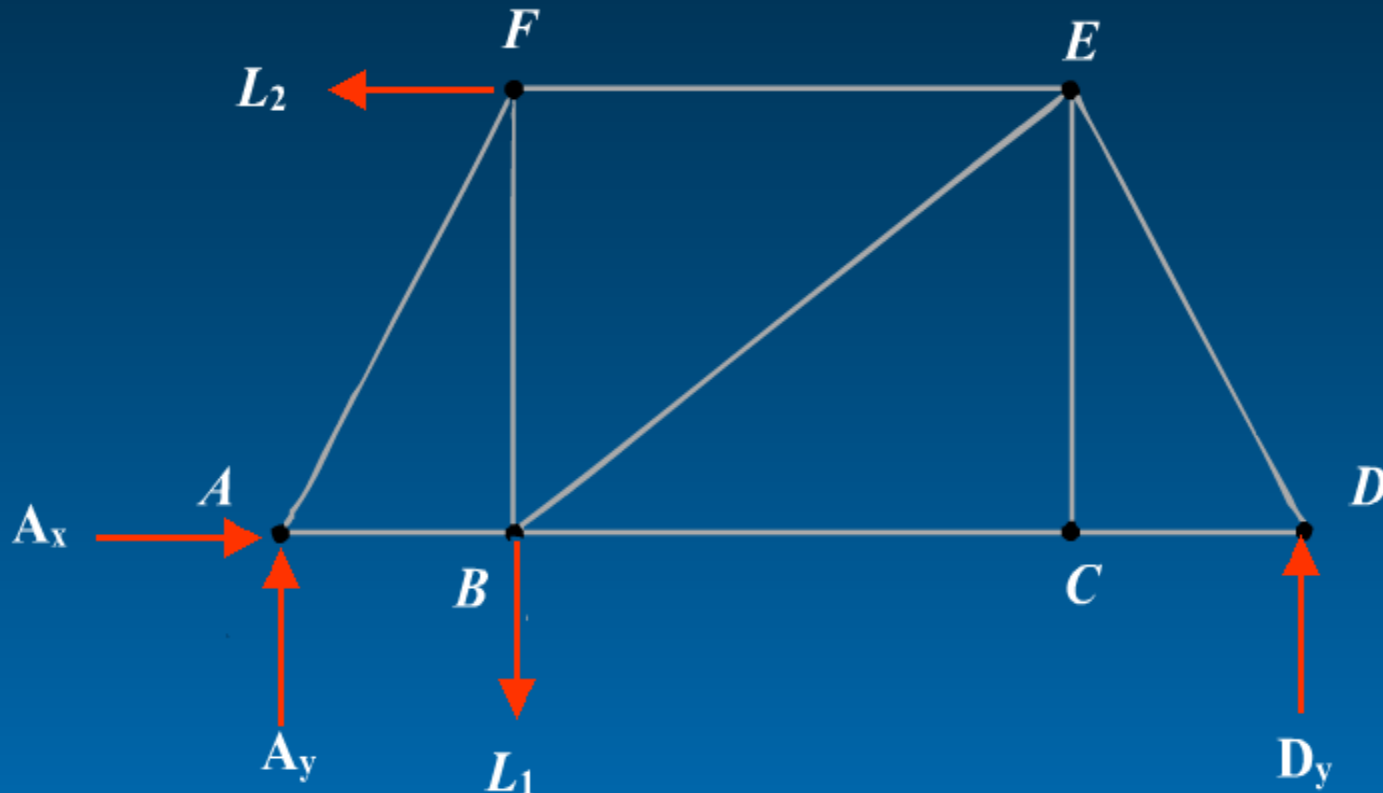
[Back](#)

Click on "Back" to go back and make another cut

2/3



General Concept of the Method of Sections



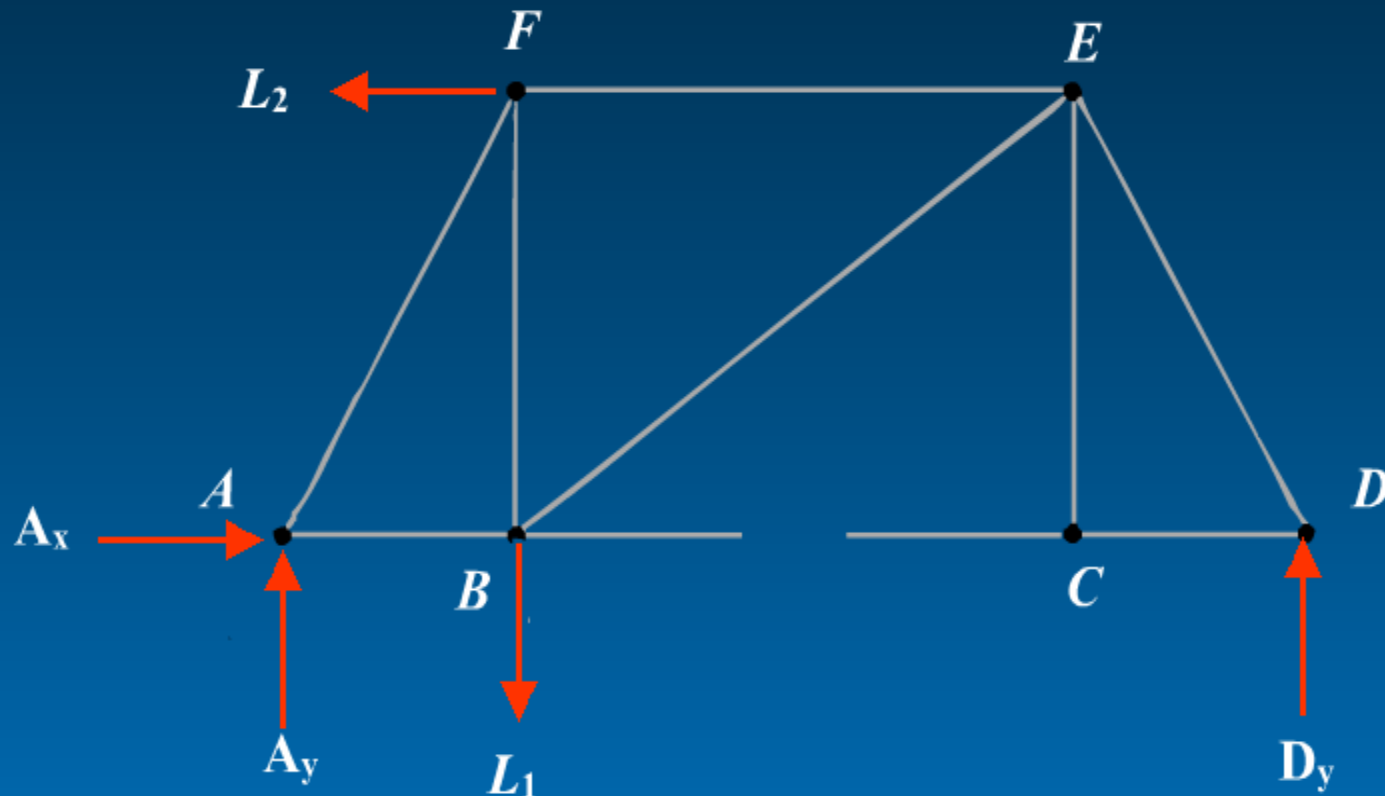
Click on the members to cut/uncut.



2/3



General Concept of the Method of Sections



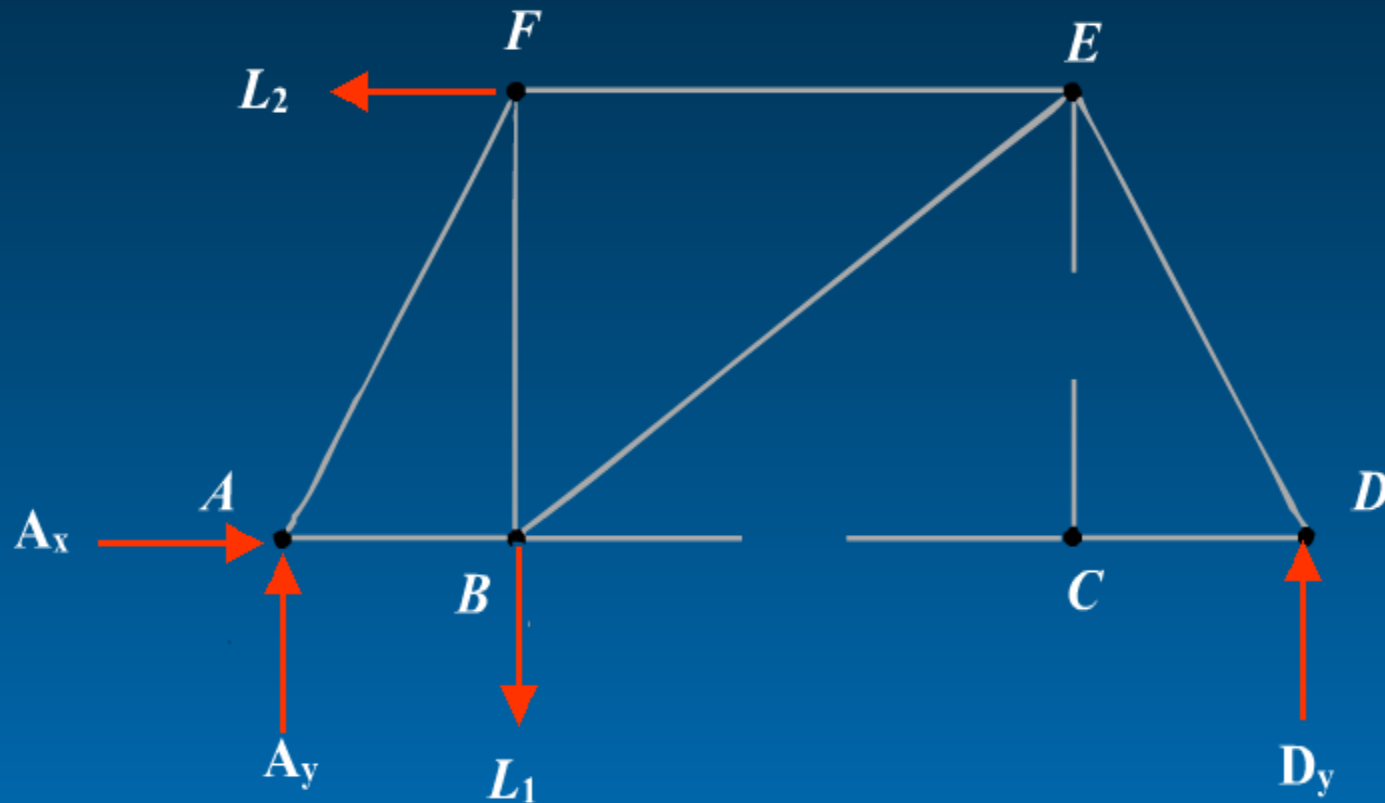
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General Concept of the Method of Sections



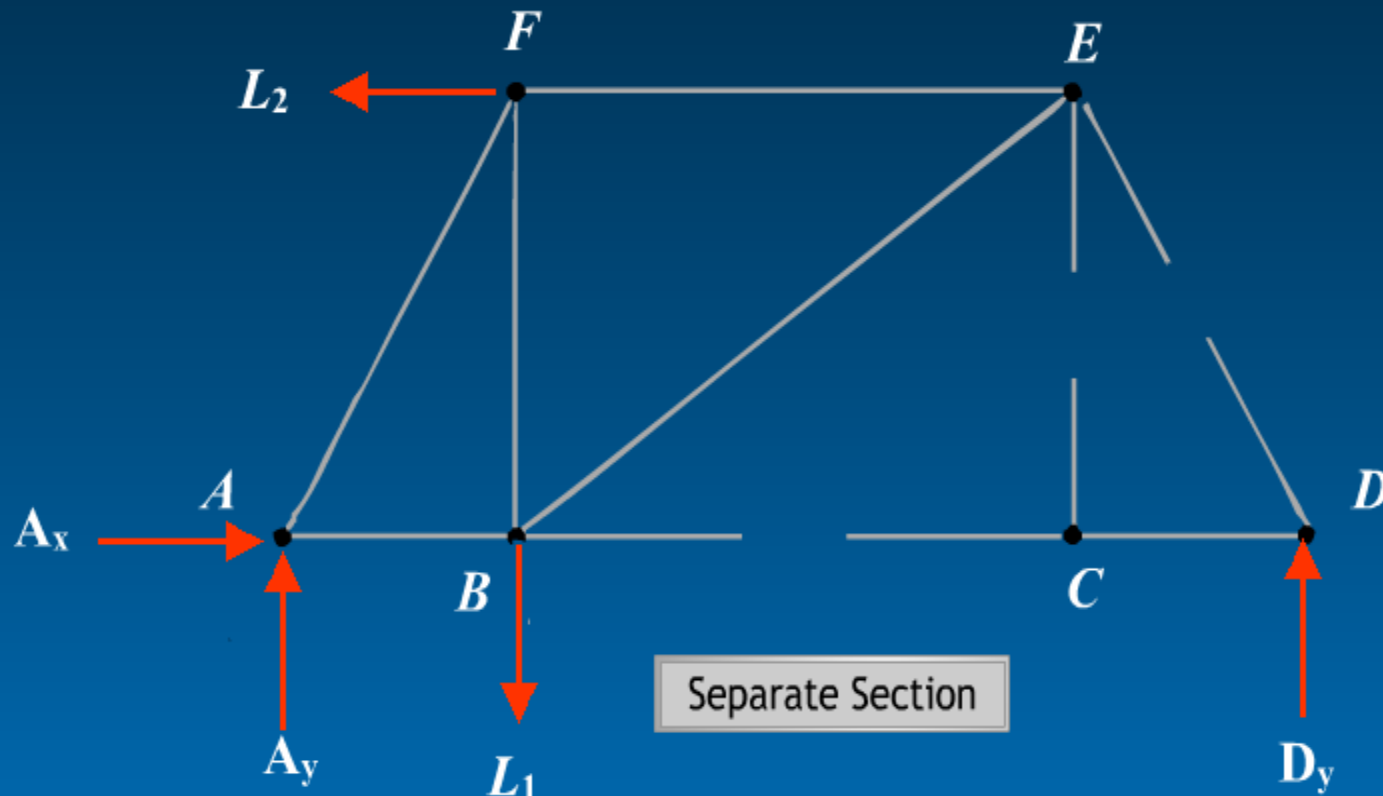
Click on the members to cut/uncut.



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General Concept of the Method of Sections

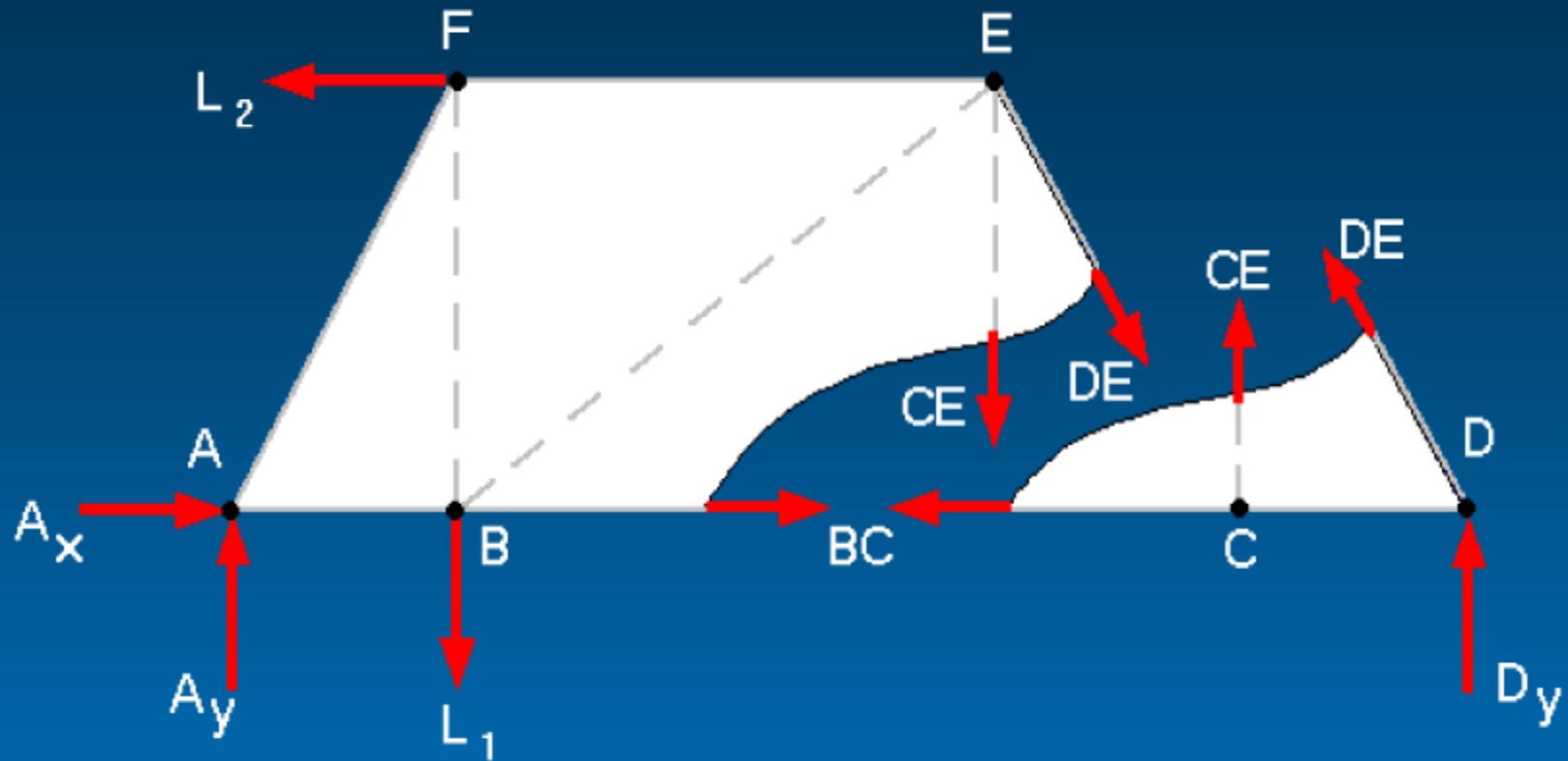


Click on the members to cut/uncut.



2/3

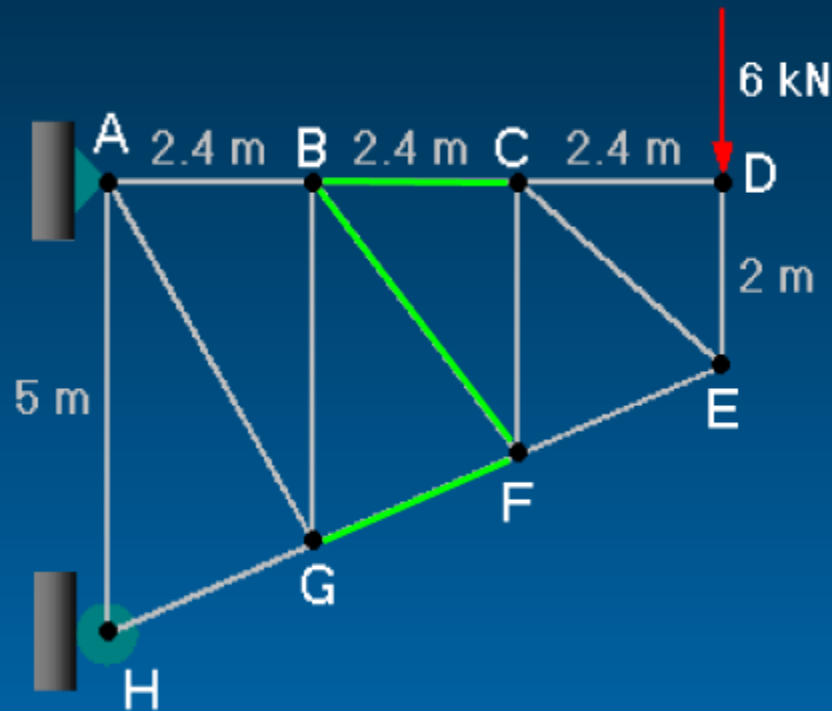


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Click on "Back" to go back and make another cut

2/3

Example: Calculate the forces in BC, BF, and GF



Next

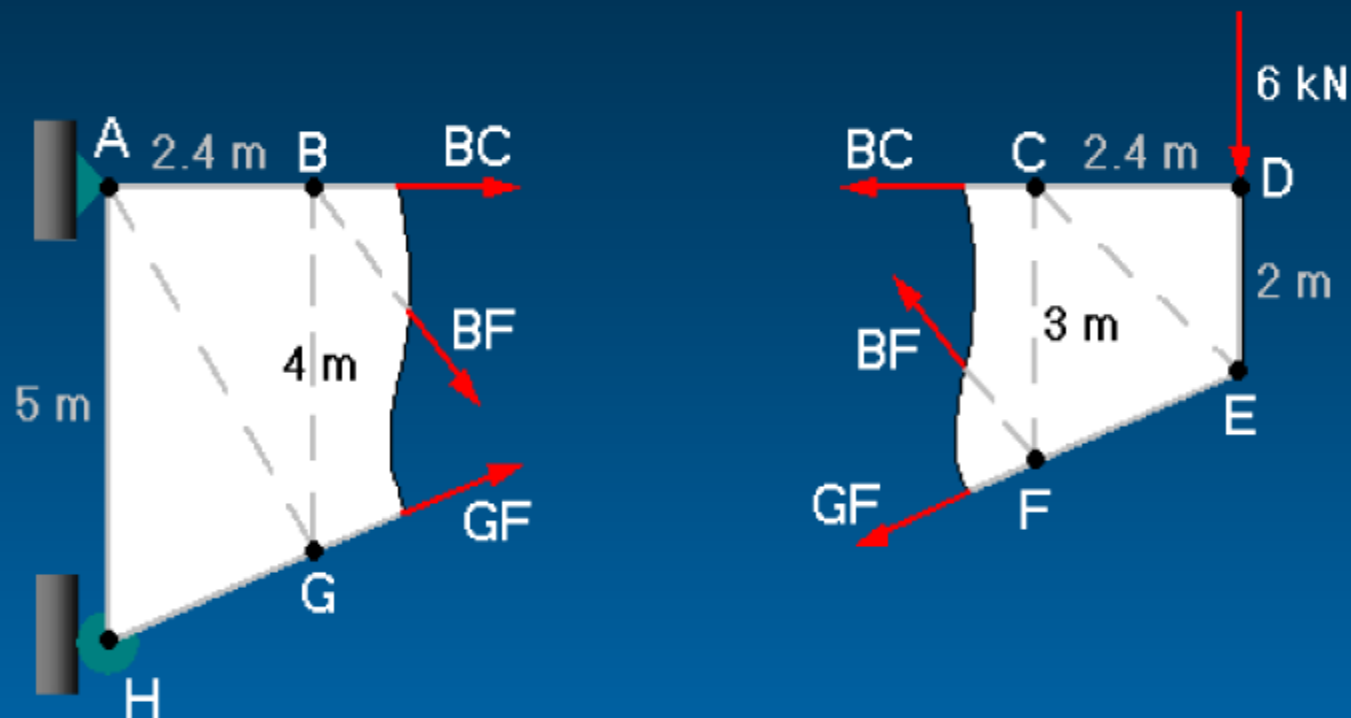
Click on "Next" to make a cut at BC, BF, and GF



3/3



Example: Calculate the forces in BC, BF, and GF

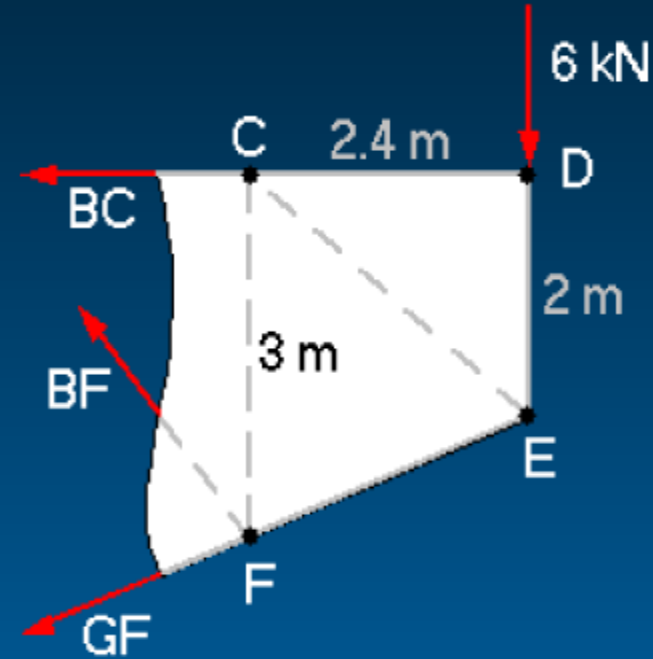
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Click on "Next" to continue



3/3





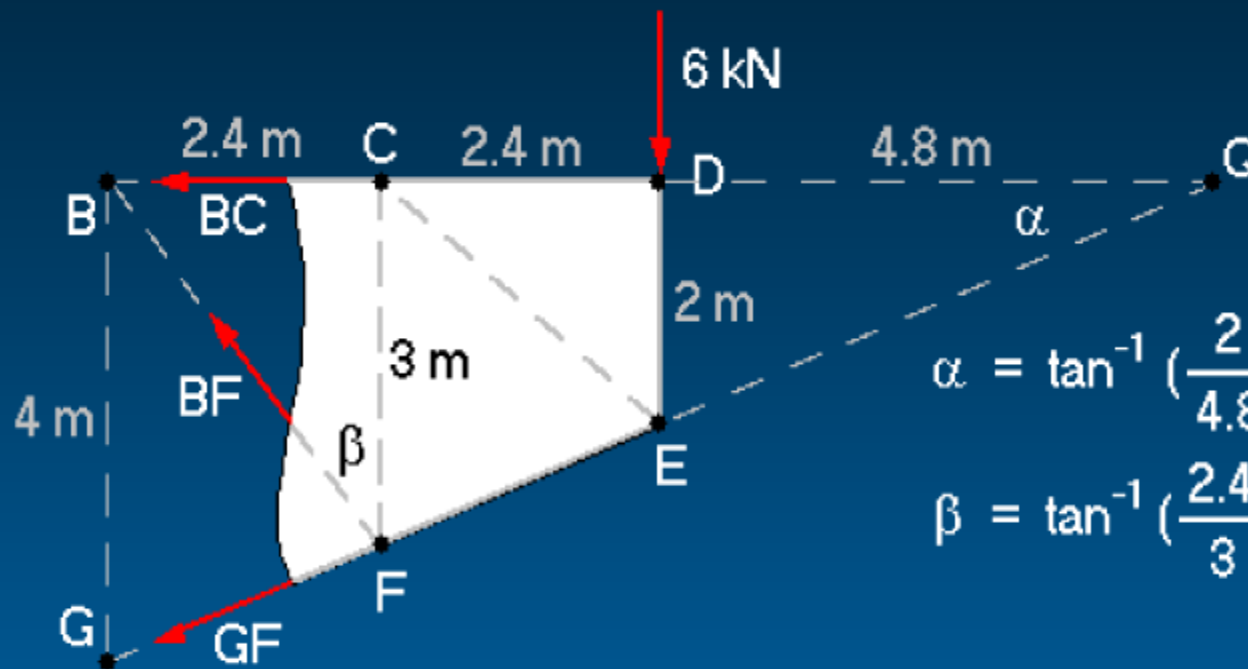
Next

Click on "Next" to continue



3/3





$$+\circlearrowleft \Sigma M_F = 0 :$$

$$+\circlearrowleft \Sigma M_B = 0 :$$

$$+\circlearrowleft \Sigma M_Q = 0 :$$

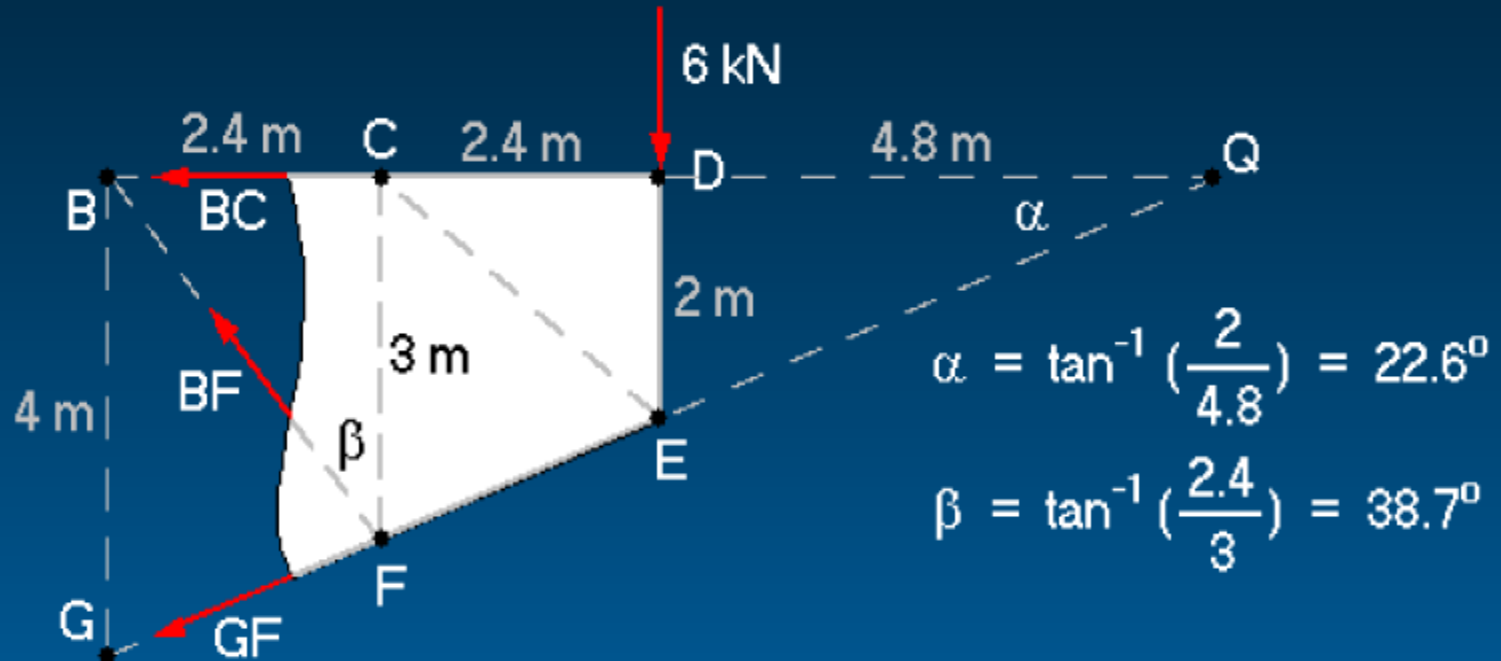
[Show solution](#)

Click on the governing equations



3/3





$$+\circlearrowleft \Sigma M_F = 0 : BC(3) - 6(2.4) = 0$$

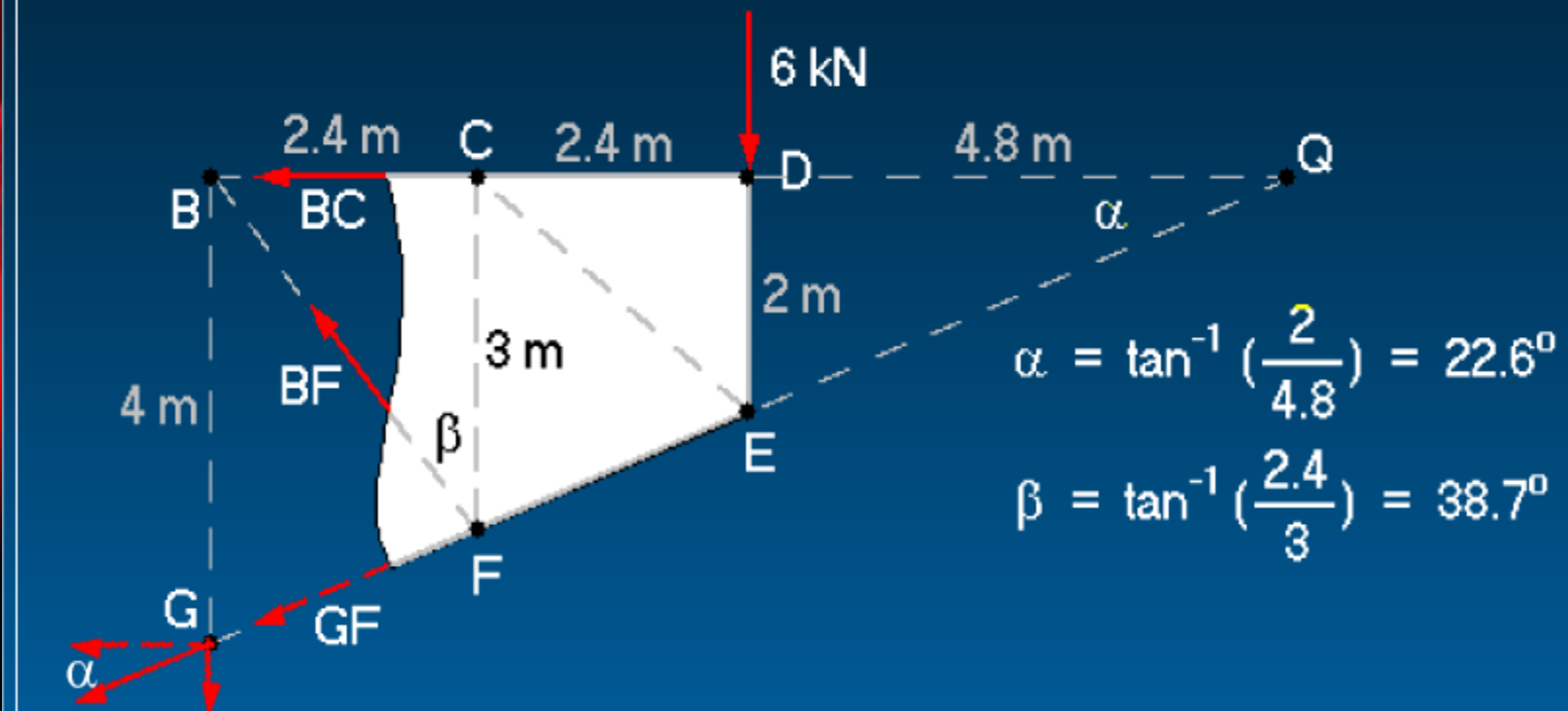
$$+\circlearrowleft \Sigma M_B = 0 :$$

$$+\circlearrowleft \Sigma M_Q = 0 :$$

[Show solution](#)
[Click on the governing equations](#)


3/3





$$GF + \curvearrowright \Sigma M_F = 0 : BC(3) - 6(2.4) = 0$$

$$+ \curvearrowright \Sigma M_B = 0 : -(GF \cos \alpha)(4) - 6(4.8) = 0$$

$$+ \curvearrowright \Sigma M_Q = 0 :$$

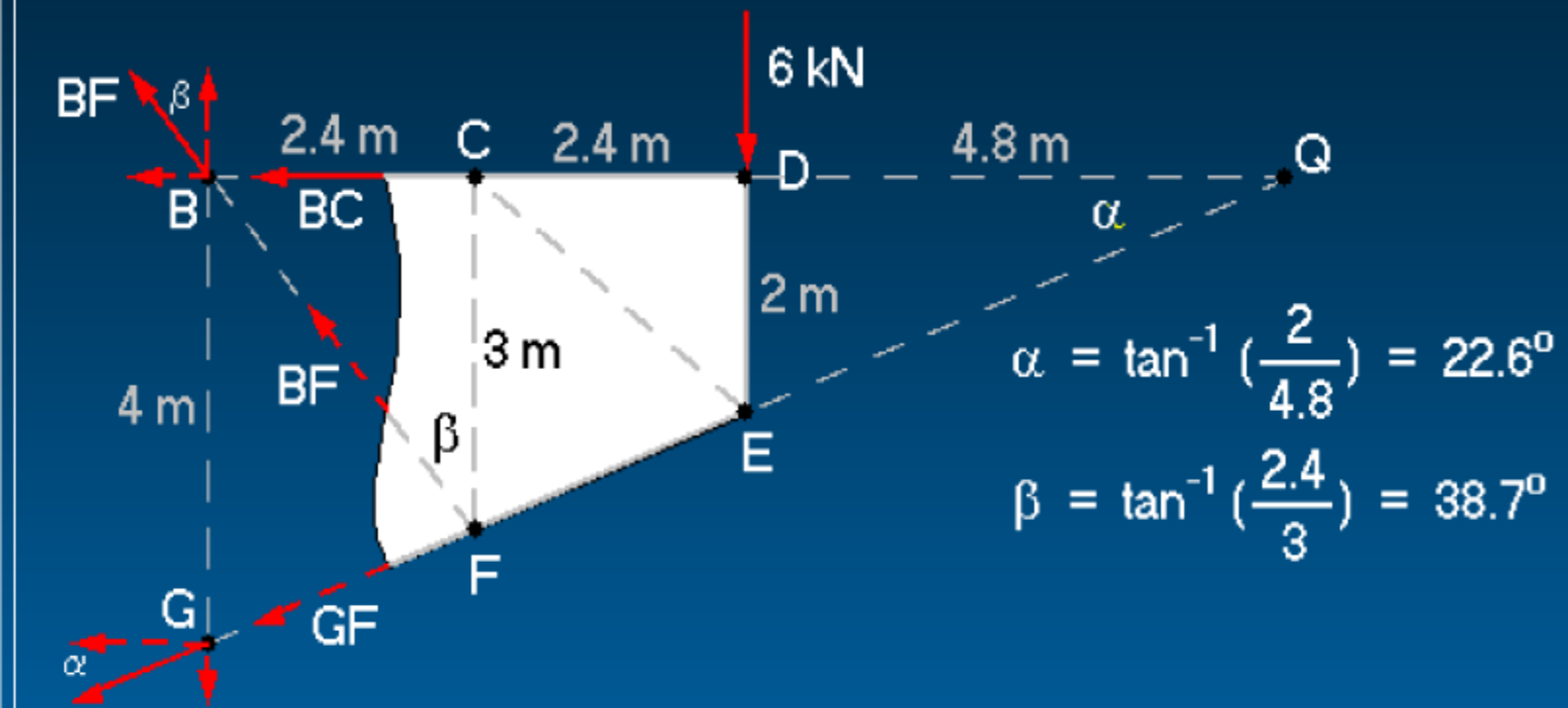
[Show solution](#)

Click on the governing equations



3/3





$$GF + \curvearrowright \Sigma M_F = 0 : BC(3) - 6(2.4) = 0$$

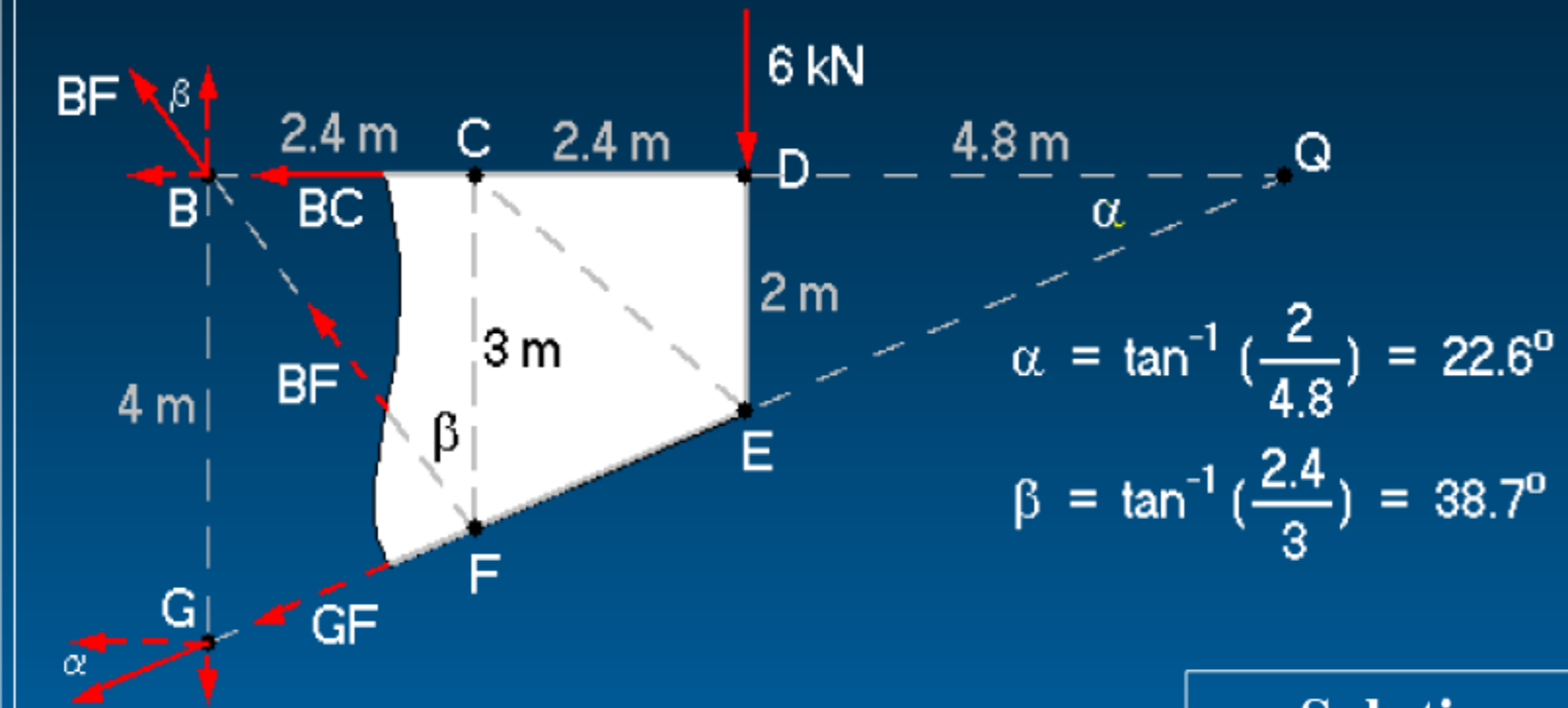
$$+ \curvearrowright \Sigma M_B = 0 : -(GF \cos \alpha)(4) - 6(4.8) = 0$$

$$+ \curvearrowright \Sigma M_Q = 0 : 6(4.8) - (BF \cos \beta)(9.6) = 0$$

[Show solution](#)


3/3





GF₊  **$\Sigma M_F = 0$** : $BC(3) - 6(2.4) = 0$

$$+\curvearrowright \Sigma M_B = 0 : -(GF \cos \alpha)(4) - 6(4.8) = 0$$

$$+\curvearrowright \Sigma \mathbf{M}_Q = \mathbf{0} : 6(4.8) - (BF \cos \beta)(9.6) = 0$$

Solution:

$$BC = 4.8 \text{ kN (T)}$$

$$GF = -7.8 \text{ kN (C)}$$

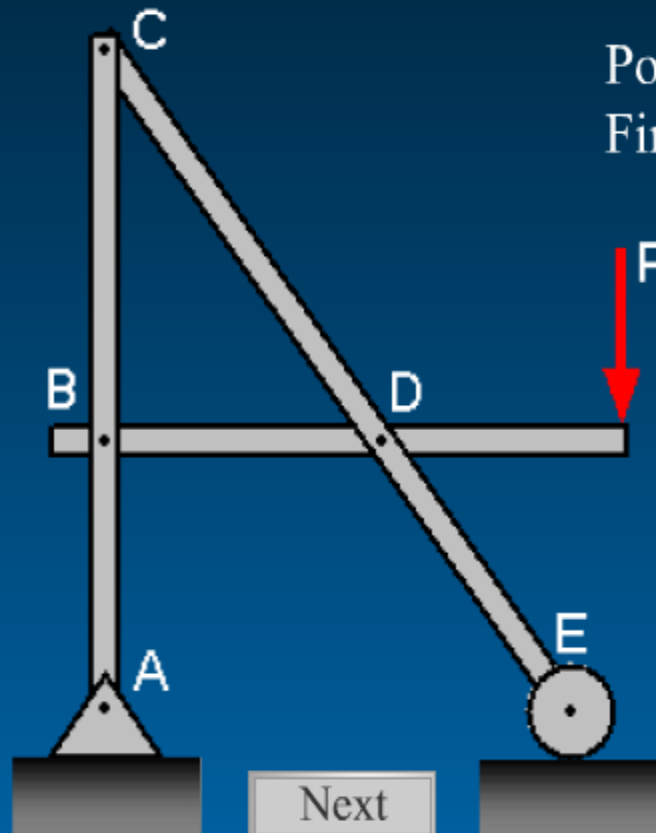
$$BF = 3.84 \text{ kN (T)}$$

Frames and Machines

1. In general, find the external reactions first (not always possible or necessary). Note that some frames will be rigid when detached from their supports and some frames will not.
2. Dismember the frames as necessary, drawing a free-body diagram of each member. Make use of Newton's Third Law for internal forces.
3. Apply equilibrium equations to each member. The number of equilibrium equations must be equal to that of unknowns.



General example



Possible question:
Find all pin reactions.

[Next](#)

Note that there are three external reactions; therefore, the frame is externally statically determinate. Note also that this frame is rigid when pins A and E are removed. Finally, note that all members are three-force members.

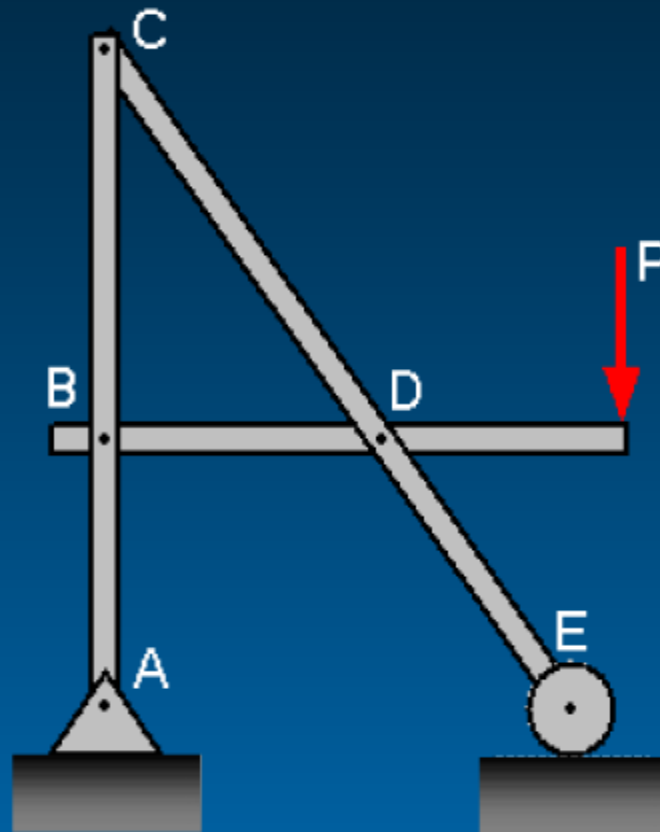
Click on "Next" to proceed



2/3



General example



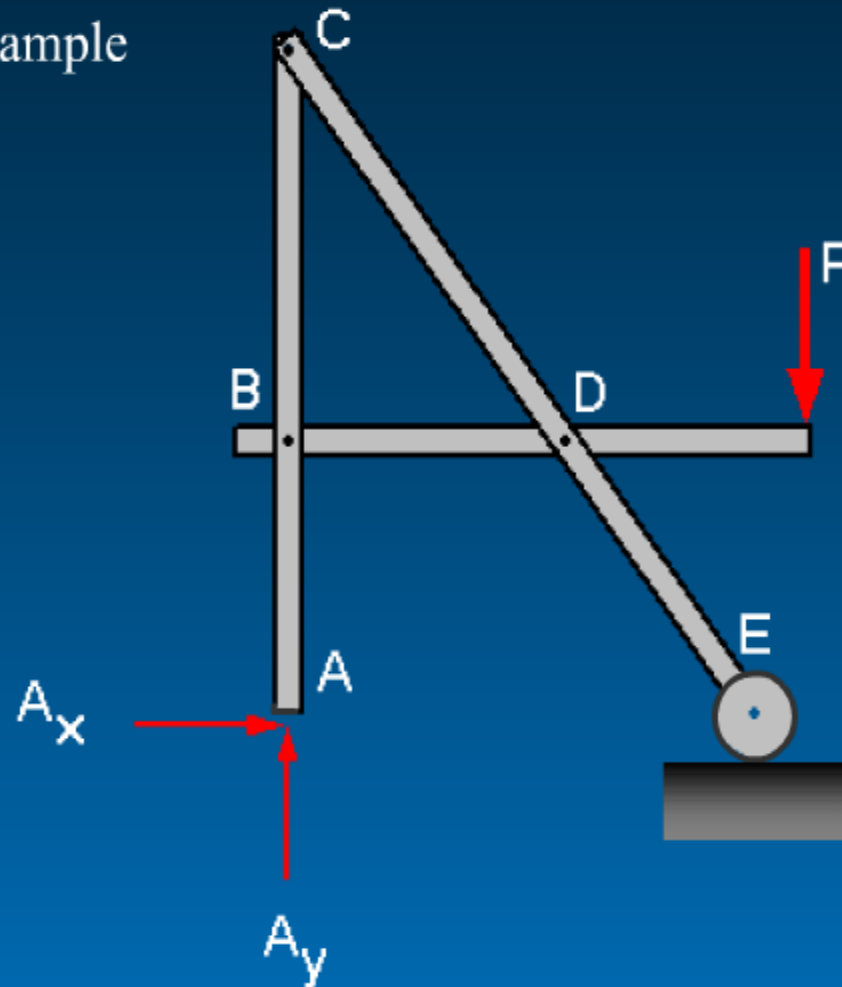
Click on support to display reactions



2/3



General example



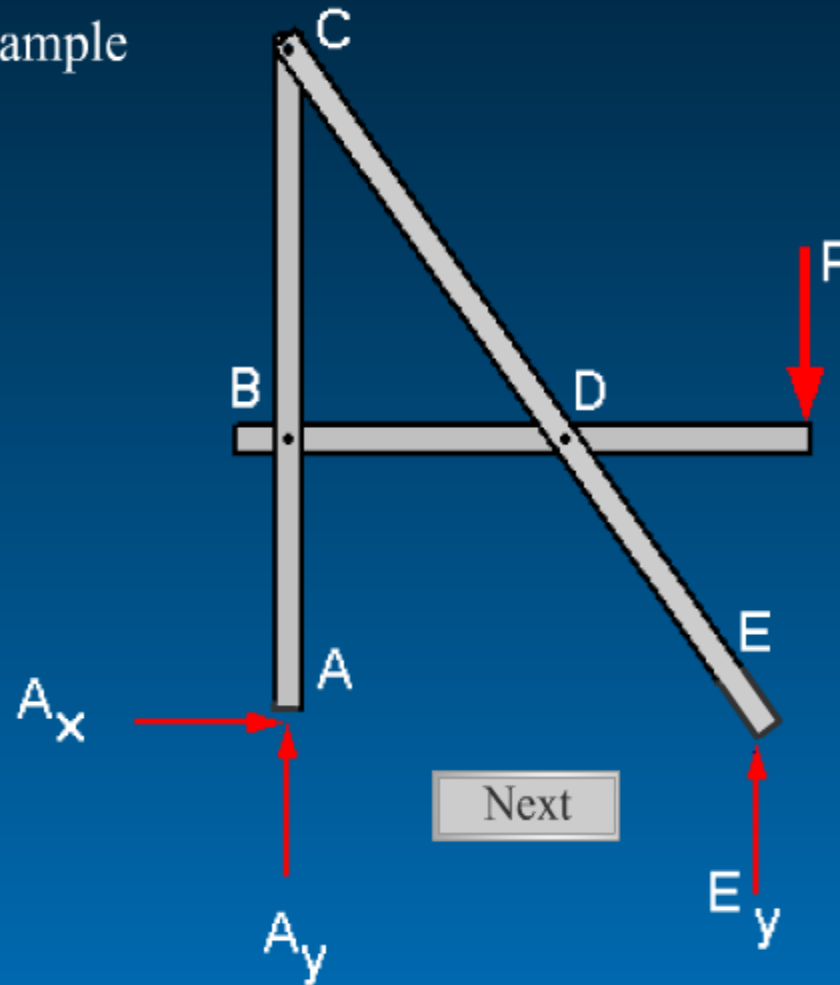
Click on support to display reactions



2/3



General example



Next

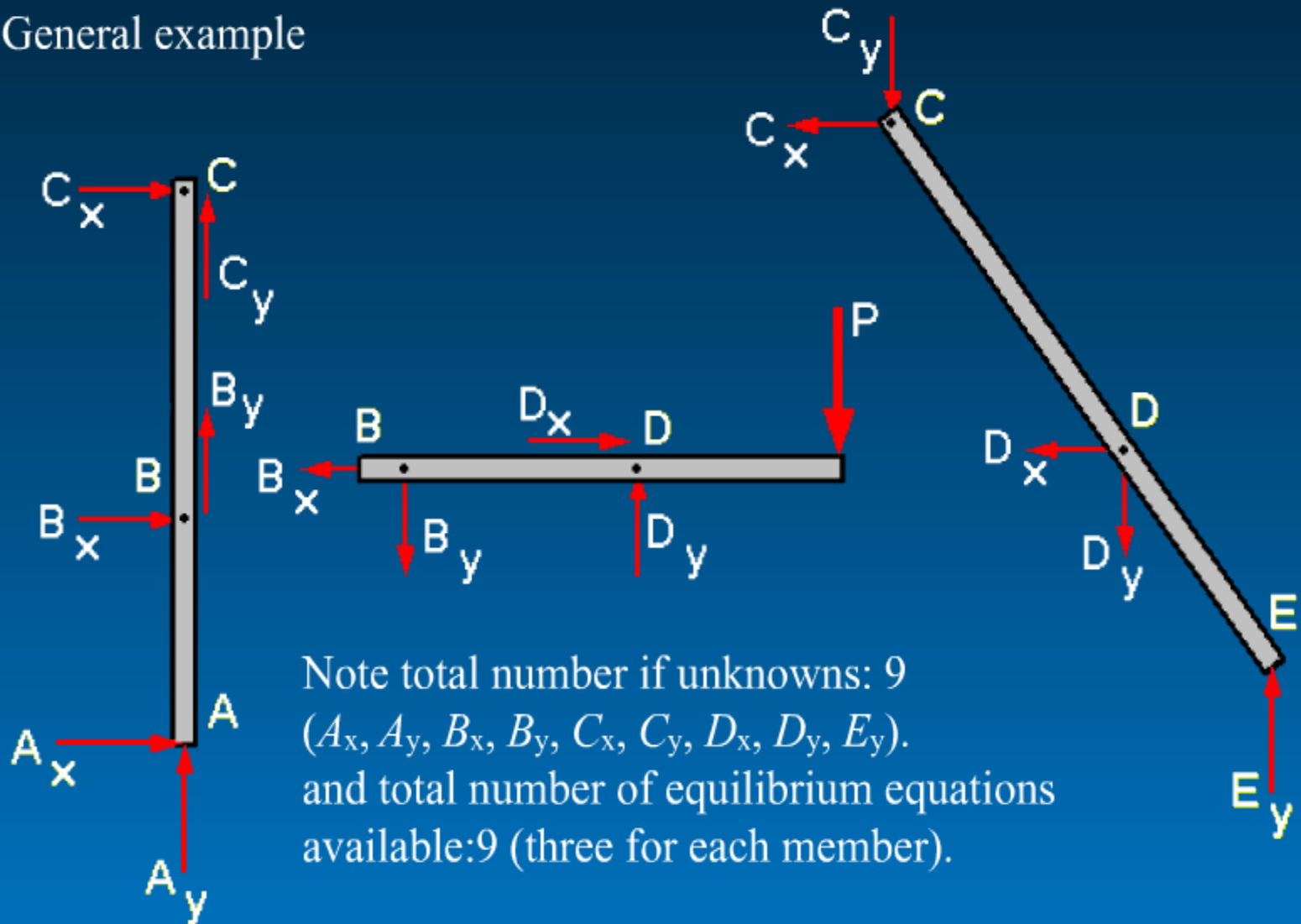
Click on "Next" to dismember frames



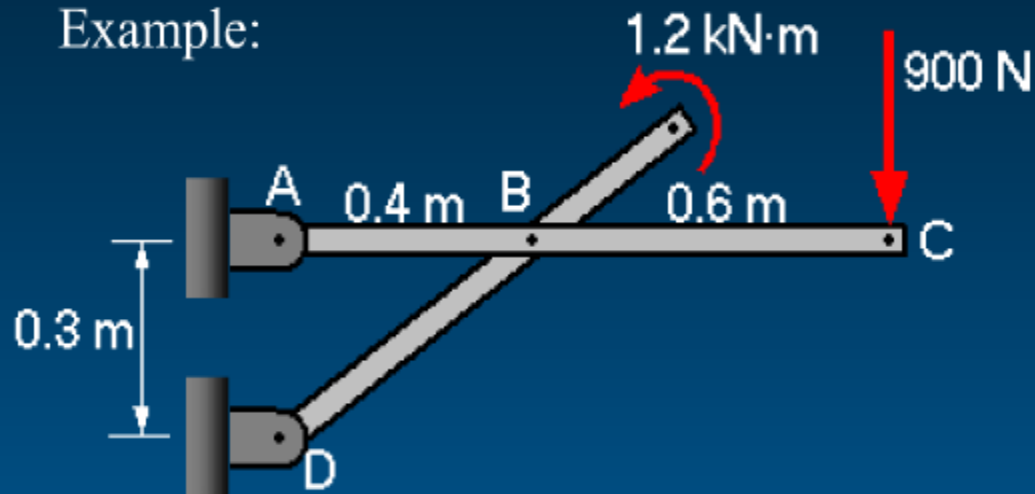
2/3



General example



Example:



Find all pin reactions.

[Next](#)

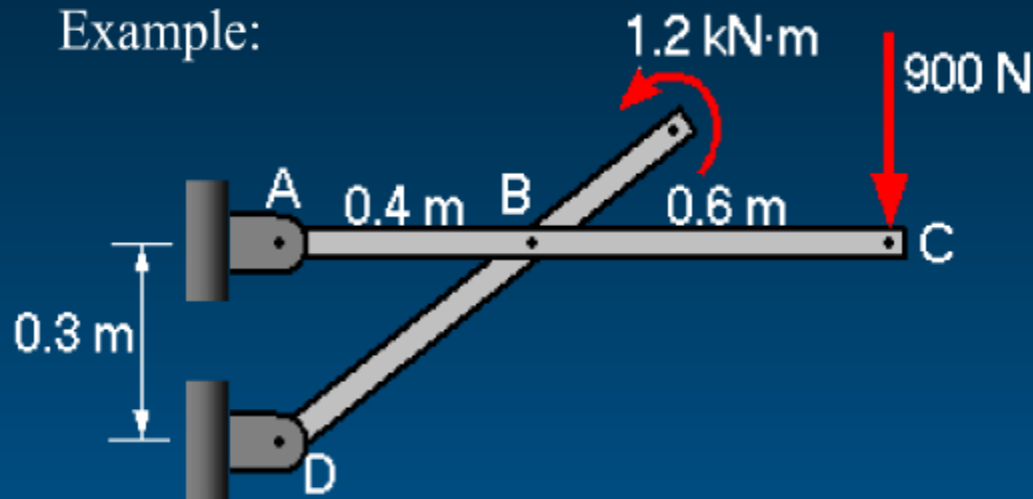
Click on "Next" to proceed



3/3



Example:



Find all pin reactions.

Note that the frame is not rigid when removed from its supports and that it is externally statically indeterminate.

[Next](#)

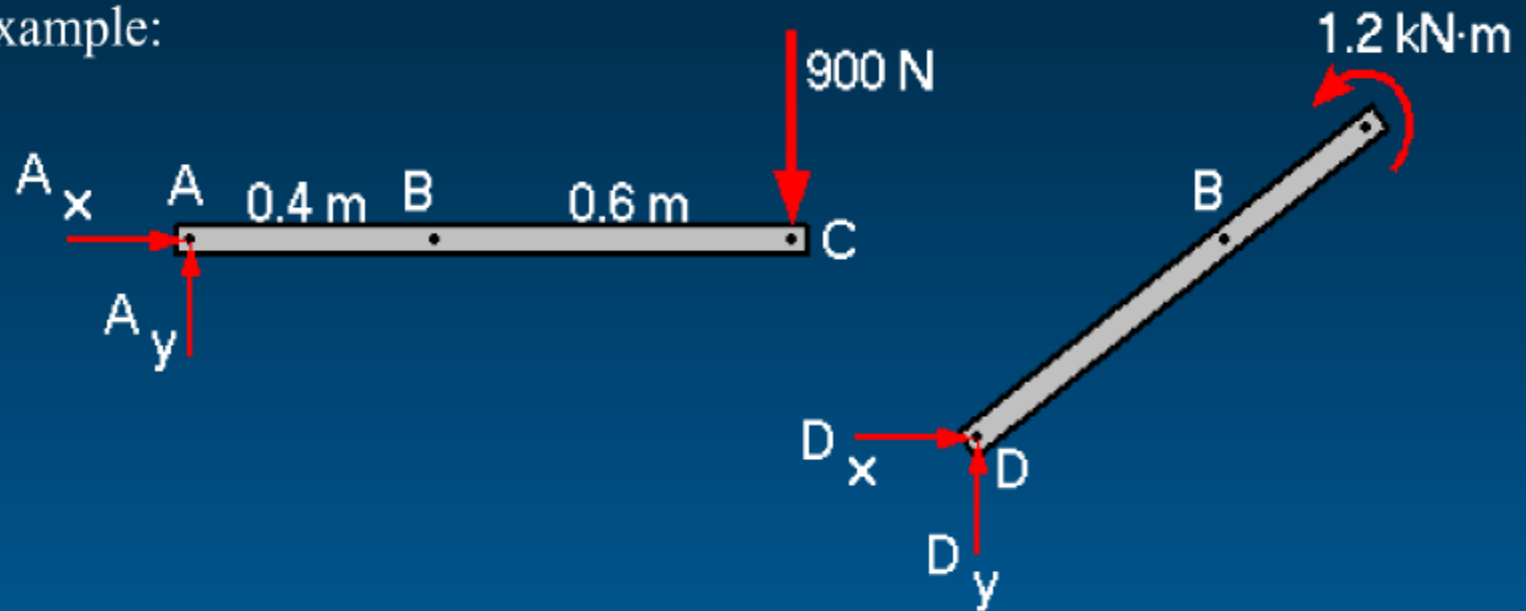
Click on "Next" to proceed



3/3



Example:

[Next](#)

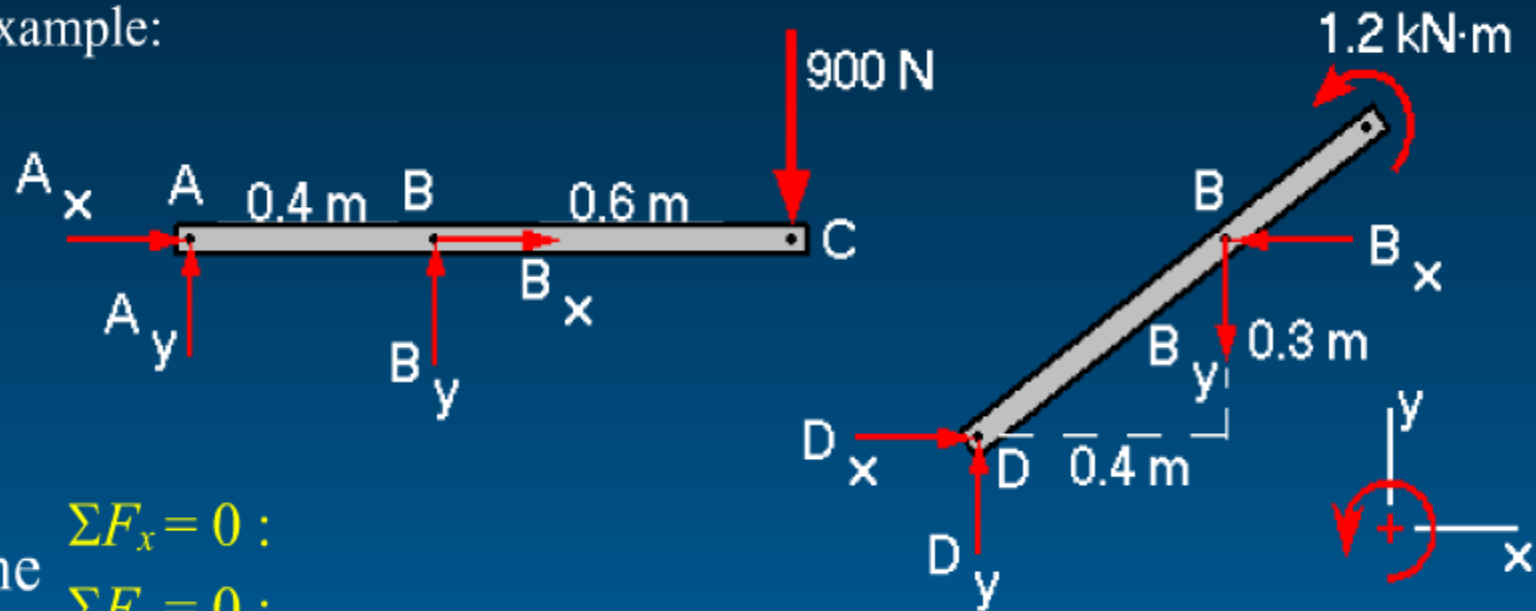
Click on "Next" to proceed



3/3



Example:



Frame
AC:

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_A = 0 :$$

Frame
BD:

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_B = 0 :$$

Show solution

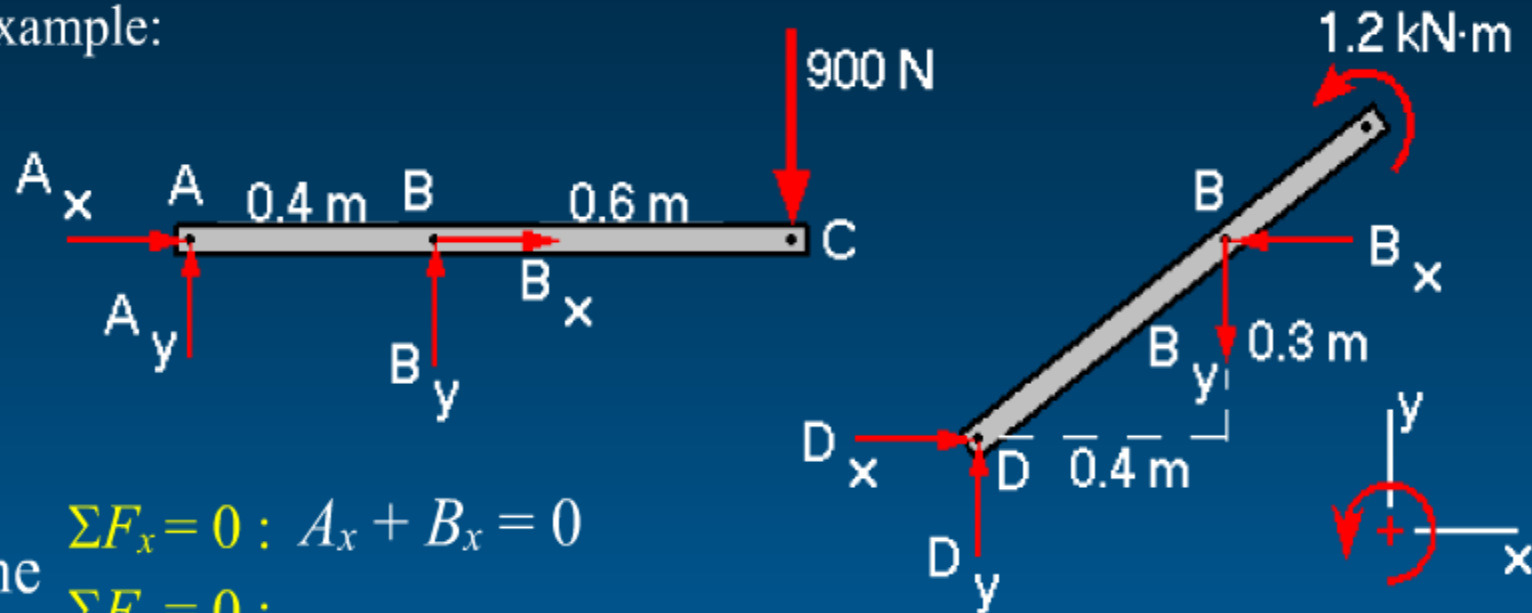
Click on equation buttons to display equations



3/3



Example:



Frame AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_A = 0 :$$

Frame BD:

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_B = 0 :$$

Show solution

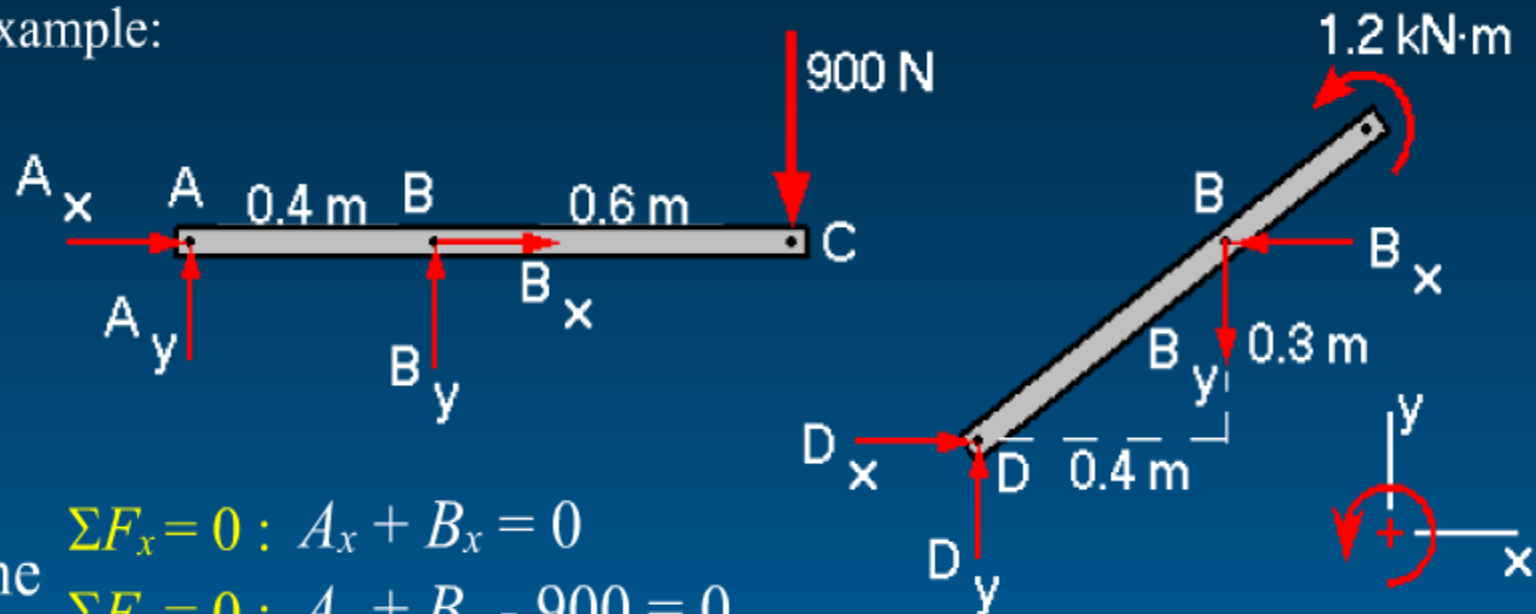
Click on equation buttons to display equations



3/3



Example:

Frame
AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 : A_y + B_y - 900 = 0$$

$$\Sigma M_A = 0 :$$

Frame
BD:

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_B = 0 :$$

Show solution

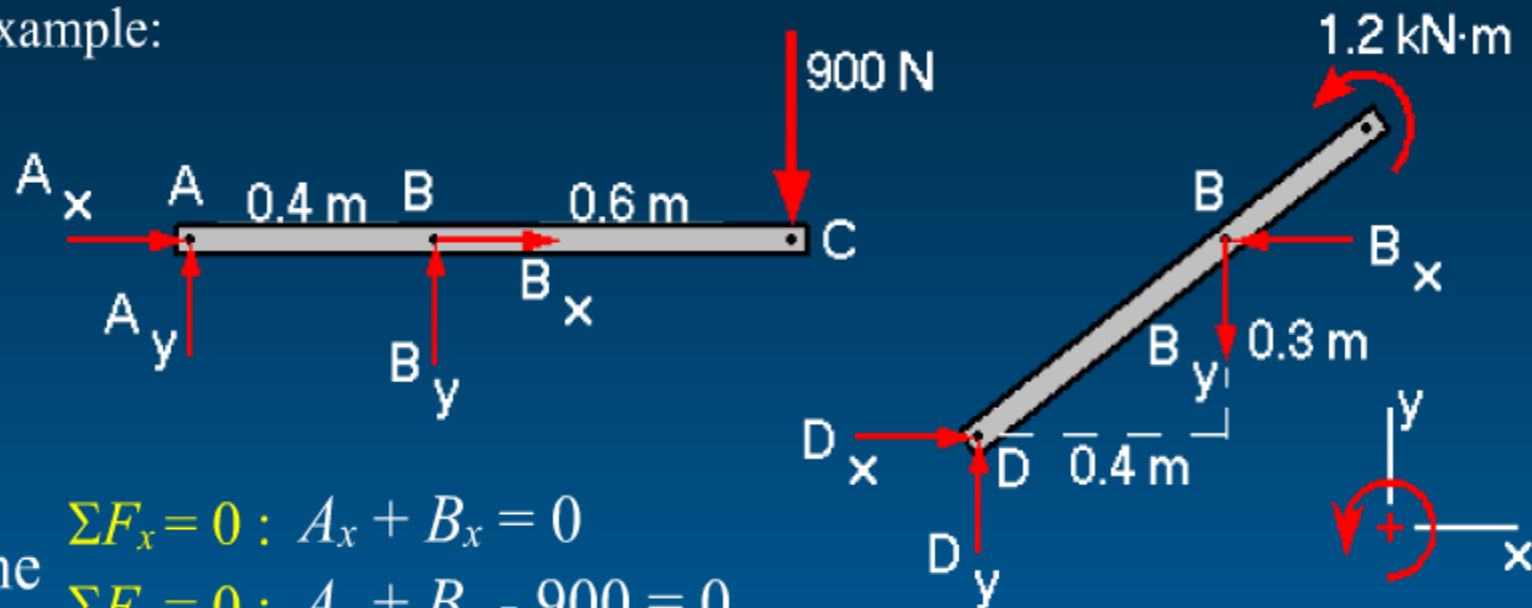
Click on equation buttons to display equations



3/3



Example:

Frame
AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 : A_y + B_y - 900 = 0$$

$$\Sigma M_A = 0 : B_y(0.4) - 900(1) = 0$$

Frame
BD:

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_B = 0 :$$

Show solution

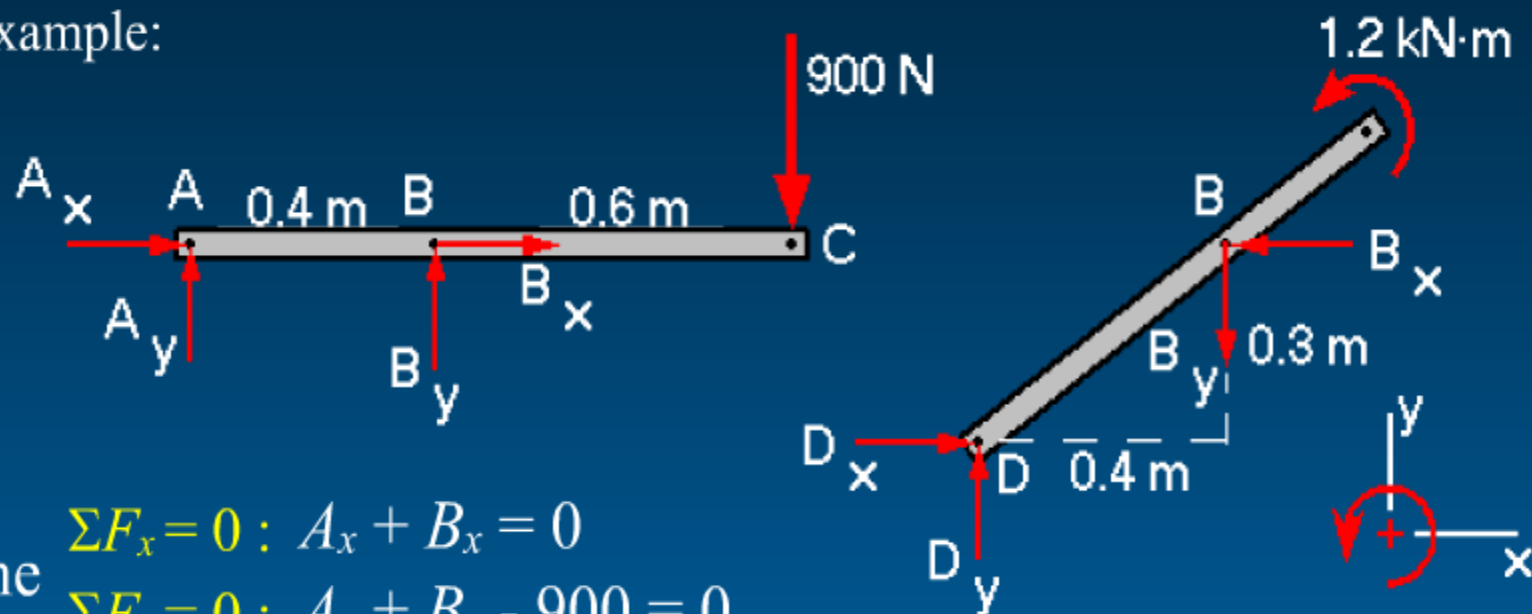
Click on equation buttons to display equations



3/3



Example:

Frame
AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 : A_y + B_y - 900 = 0$$

$$\Sigma M_A = 0 : B_y(0.4) - 900(1) = 0$$

Frame
BD:

$$\Sigma F_x = 0 : D_x - B_x = 0$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_B = 0 :$$

Show solution

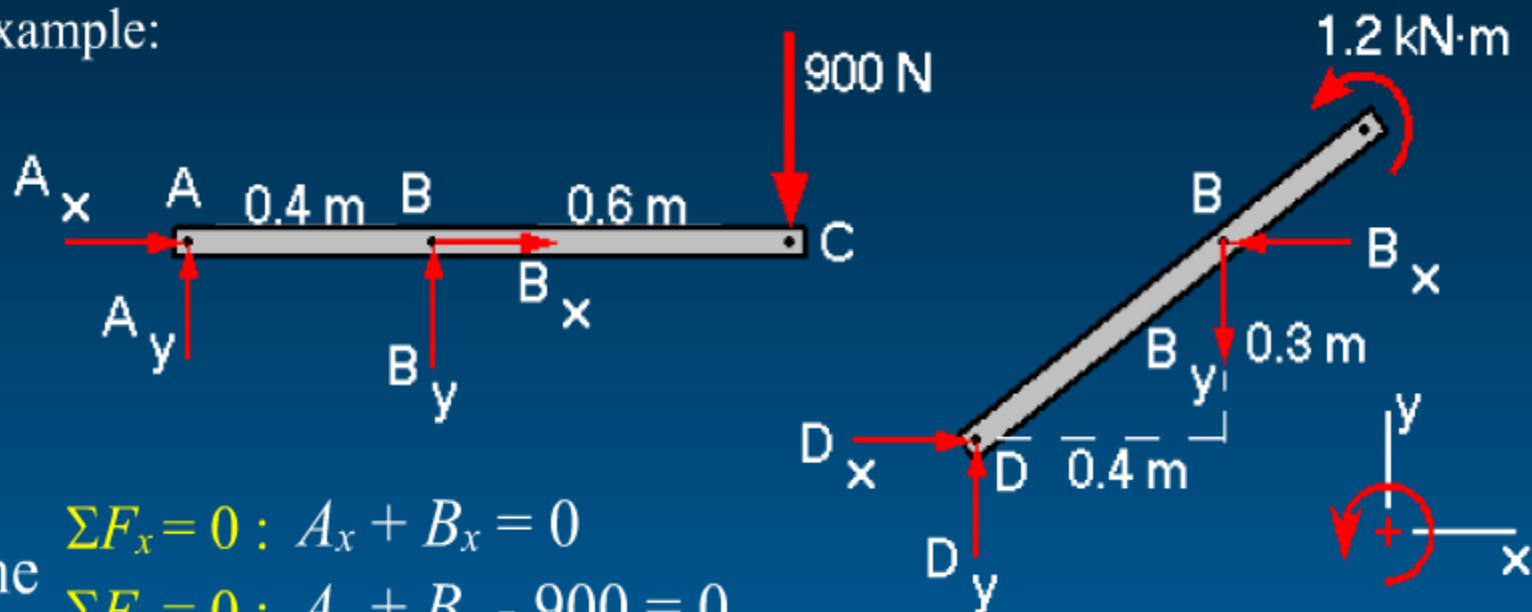
Click on equation buttons to display equations



3/3



Example:

Frame
AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 : A_y + B_y - 900 = 0$$

$$\Sigma M_A = 0 : B_y(0.4) - 900(1) = 0$$

Frame
BD:

$$\Sigma F_x = 0 : D_x - B_x = 0$$

$$\Sigma F_y = 0 : D_y - B_y = 0$$

$$\Sigma M_B = 0 :$$

Show solution

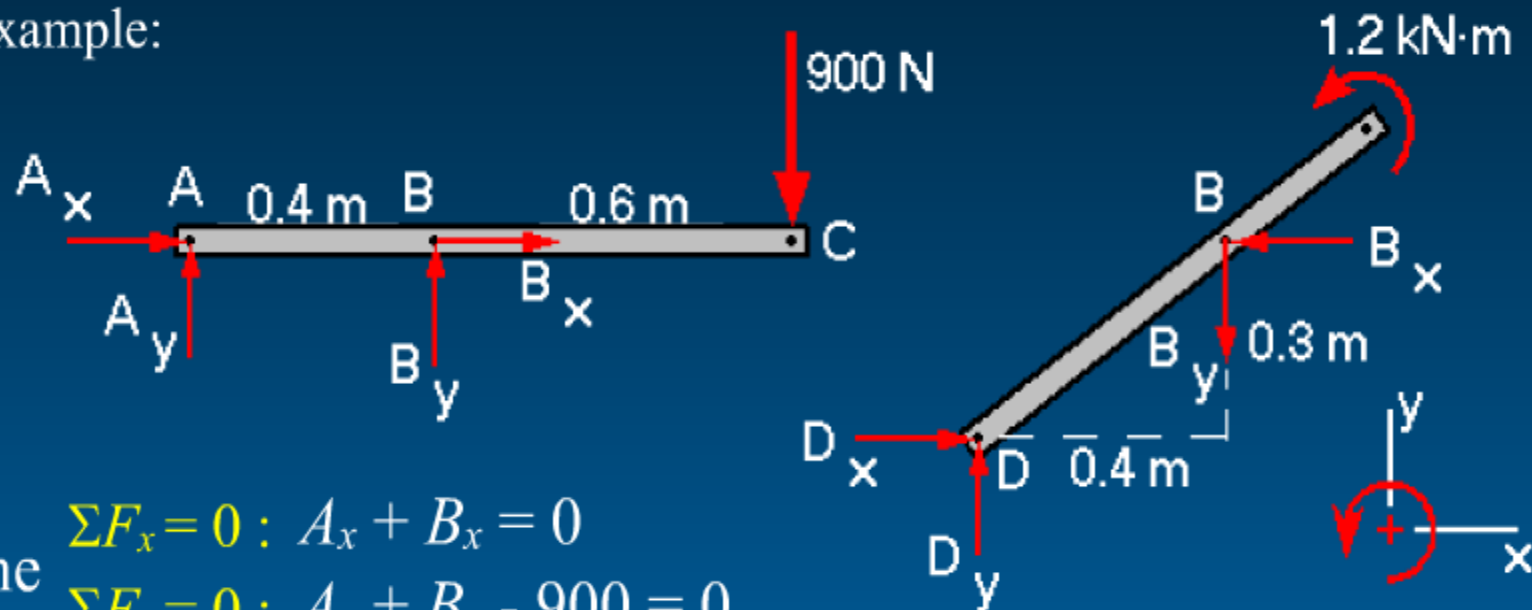
Click on equation buttons to display equations



3/3



Example:

Frame
AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 : A_y + B_y - 900 = 0$$

$$\Sigma M_A = 0 : B_y(0.4) - 900(1) = 0$$

Frame
BD:

$$\Sigma F_x = 0 : D_x - B_x = 0$$

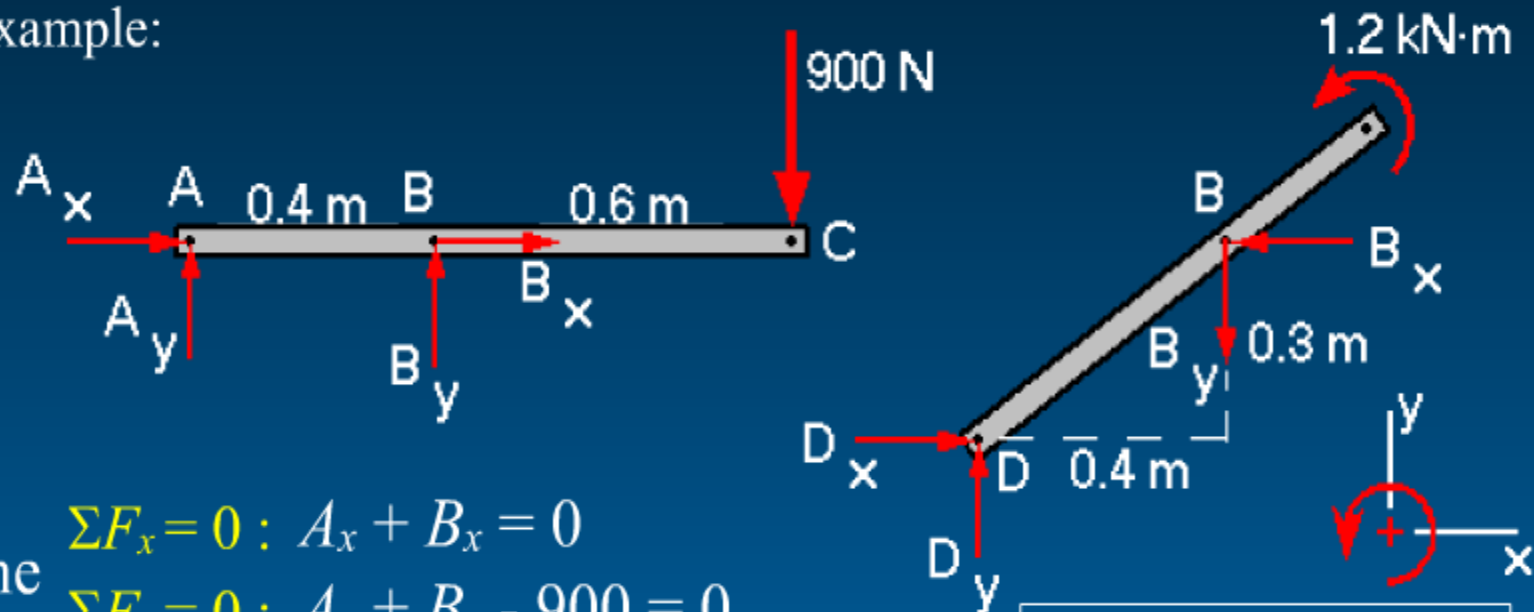
$$\Sigma F_y = 0 : D_y - B_y = 0$$

$$\Sigma M_B = 0 : D_x(0.3) - D_y(0.4) + 1200 = 0$$

Show solution

Click on equation buttons to display equations

Example:

Frame
AC:

$$\Sigma F_x = 0 : A_x + B_x = 0$$

$$\Sigma F_y = 0 : A_y + B_y - 900 = 0$$

$$\Sigma M_A = 0 : B_y(0.4) - 900(1) = 0$$

Frame
BD:

$$\Sigma F_x = 0 : D_x - B_x = 0$$

$$\Sigma F_y = 0 : D_y - B_y = 0$$

$$\Sigma M_B = 0 : D_x(0.3) - D_y(0.4) + 1200 = 0$$

$$A_x = 1000 \text{ N}$$

$$A_y = -1350 \text{ N}$$

$$B_x = D_x = -1000 \text{ N}$$

$$B_y = D_y = 2250 \text{ N}$$

Show solution

Click on equation buttons to display equations

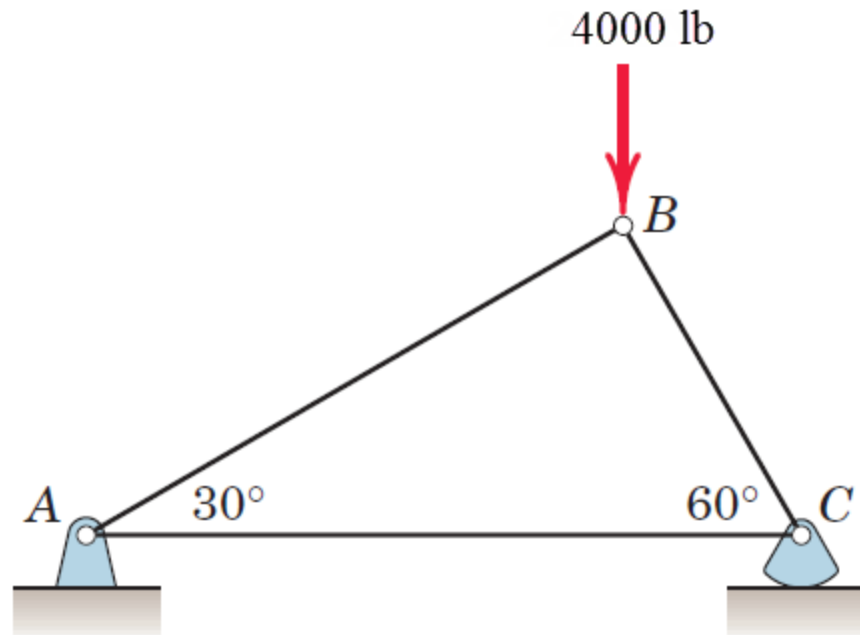


3/3



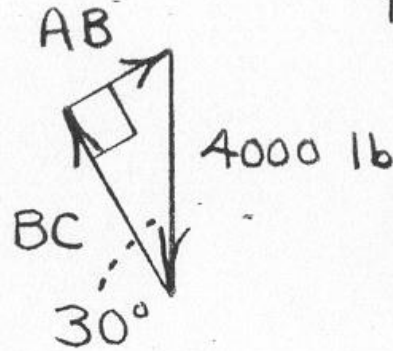
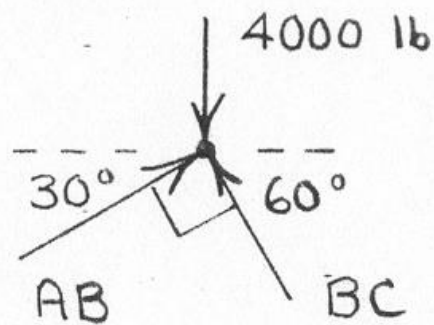
METHOD OF JOINTS

- 4/1** Determine the force in each member of the loaded truss. Explain why knowledge of the lengths of the members is unnecessary.



4/1

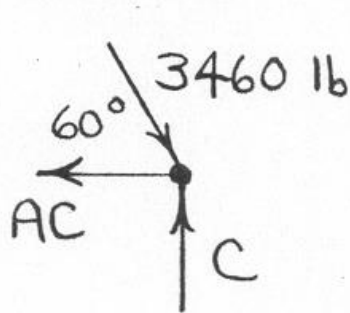
Joint B:



$$AB = 4000 \left(\frac{1}{2} \right) = \underline{2000 \text{ lb C}}$$

$$BC = 4000 \frac{\sqrt{3}}{2} = \underline{3460 \text{ lb C}}$$

Joint C:

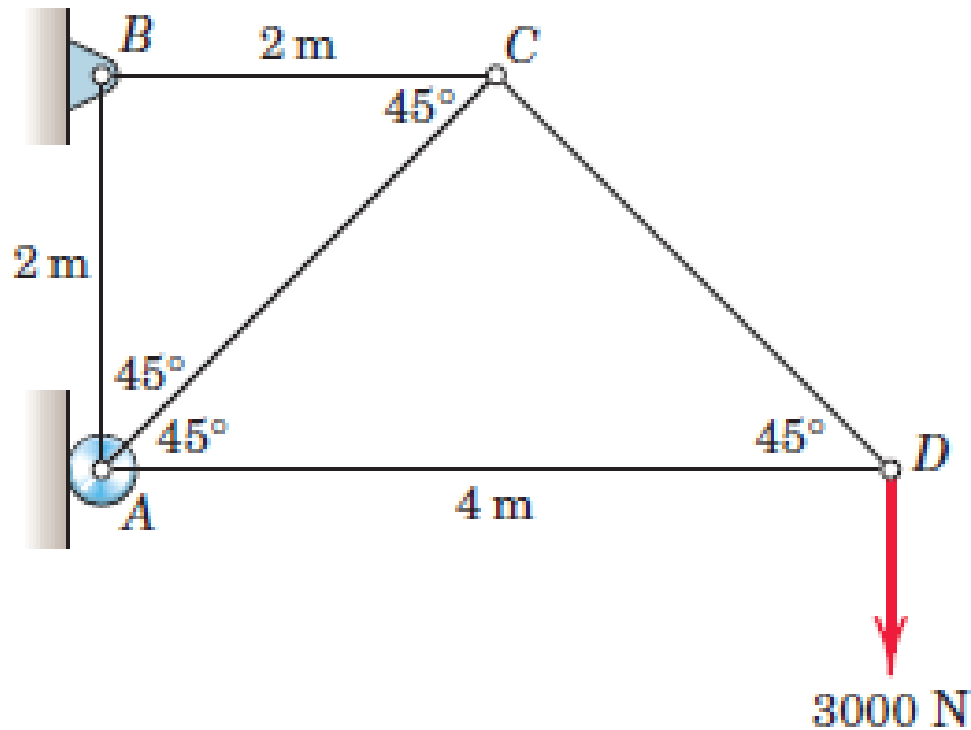


$$\rightarrow \sum F_x = 0: -AC + 3460 \cos 60^\circ = 0$$

$$\underline{AC = 1732 \text{ lb T}}$$

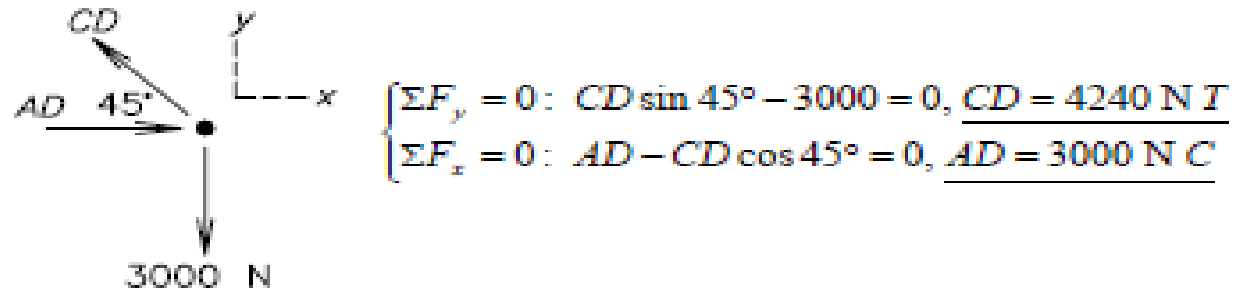
4/3 Determine the force in each member of the loaded truss.

Ans. $AB = 3000 \text{ N T}$, $AC = 4240 \text{ N C}$, $CD = 4240 \text{ N T}$
 $AD = 3000 \text{ N C}$, $BC = 6000 \text{ N T}$

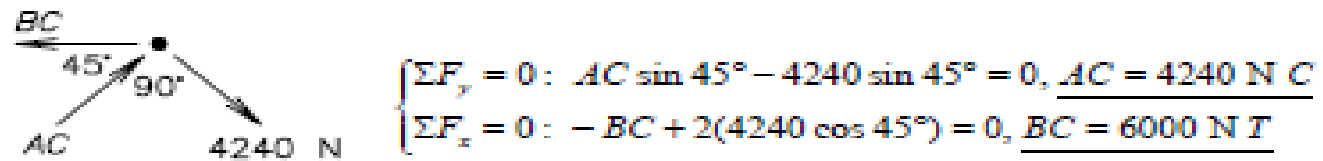


We can begin at joint D without finding the external reactions.

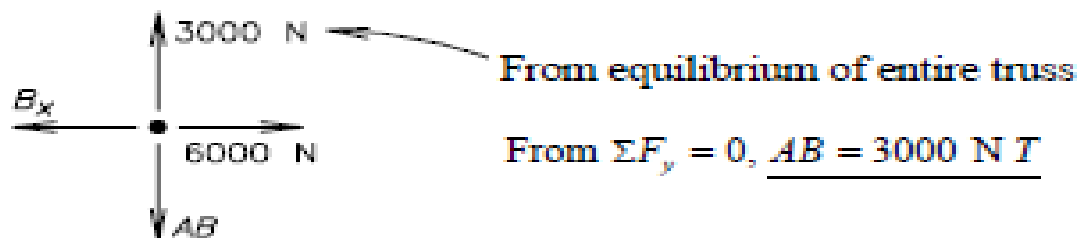
Joint D:



Joint C:

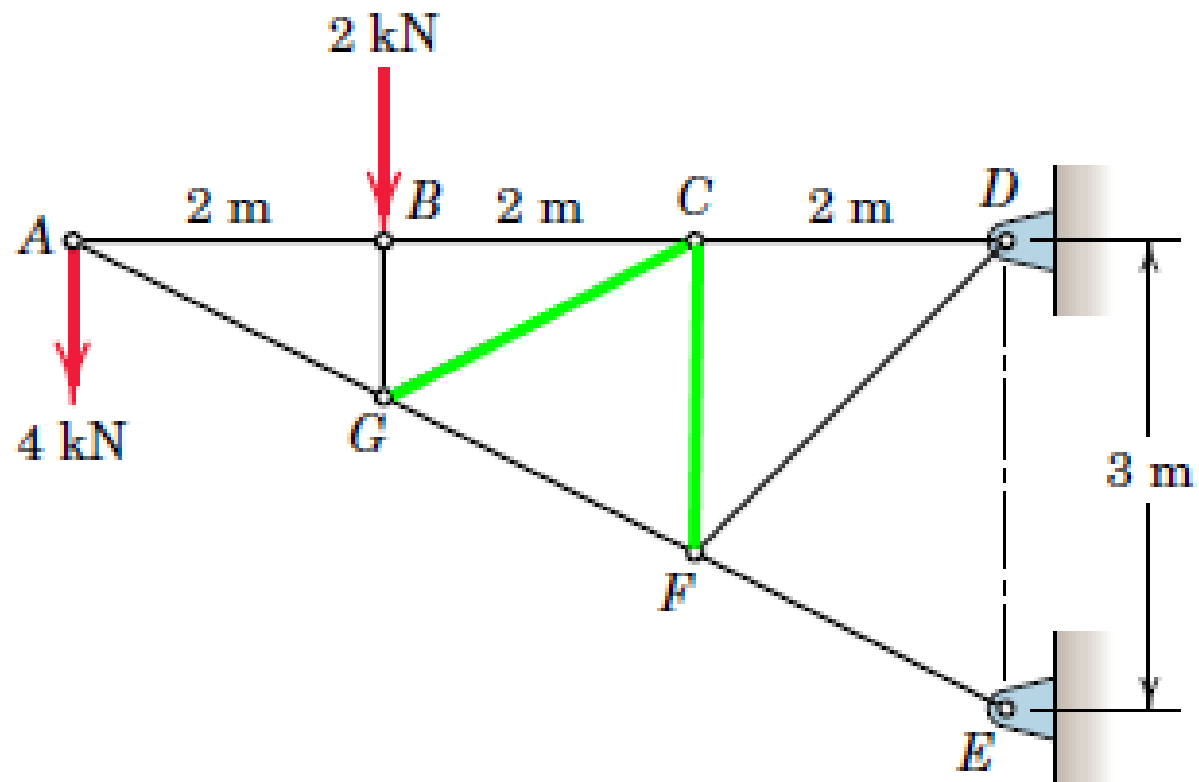


Joint B:



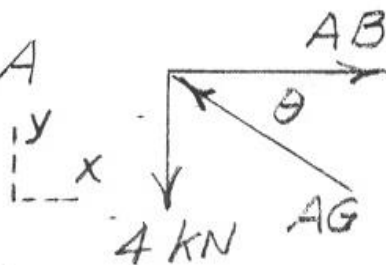
4/11 Calculate the forces in members CG and CF for the truss shown.

Ans. $CG = 2.24 \text{ kN T}$, $CF = 1 \text{ kN C}$



4/11

Joint A



$$\theta = \tan^{-1} 1/2 = 26.57^\circ$$

$$\sin \theta = 1/\sqrt{5}, \cos \theta = 2/\sqrt{5}$$

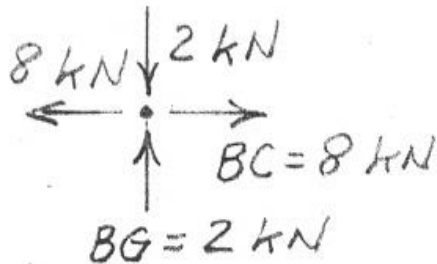
$$\sum F_y = 0; AG/\sqrt{5} - 4 = 0$$

$$AG = 4\sqrt{5} \text{ kN C}$$

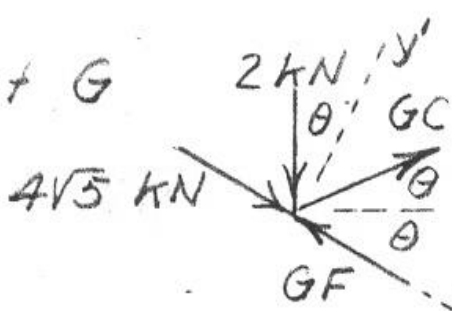
$$\sum F_x = 0; AB - 4\sqrt{5}(2/\sqrt{5}) = 0$$

$$AB = 8 \text{ kN T}$$

Joint B



Joint G



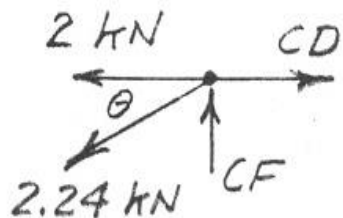
$$\sum F_{y'} = 0; 2\left(\frac{2}{\sqrt{5}}\right) - CG \sin 2\theta = 0$$

$$CG = \frac{4}{\sqrt{5}} \cdot \frac{1}{0.8} = 2.24 \text{ kN T}$$

$$\sum F_{x'} = 0; 2.24 \cos 2\theta + 2/\sqrt{5} + 4\sqrt{5} - GF = 0$$

$$GF = 5\sqrt{5} = 11.18 \text{ kN C}$$

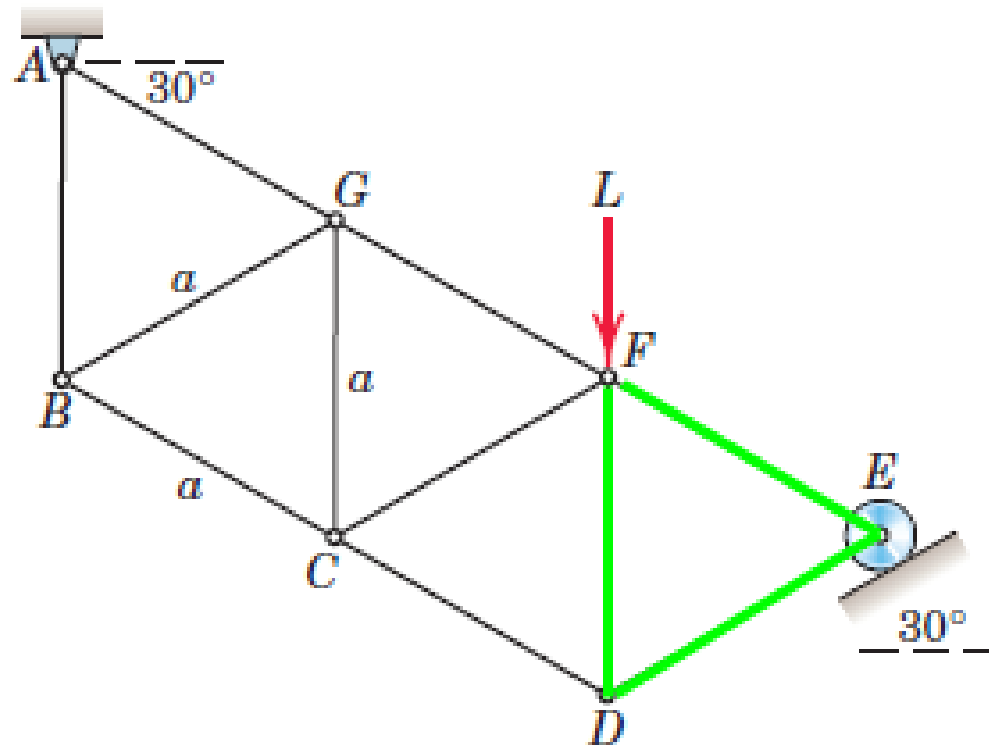
Joint C



$$\sum F_y = 0; CF - 2.24 \sin \theta = 0$$

$$CF = 1.00 \text{ kN C}$$

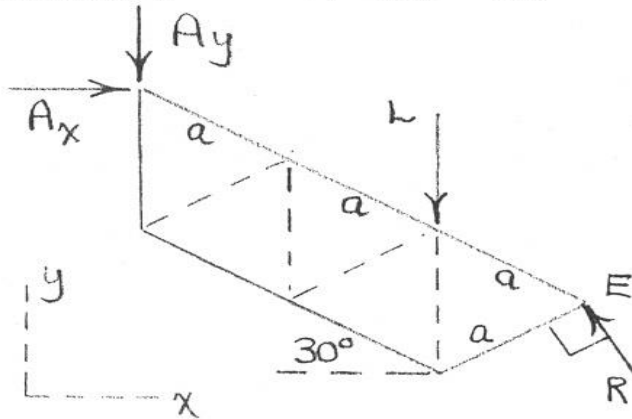
4/14 The truss is composed of equilateral triangles of sides a and is loaded and supported as shown. Determine the forces in members EF , DE , and DF .



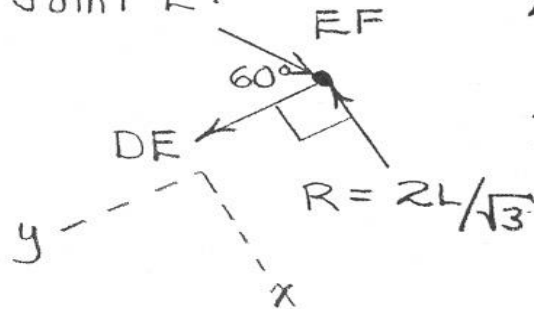
4/14

$$+\circlearrowleft \sum M_A = 0 : L(2a \cos 30^\circ) - R(3a \sin 30^\circ) = 0$$

$$R = \frac{2L}{\sqrt{3}}$$



Joint E:



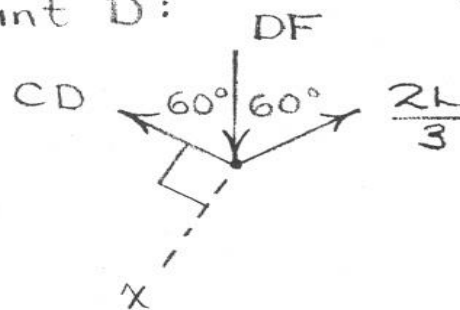
$$\sum F_x = 0 : EF \sin 60^\circ - \frac{2L}{\sqrt{3}} = 0$$

$$EF = \frac{4L}{3} \quad C$$

$$\sum F_y = 0 : DE - \frac{4L}{3} \cos 60^\circ = 0$$

$$DE = \frac{2L}{3} \quad T$$

Joint D:



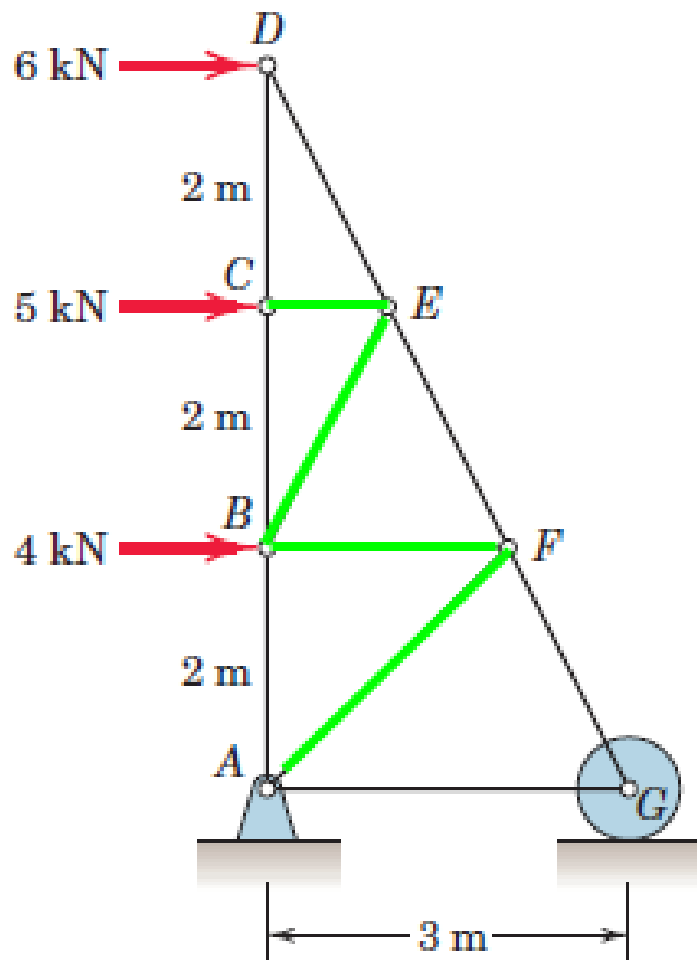
$$\sum F_x = 0 : DF \cos 30^\circ - \frac{2L}{3} \cos 30^\circ = 0$$

$$DF = \frac{2L}{3} \quad C$$

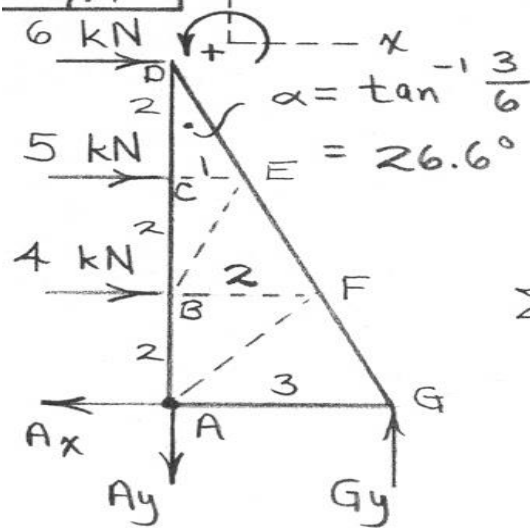
4/17 Determine the forces in members AF , BE , BF , and CE of the loaded truss.

Ans. $AF = 6.13 \text{ kN T}$, $BE = 5.59 \text{ kN T}$

$BF = 6.50 \text{ kN C}$, $CE = 5 \text{ kN C}$



4/17



$$\sum M_A = 0: -6(6) - 5(4) - 4(2) + G_y(3) = 0$$

$$G_y = 21.3 \text{ kN}$$

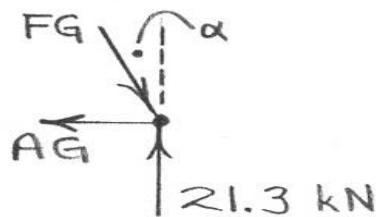
$$\sum F_y = 0: -A_y + G_y = 0$$

$$A_y = 21.3 \text{ kN}$$

$$\sum F_x = 0: 6 + 5 + 4 - A_x = 0$$

$$A_x = 15 \text{ kN}$$

Joint G:



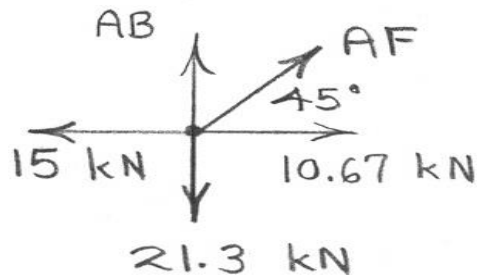
$$\sum F_y = 0: 21.3 - FG \cos \alpha = 0$$

$$FG = 23.9 \text{ kN C}$$

$$\sum F_x = 0: 23.9 \sin \alpha - AG = 0$$

$$AG = 10.67 \text{ kN T}$$

Joint A:



$$\sum F_x = 0: 10.67 - 15 + AF \cos 45^\circ = 0$$

$$AF = 6.13 \text{ kN T}$$

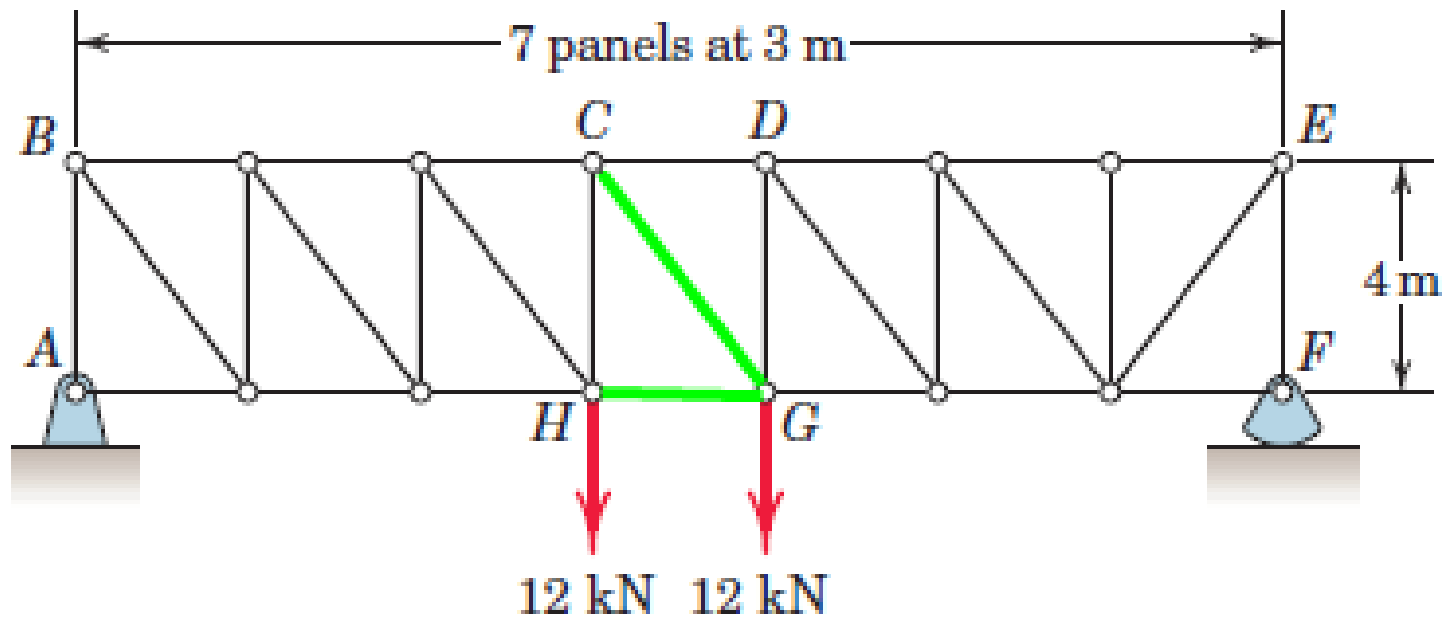
$$\sum F_y = 0: AB + 6.13 \sin 45^\circ - 21.3 = 0$$

$$AB = 17 \text{ kN T}$$

METHOD OF SECTIONS

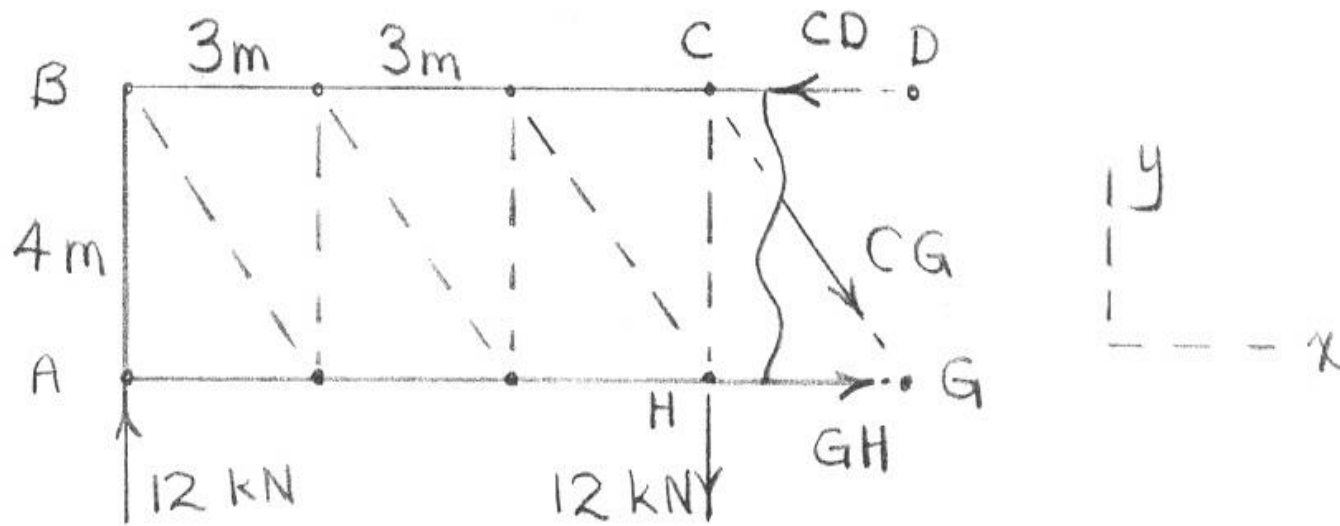
4/29 Determine the forces in members CG and GH .

Ans. $CG = 0$, $GH = 27 \text{ kN T}$



4/29

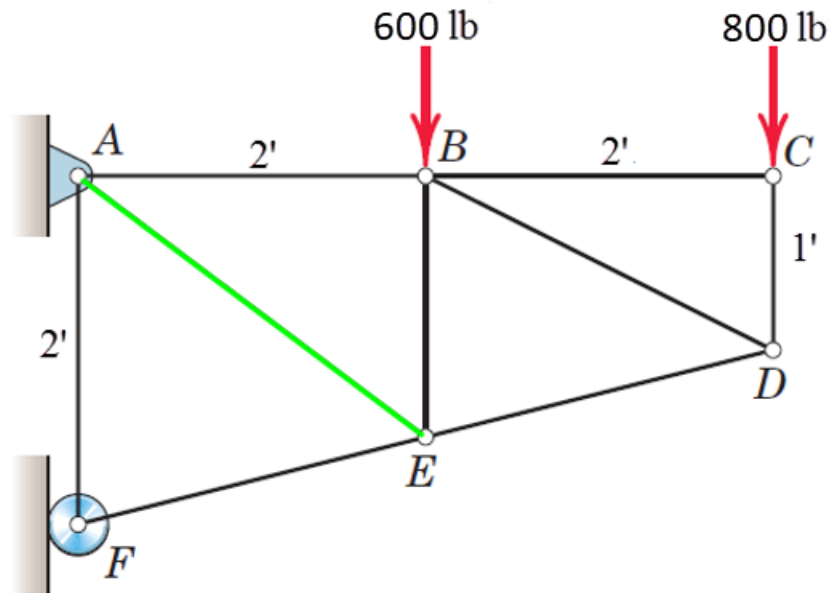
From truss as a whole, $A_y = F_y = 12 \text{ kN}$ and $A_x = 0$. Left section:

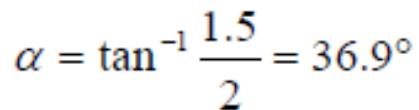


By inspection of $\sum F_y = 0$, $CG = 0$.

$$\curvearrowleft \sum M_C = 0 : GH(4) - 12(9) = 0, \quad \underline{GH = 27 \text{ kN T}}$$

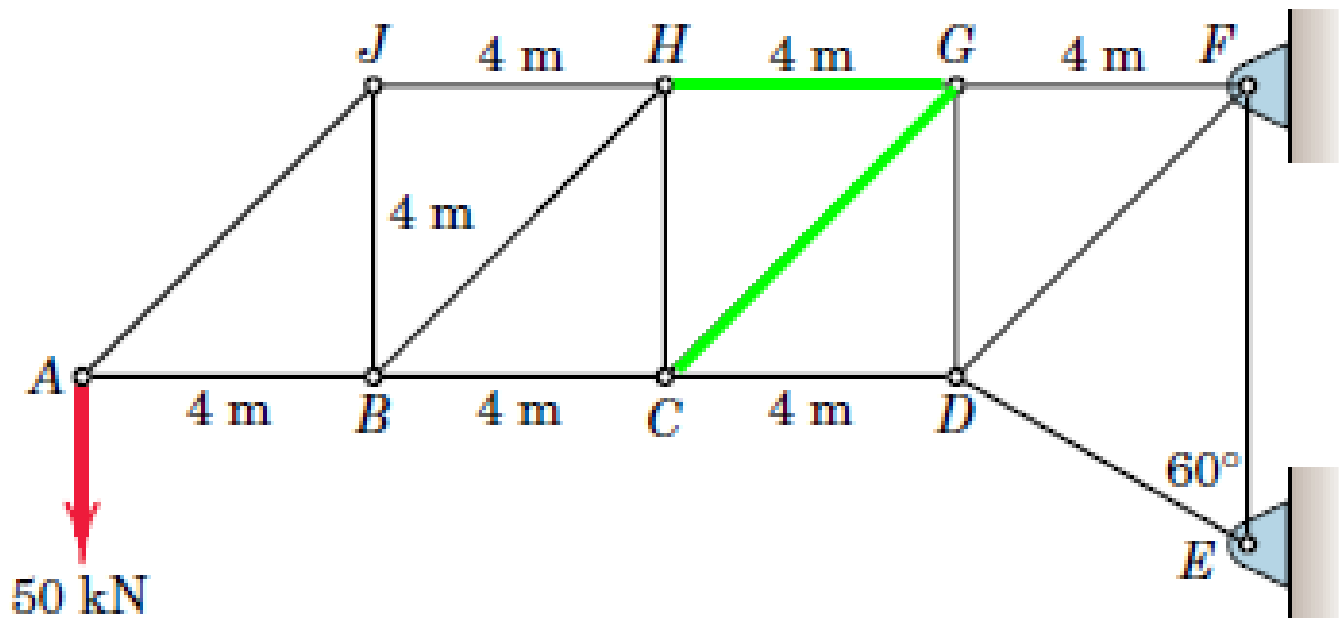
4/30 Determine the force in member AE of the loaded truss.



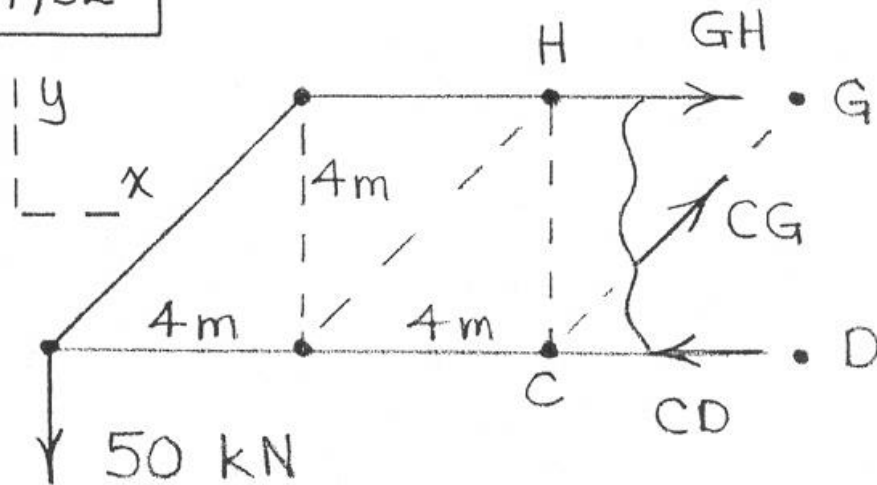


$$AE = 1417 \text{ lb } T$$

4/32 Determine the forces in members GH and CG for the truss loaded and supported as shown. Does the statical indeterminacy of the supports affect your calculation?



4/32

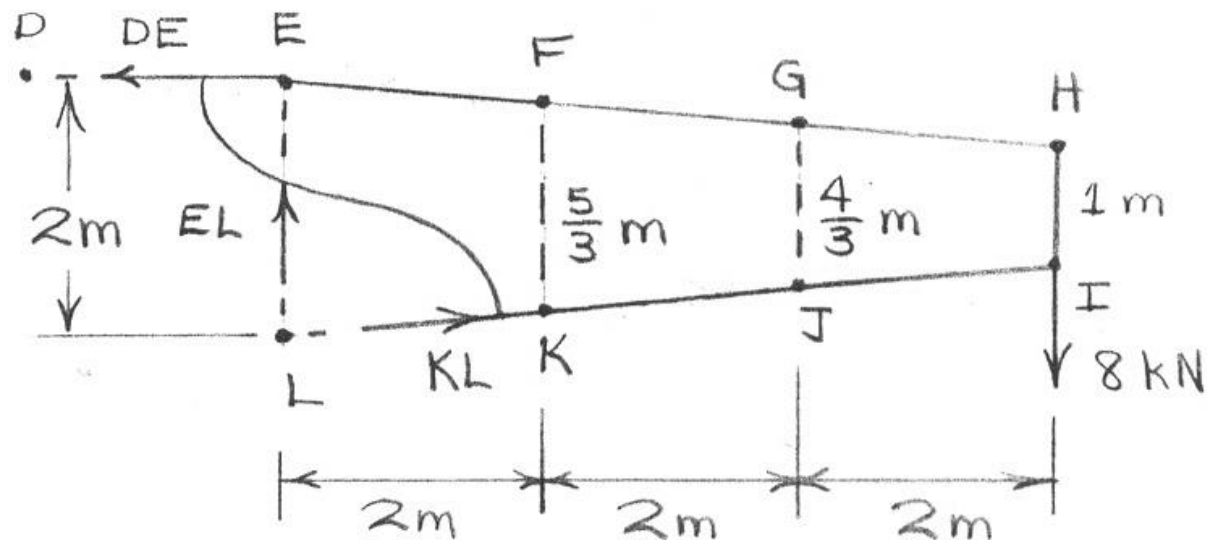


$$\sum F_y = 0 : CG \sin 45^\circ - 50 = 0, \quad \underline{CG = 70.7 \text{ kN T}}$$

$$\sum M_C = 0 : GH(4) - 50(8) = 0, \quad \underline{GH = 100 \text{ kN T}}$$

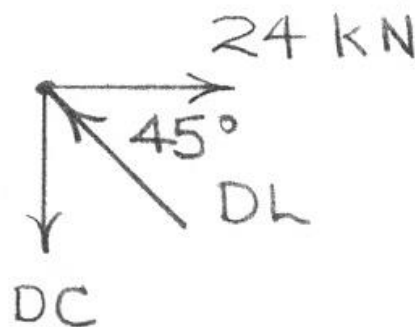
All members except EF are statically determinate, so above solution is unaffected by the redundant support.

4/35



$$\sum M_L = 0: DE(2) - 8(6) = 0, \quad \underline{DE = 24 \text{ kN T}}$$

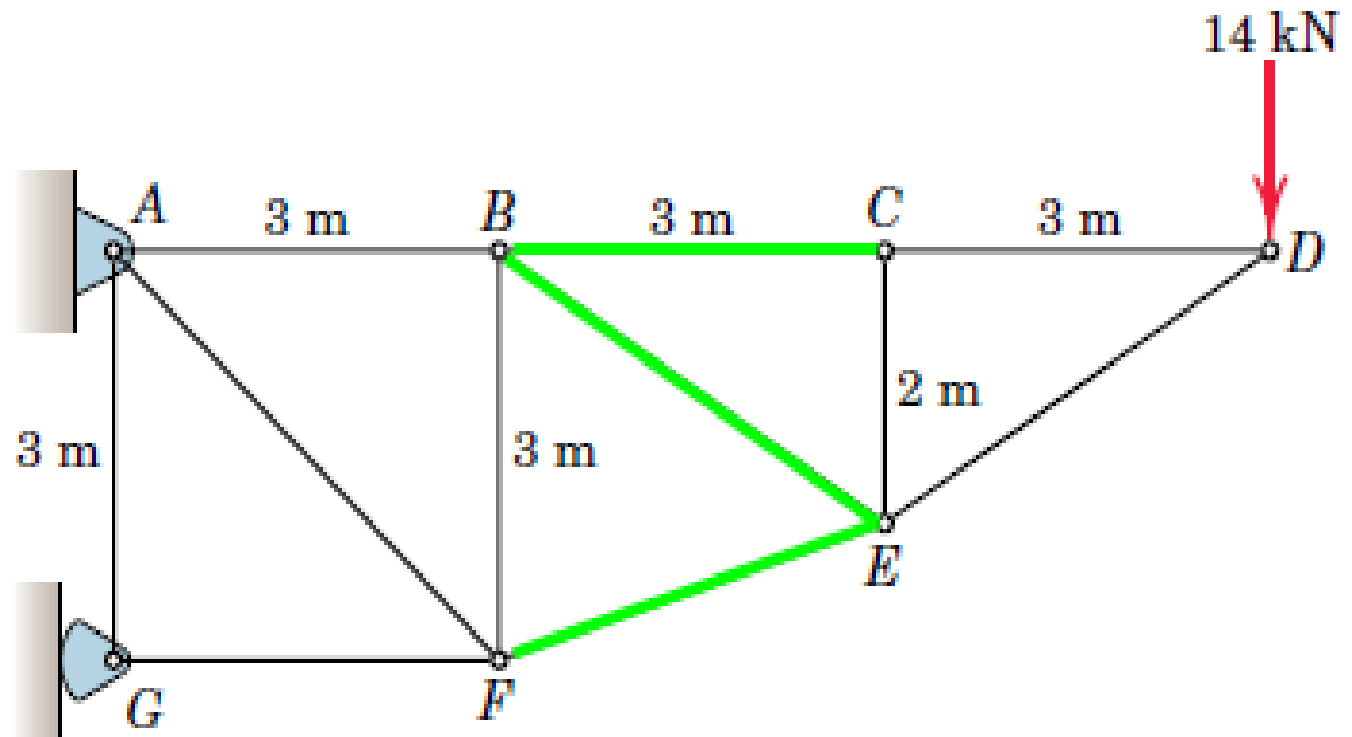
Joint D:



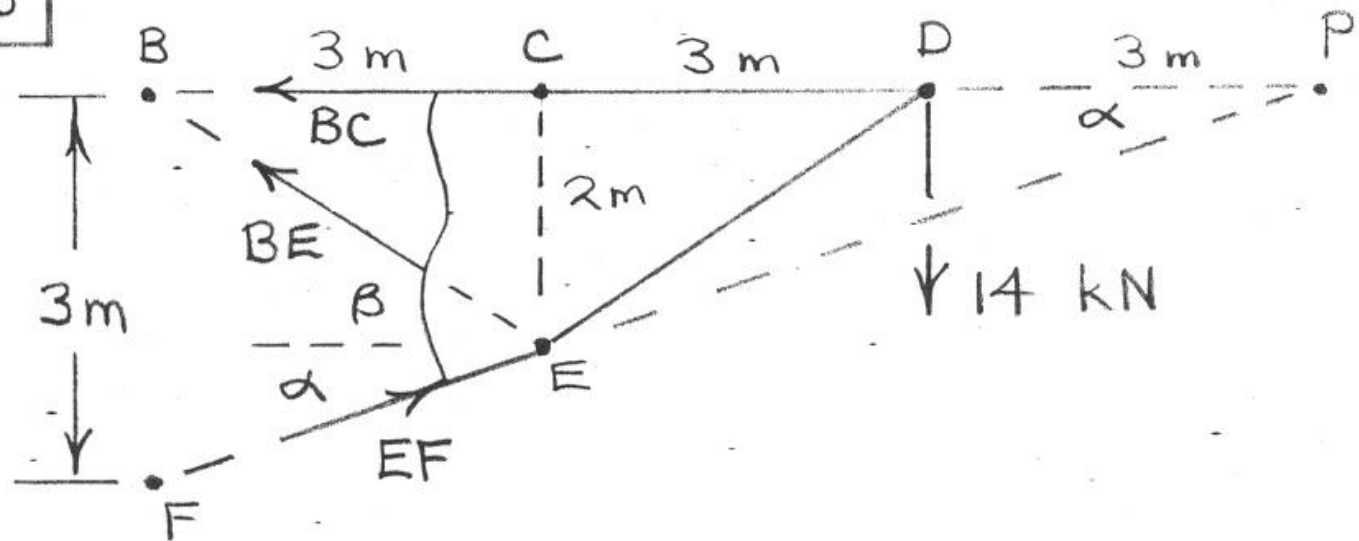
$$\Rightarrow \sum F_x = 0: 24 - DL \cos 45^\circ = 0$$

$$\underline{DL = 33.9 \text{ kN C}}$$

4/36 Calculate the forces in members BC , BE , and EF . Solve for each force from an equilibrium equation which contains that force as the only unknown.



4/36



$$\alpha = \tan^{-1}\left(\frac{2}{6}\right) = 18.43^\circ, \quad \beta = \tan^{-1}\frac{2}{3} = 33.7^\circ$$

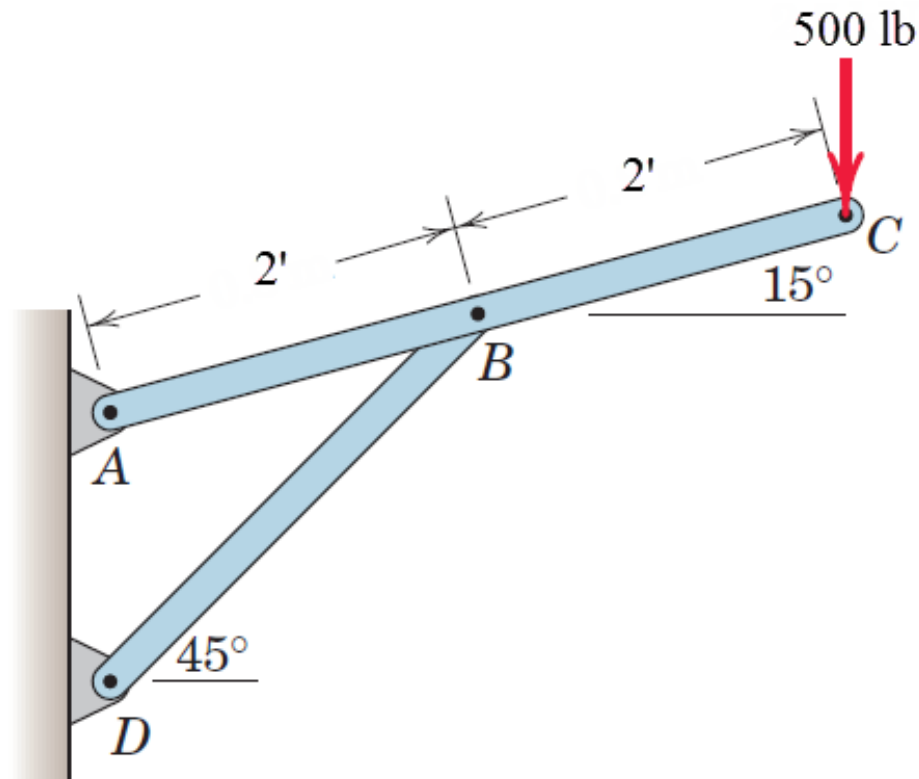
$$\uparrow + \sum M_E = 0: BC(2) - 14(3) = 0, \quad \underline{BC = 21 \text{ kN T}}$$

$$\uparrow + \sum M_P = 0: -BE \sin \beta (9) + 14(3) = 0, \quad \underline{BE = 8.41 \text{ kN T}}$$

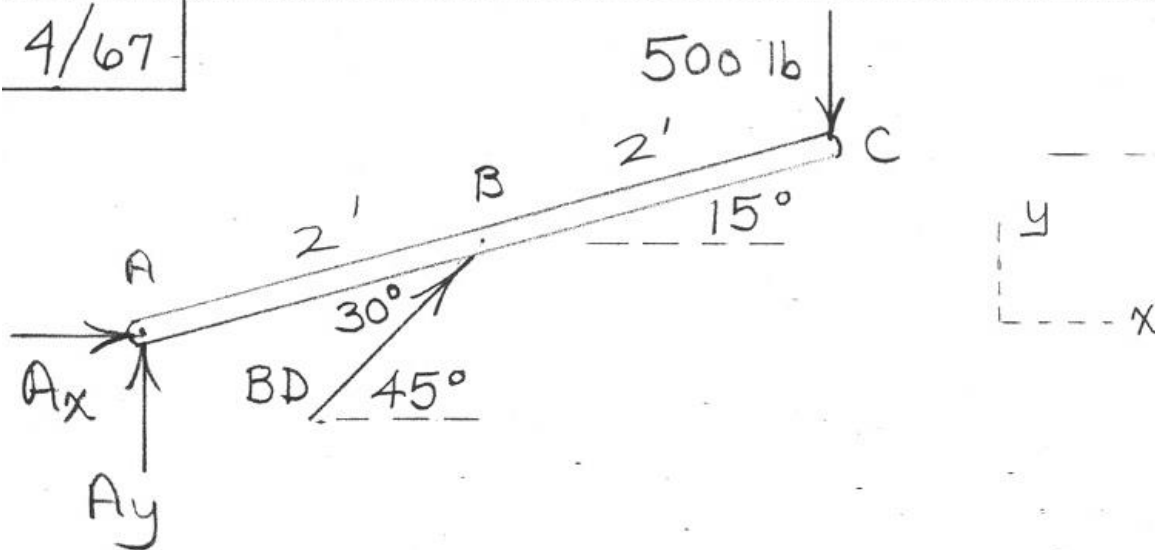
$$\uparrow + \sum M_B = 0: EF \cos \alpha (3) - 14(6) = 0, \quad \underline{EF = 29.5 \text{ kN C}}$$

FRAMES AND MACHINES

- 4/67** Determine the force in member BD and the forces supported by all pins in the loaded frame.



4/67



$$\uparrow + \sum M_A = 0: BD \sin 30^\circ (2) - 500 (4 \cos 15^\circ) = 0$$

$$\underline{BD = 1932 \text{ lb}}$$

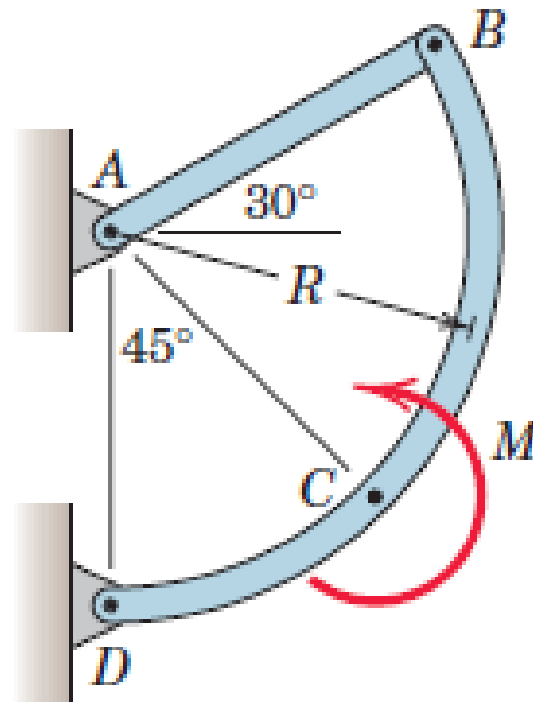
Forces at B and D = 1932 lb

$$\sum F_x = 0: A_x + 1932 \cos 45^\circ = 0, \quad A_x = -1366 \text{ lb}$$

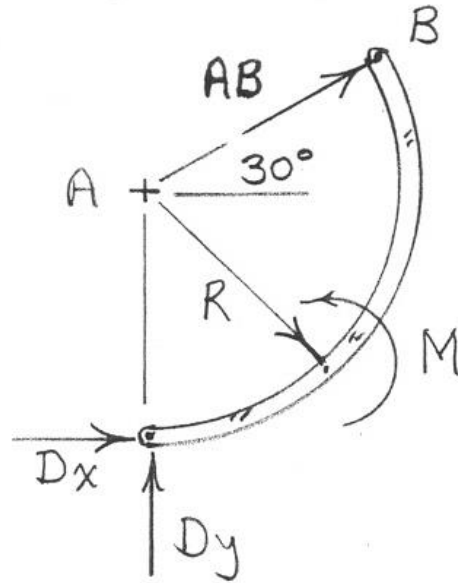
$$\sum F_y = 0: A_y + 1932 \sin 45^\circ - 500 = 0, \quad A_y = -866 \text{ lb}$$

$$A = \sqrt{1366^2 + 866^2} = \underline{1617 \text{ lb}}$$

4/68 Determine the force supported by each pin of the loaded truss.



4/68



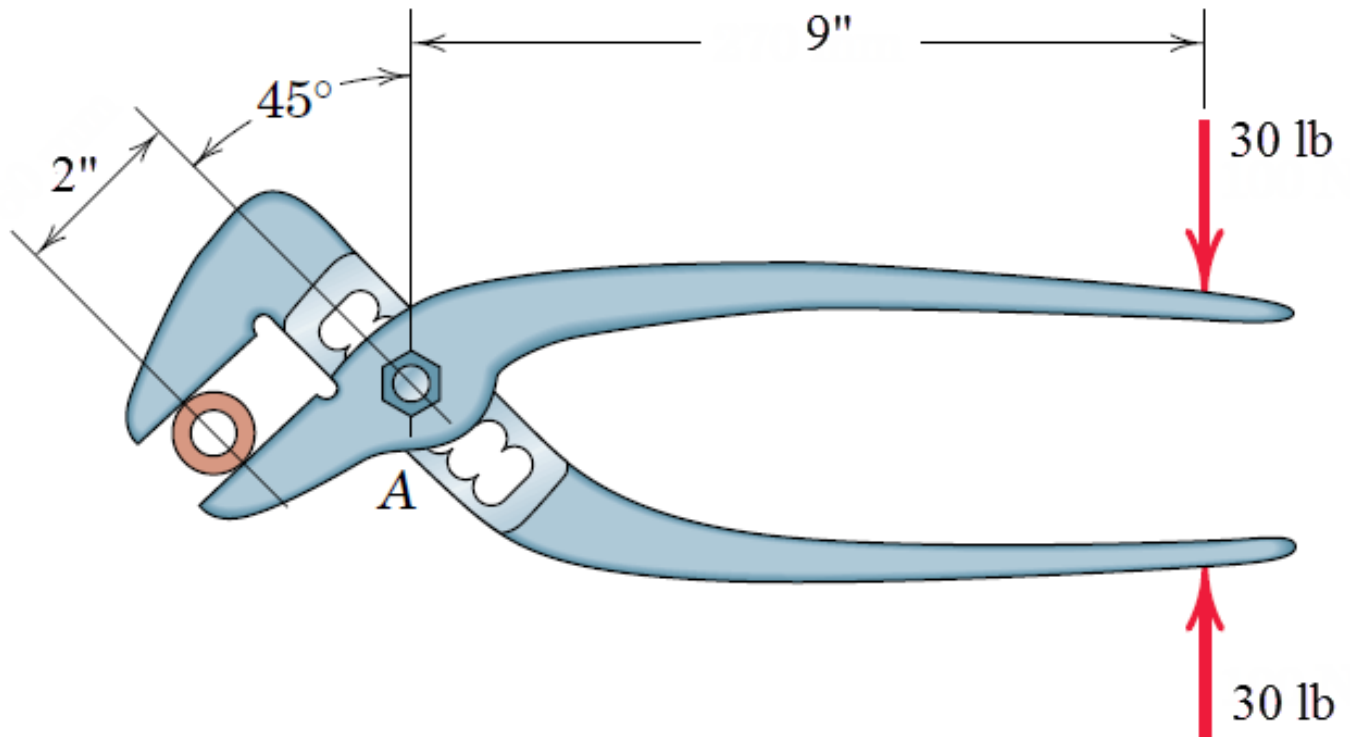
$$\sum M_D = 0 : -AB \cos 30^\circ (R) + M = 0$$

$$AB = \frac{M}{R \cos 30^\circ} = \frac{M}{R \sqrt{3}/2} = \frac{2\sqrt{3} M}{3R}$$

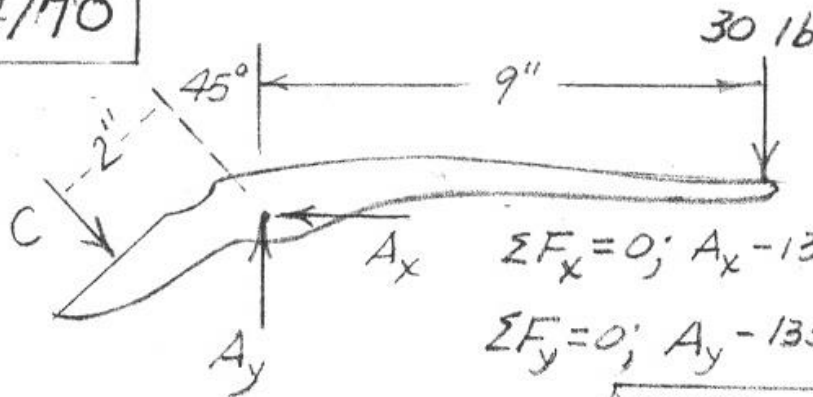
Load is a couple, so reactions form a couple:

$$A = D = -\frac{2\sqrt{3} M}{3R}$$

4/70 Compute the force supported by the pin at A for the slip-joint pliers under a grip of 30 lb.



4/70



$$\Sigma M_A = 0;$$

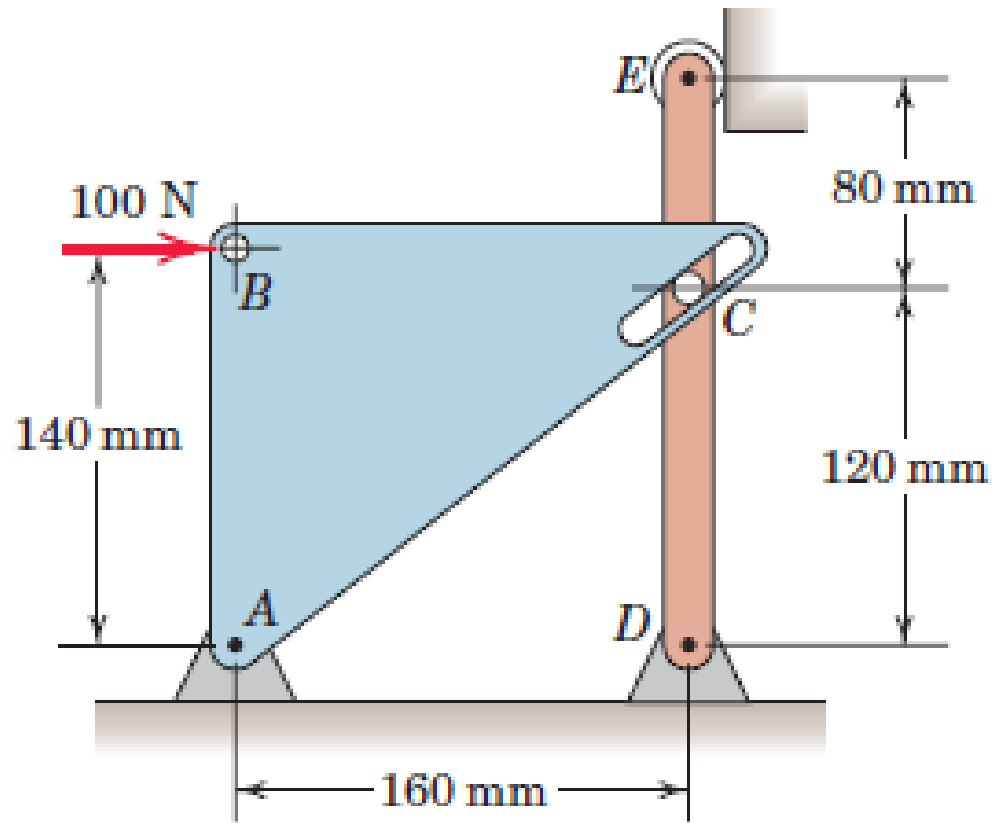
$$30(9) - 2C = 0, C = 135 \text{ lb}$$

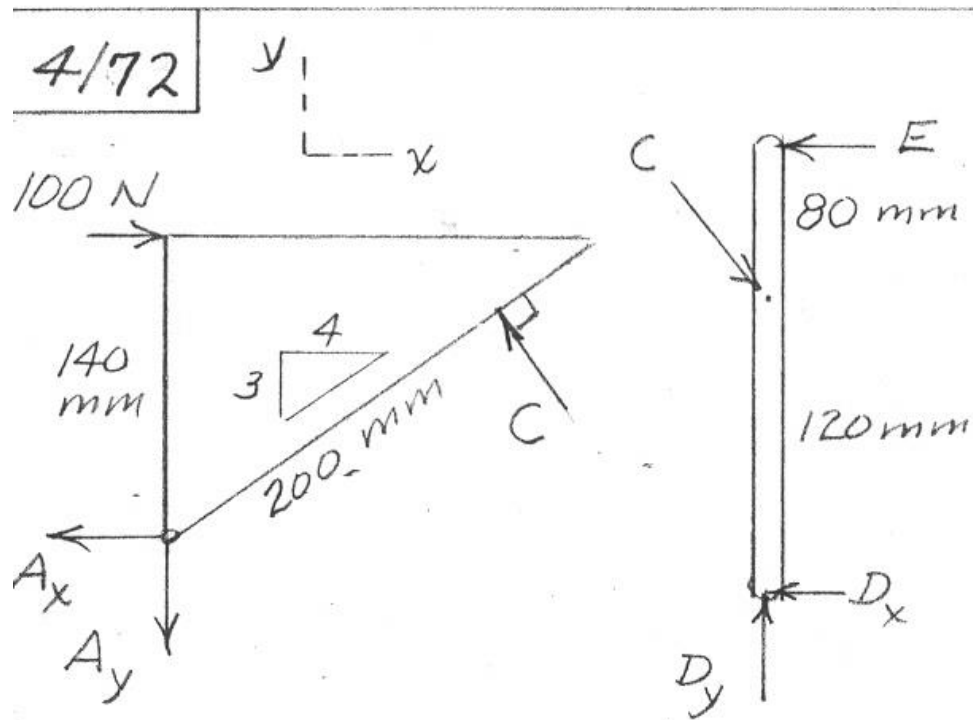
$$\Sigma F_x = 0; A_x - 135/\sqrt{2} = 0, A_x = 95.4 \text{ lb}$$

$$\Sigma F_y = 0; A_y - 135/\sqrt{2} - 30 = 0, A_y = 125.4 \text{ lb}$$

$$A = \sqrt{95.4^2 + 125.4^2} = \underline{157.6 \text{ lb}}$$

4/72 Calculate the magnitude of the force acting on the pin at D . Pin C is fixed in DE and bears against the smooth slot in the triangular plate.





Triangle :

$$\sum M_A = 0; 200C - 140(100) = 0, \quad C = 70 \text{ N}$$

Bar:

$$\sum M_D = 0; 200E - 70\left(\frac{3}{5}\right)120 = 0; \quad E = 25.2 \text{ N}$$

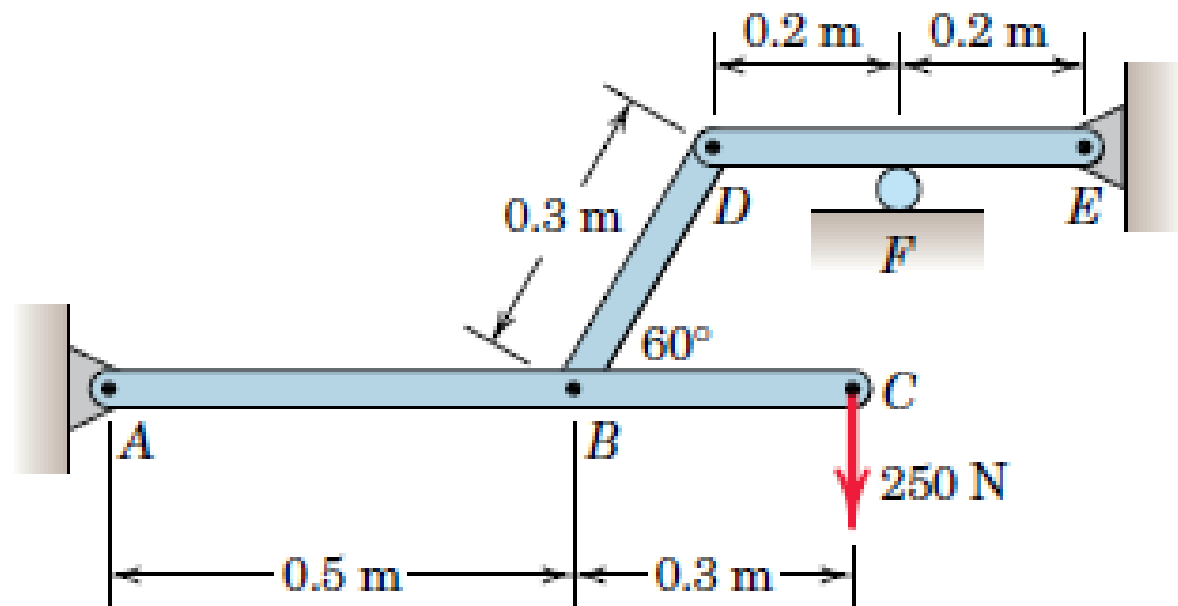
$$\sum F_x = 0; D_x + 25.2 - 70\left(\frac{3}{5}\right) = 0, \quad D_x = 16.8 \text{ N}$$

$$\sum F_y = 0; D_y - 70\left(\frac{4}{5}\right) = 0, \quad D_y = 56 \text{ N}$$

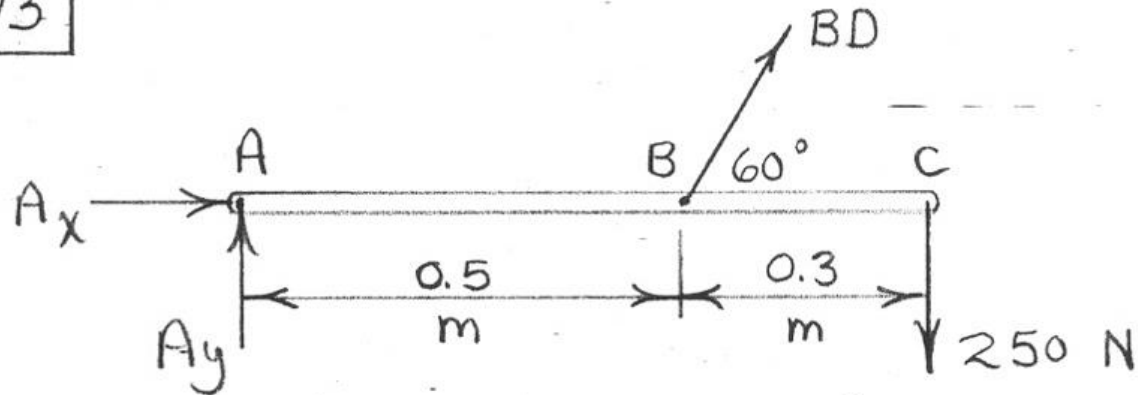
$$D = \sqrt{D_x^2 + D_y^2} = \sqrt{16.8^2 + 56^2} = \underline{58.5 \text{ N}}$$

4/73 Determine the reaction at the roller F for the frame loaded as shown.

Ans. $F = 800\text{ N}$

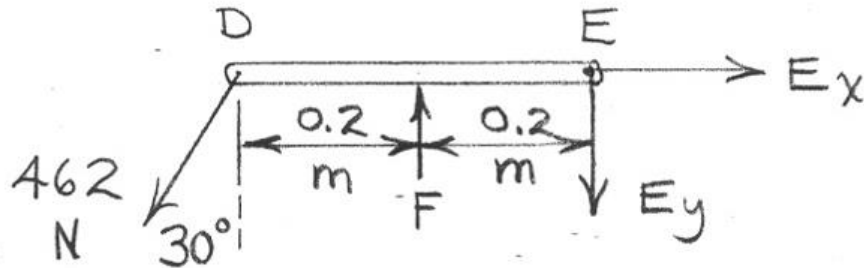


4/73



$$\curvearrowright + \sum M_A = 0 : (BD \sin 60^\circ)(0.5) - 250(0.8) = 0$$

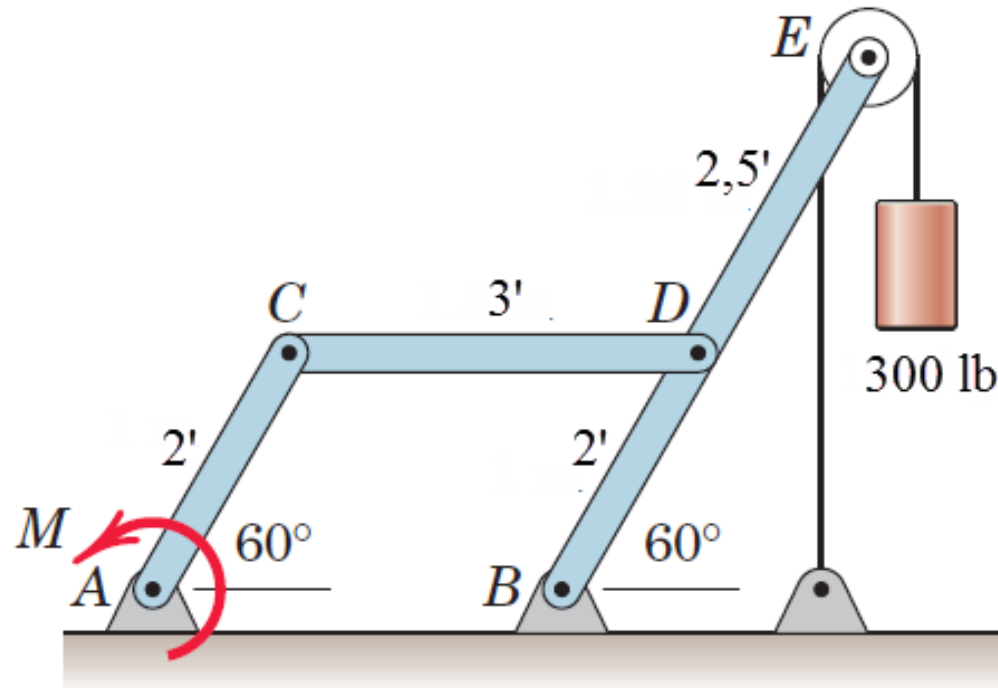
$$BD = 462 \text{ N}$$



$$\curvearrowright + \sum M_E = 0 : 462 \cos 30^\circ (0.4) - F(0.2) = 0$$

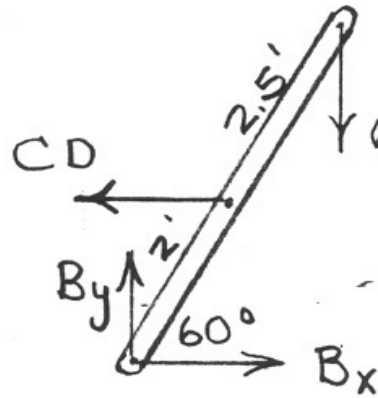
$$\underline{F = 800 \text{ N}}$$

- 4/78** Determine the moment M which must be applied at A to keep the frame in static equilibrium in the position shown. Also calculate the magnitude of the pin reaction at A .



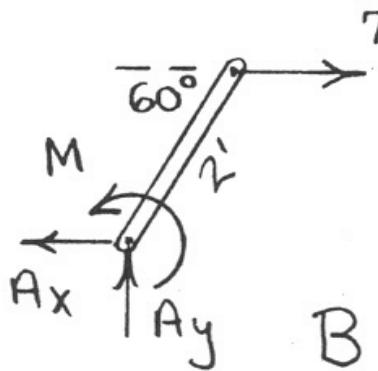
4/78

Member BE :



$$\downarrow + \sum M_B = 0: -600(4.5 \cos 60^\circ) + CD(2 \sin 60^\circ) = 0, \quad CD = 779 \text{ lb}$$

Member AC :



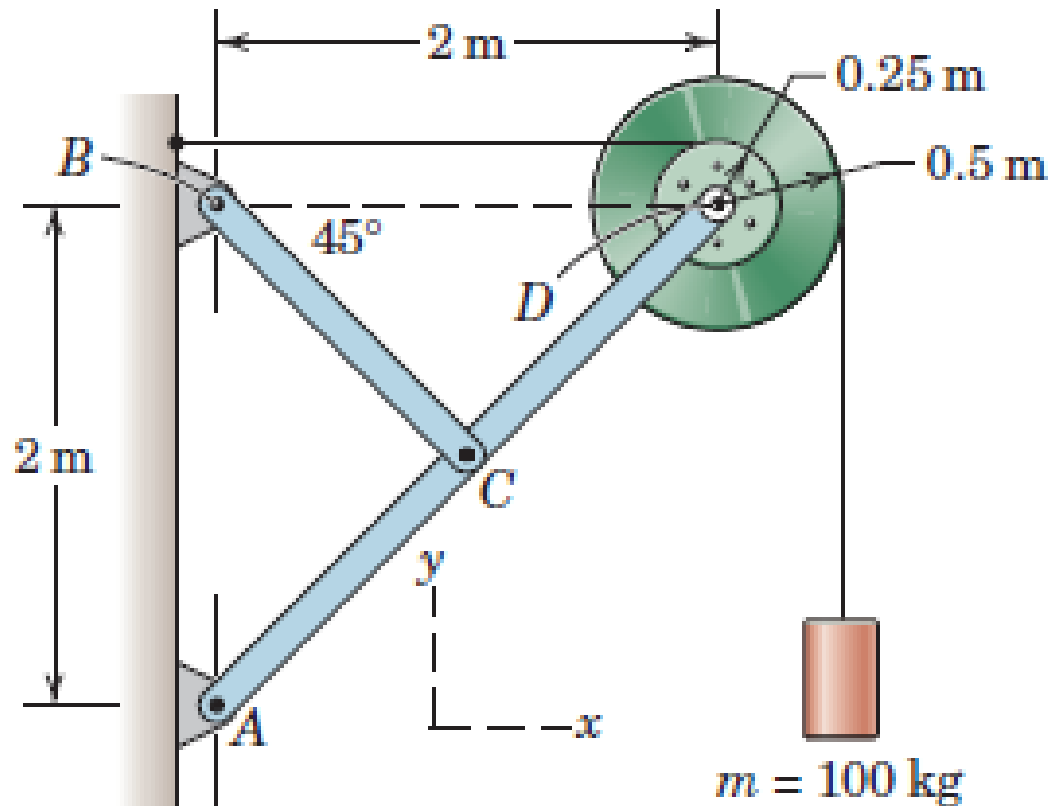
$$\sum M_A = 0: -779(2 \sin 60^\circ) + M = 0$$

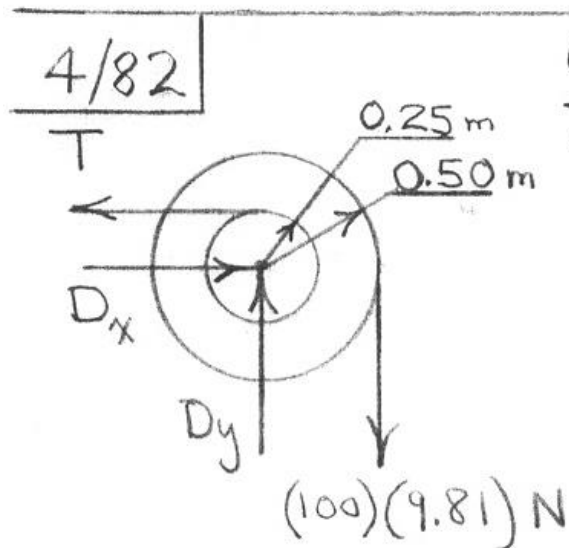
$$\underline{M = 1350 \text{ lb-ft} \quad \curvearrowleft}$$

By inspection, $A_x = 779 \text{ lb}, A_y = 0$

$$\therefore A = \sqrt{A_x^2 + A_y^2} = \underline{779 \text{ lb}}$$

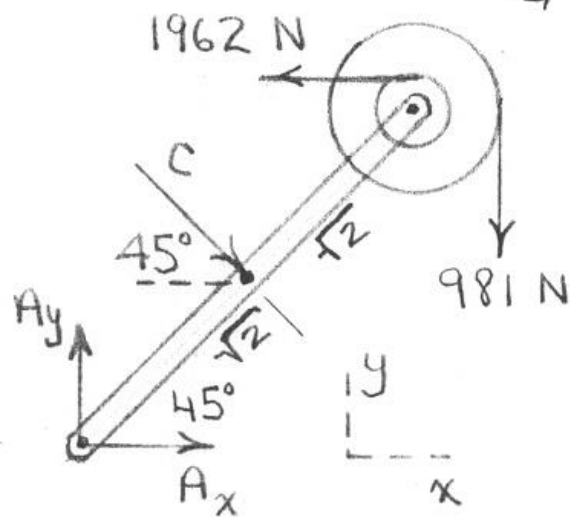
4/82 Calculate the x - and y -components of the force C which member BC exerts on member ACD . The cables are wrapped securely around the two pulleys, which are fastened together.





$$\sum M_D = 0 : T(0.25) - 981(0.5) = 0$$

$$T = 1962 \text{ N}$$



$$\sum M_A = 0 : -C\sqrt{2} + 1962(2 + 0.25) - 981(2 + 0.5) = 0$$

$$C = 1390 \text{ N}$$

$$\therefore \begin{aligned} C_x &= 981 \text{ N} \\ C_y &= -981 \text{ N} \end{aligned}$$



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